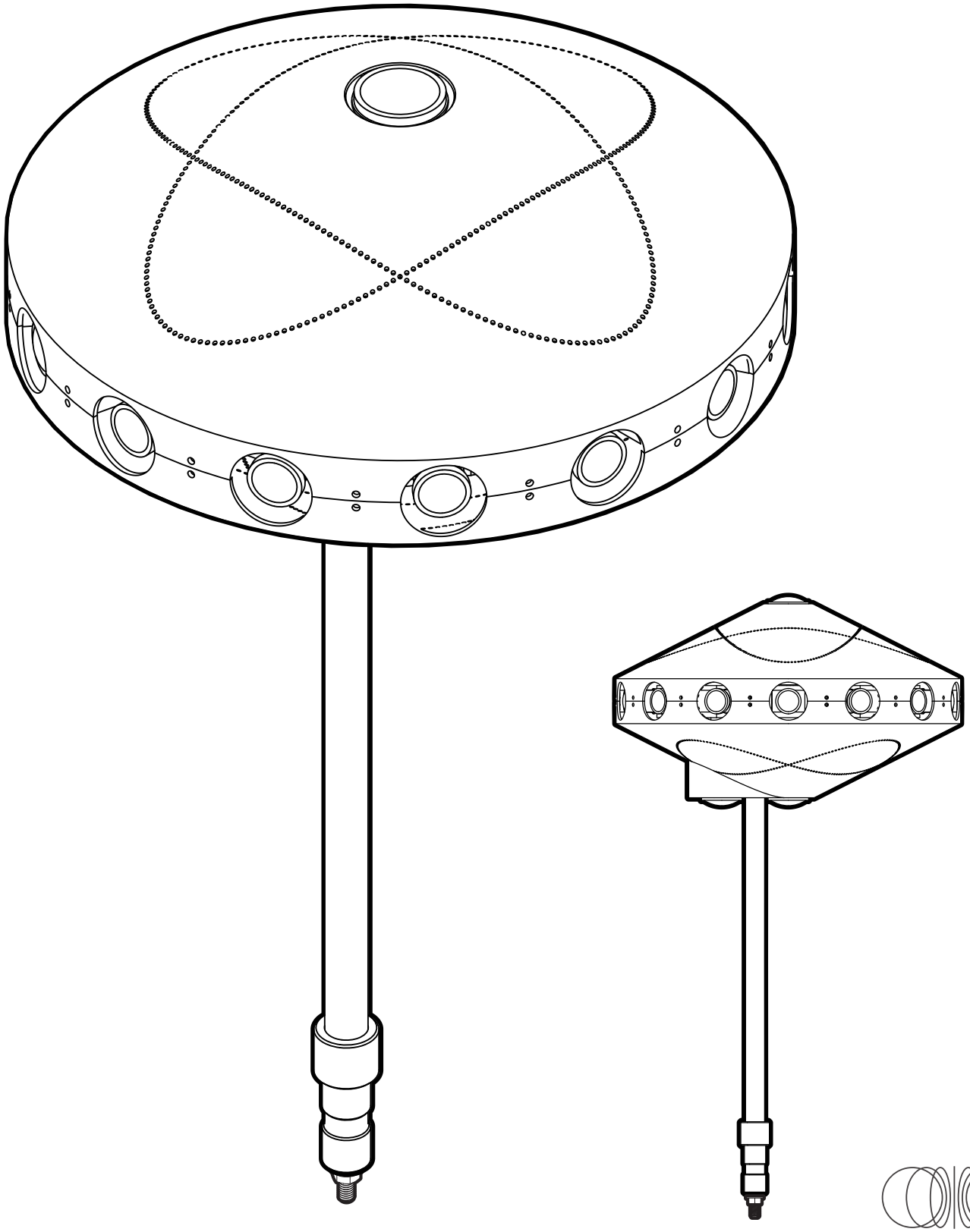
facebook

SURROUND 360



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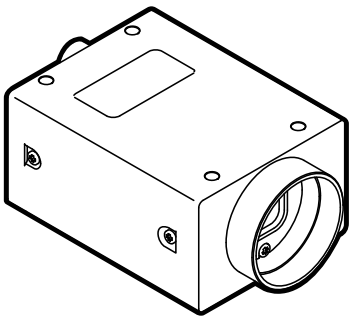
1

Parts List

1

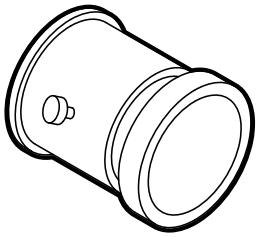
1.
Camera & Lenses

17x



Camera
Point Grey GS3-U3-41C6C-C

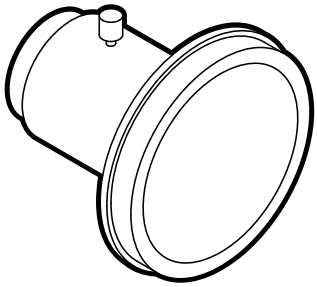
14x



Wide-Angle Lens
with custom focusing barrel
(FB360_V1_05, _33)
Sunex DSL318B-650-F2.4

OPTIONAL:
M3 brass jam nuts (14x)

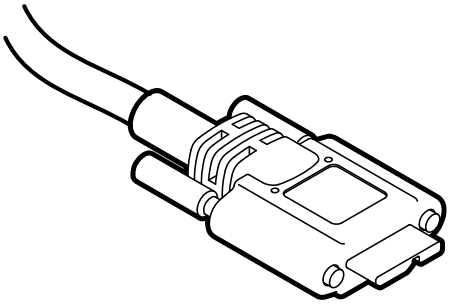
3x



Fisheye Lens
(FB360_V1_06)
Fujinon FE185C086HA-1

2.
Wiring

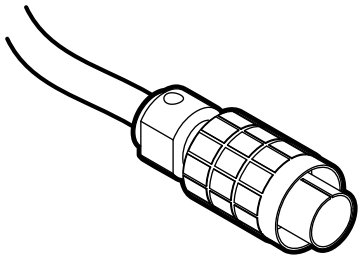
17x



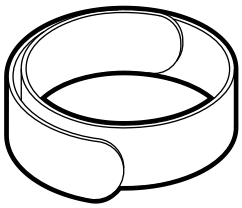
USB Cable
Type A to Micro-B

OPTIONAL:
M2 thread, 0.4mm pitch, 20mm long replacement screw

17x



GPIO Cable
(FB360_V1_05, _33)
Hirose HR25 circular connector, 8 pins

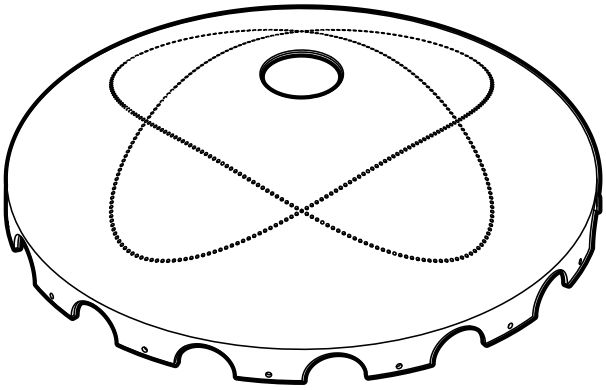


Velcro Straps

1 | 3.

Machined Parts

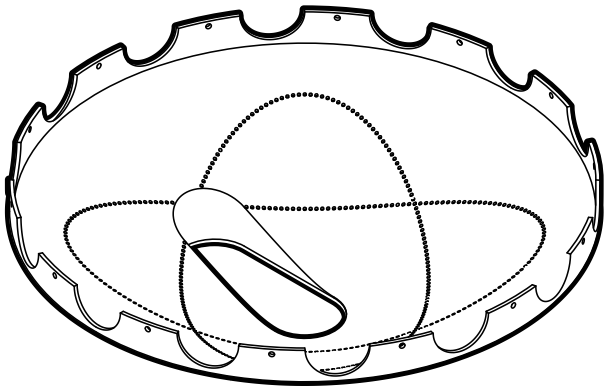
1x



Top Cover
(FB360_V1_30)

mild steel,
painted finish

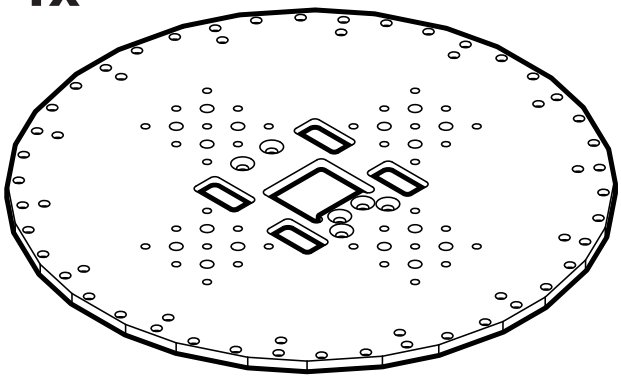
1x



Bottom Cover
(FB360_V1_29)

mild steel,
painted finish

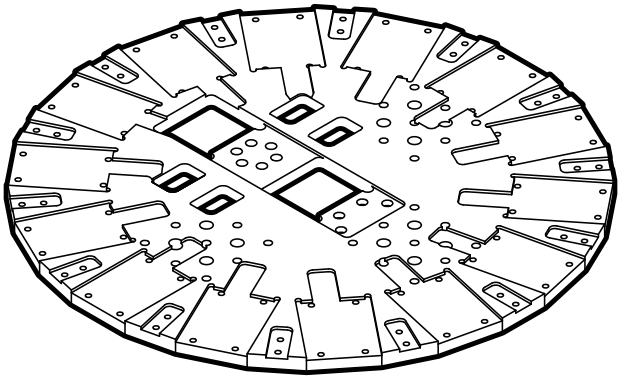
1x



Top Plate
(FB360_V1_22)

1/4″ thick Mic-6 aluminum,
black anodize finish

1x



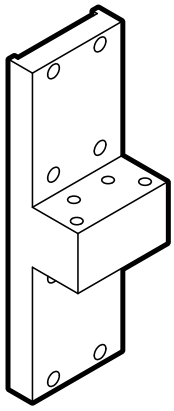
Base Plate
(FB360_V1_21)

3/8″ thick Mic-6 aluminum,
black anodize finish

1 | 4.

Machined Parts

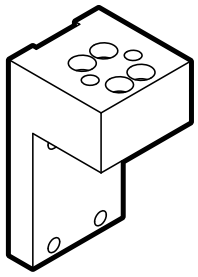
1x



Upright
(FB360_V1_23)

6061 aluminum,
black anodize finish

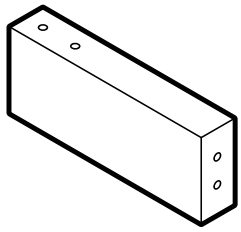
1x



Camera Bracket
(FB360_V1_24)

6061 aluminum,
black anodize finish

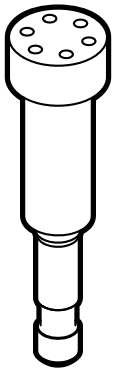
14x



Shell Support
(FB360_V1_28)

6061 aluminum,
black anodize finish

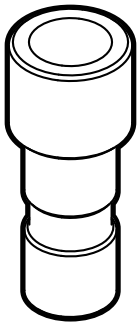
1x



Post
(FB360_V1_25)

1144 carbon steel,
black oxide finish

1x



Adapter
(FB360_V1_27)

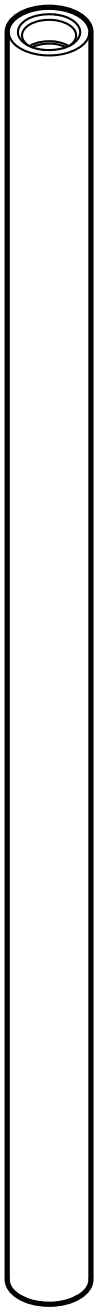
1144 carbon steel,
black oxide finish

1x

Stop Nut
(FB360_V1_31)

Brass

1x



Support Tube
(FB360_V1_26)

4130 CR tubing,
black oxide finish

1 | 5.

Hardware

152x

M3 Socket-Head Cap Screw

Type 18-8 stainless steel
M3 thread, 6mm length,
0.5mm pitch

152x

M3 Belleville Spring Lock Washer

Type 18-8 stainless steel
M3 screw size

28x

M3 Flanged Button-Head Socket Cap Screw

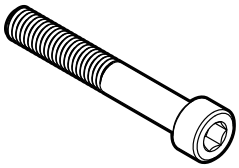
Black-oxide,
M3 thread, 10mm length,
0.5mm pitch

8x

M6 Flat-Head Screw

Type 18-8 stainless steel
M6 thread, 14mm length,
1.0mm pitch

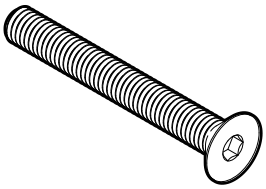
2x



M6 Socket-Head Cap Screw

Type 18-8 stainless steel
M6 thread, 40mm length,
1.0mm pitch

4x



M6 Flat-Head Screw

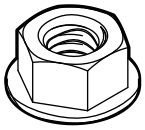
Type 18-8 stainless steel
M6 thread, 50mm length,
1.0mm pitch

16x

M6 Belleville Spring Lock Washer

Type 18-8 stainless steel
M6 screw size

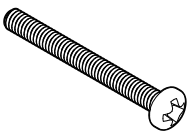
1x



Steel Rotating Flanged Nut

Black-oxide steel,
5/16″-18 thread size
3/4″ flange diameter
7/16″ overall height

Optional



M2 Socket-Head Cap Screw

Type 18-8 stainless steel
M2 thread, 20mm length,
0.4mm pitch

(replace USB locking screws)

M3 Jam Nut

Brass, M3 thread size,
5.5mm Wide, 1.8mm
High

(lock nut for Sunex barrel)

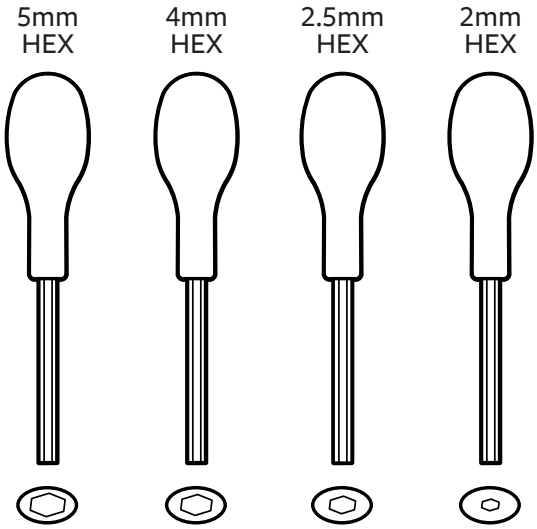
1x

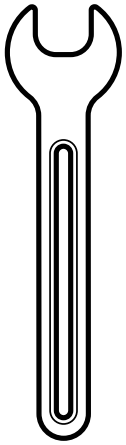
Steel Threaded Rod

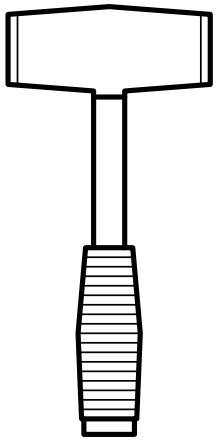
(FB360_V1_32)

ASTM A193 grade B7 steel
5/16″-18 thread
20-1/4″ long
fully threaded

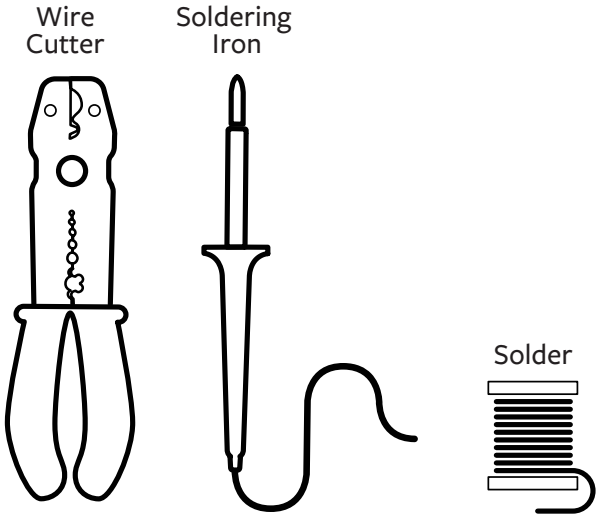
Hand Tools



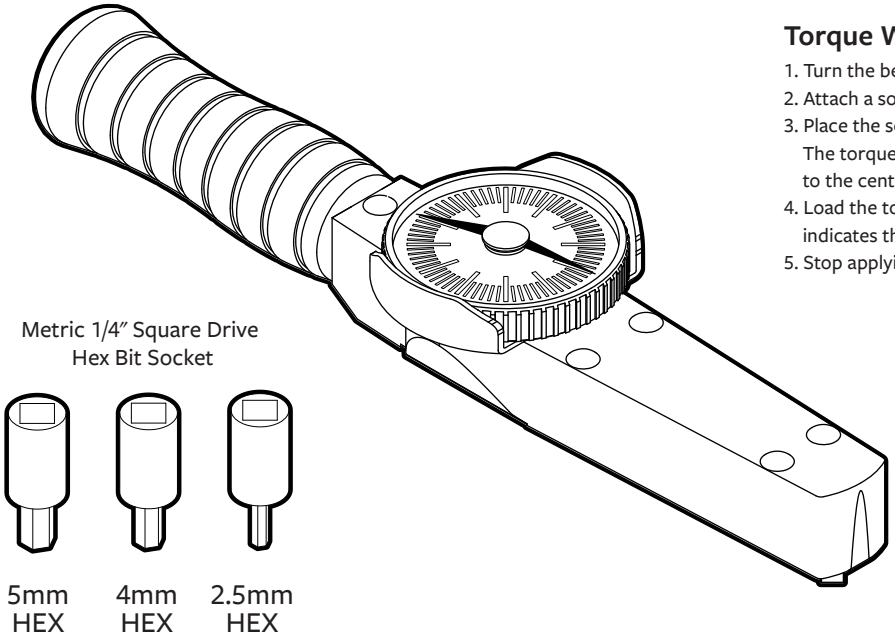




Wiring Tools



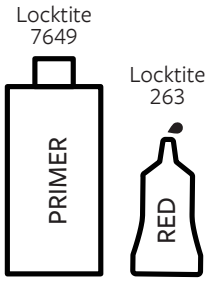
Torque Wrench



Torque Wrench Instructions:

1. Turn the bezel until pointer is resting on zero.
2. Attach a socket or attachment to the square drive.
3. Place the socket or attachment over the fastener to be torqued. The torque wrench handle must be perpendicular (90 degrees) to the centerline of the drive, socket and fastener.
4. Load the torque wrench at a slow and steady rate until the pointer indicates the desired torque.
5. Stop applying force. The pointer should return to zero.

Threadlockers & Adhesive

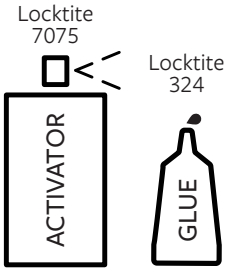


Permanent Threadlocker

Removable Threadlocker

Threadlocker Instructions:

1. Apply primer to surface of bolt and nut.
2. Apply several drops of the threadlocker onto the bolt at the nut engagement area.
3. Assemble parts and tighten as required. Sets in approximately 10 minutes and fully cures in 24 hours.



Cyanoacrylate Adhesive

Adhesive Instructions:

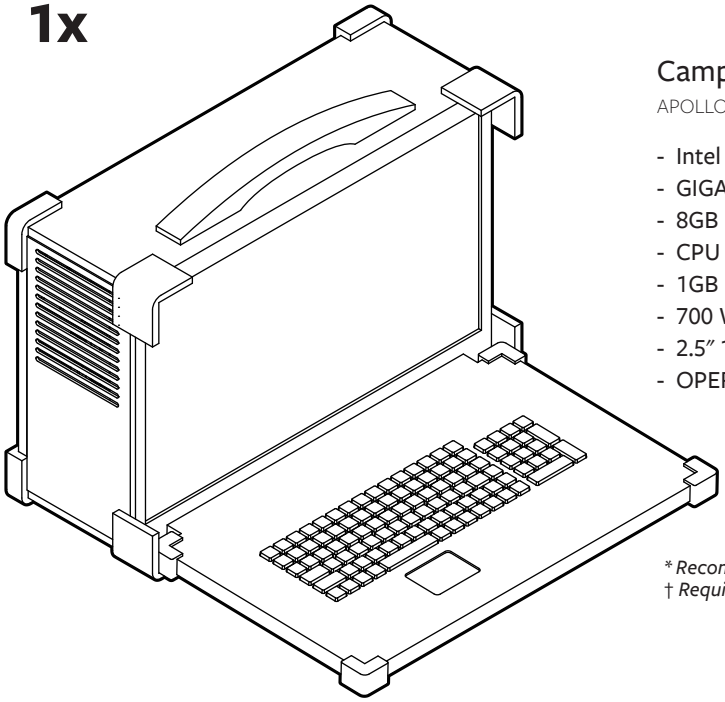
1. Apply the surface activator to both surfaces.
2. Wait 20 seconds for the activator to completely dry.
3. Apply glue sparingly to one side only using approximately one drop per square inch of surface.
4. Press parts together immediately.
5. Hold in place for 30 seconds or until bond sets.

1 | 8.

Computer Hardware

Assembly Note
See system setup instructions for details

1x

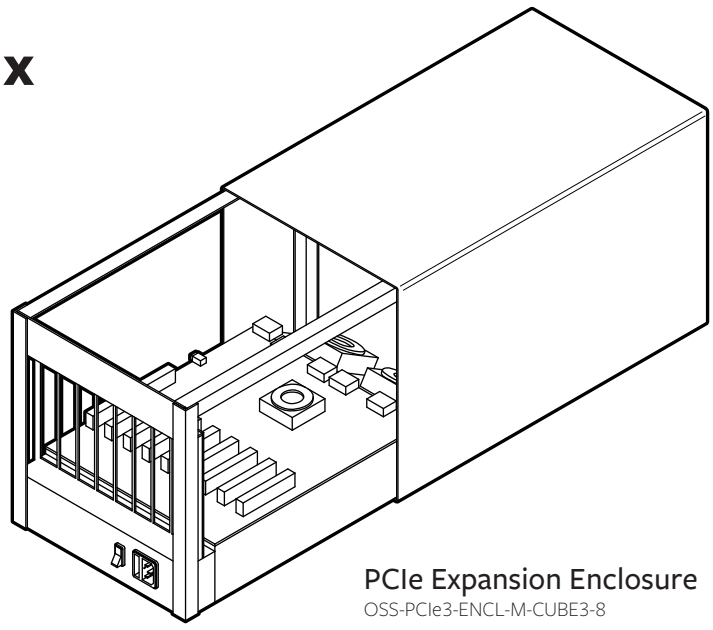


Camputer - Ruggedized “Lunchbox” Computer
APOLLO-A1 MODEL

- Intel Core i7-5960X Haswell-E 8-Core 3.0 GHz LGA 2011-v3*
- GIGABYTE GA-X99P-SLI (rev. 1.0) LGA 2011-v3 Intel X99 Motherboard*
- 8GB DDR4 2400 288-PIN Memory = (64GB of Memory Installed)*
- CPU COOLING FAN FOR LGA 2011-v3*
- 1GB NVIDIA PCIe x16 VIDEO CARD*
- 700 WATT POWER SUPPLY*
- 2.5″ 128GB SSD*
- OPERATING SYSTEM - UBUNTU 14.04 LTS†

** Recommended*
† Required

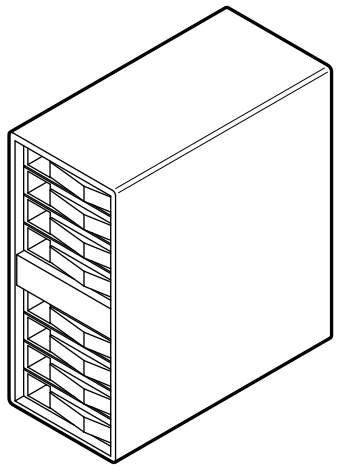
1x



PCIe Expansion Enclosure
OSS-PCIe3-ENCL-M-CUBE3-8

Expansion Backplane
OSS-BP-452

1x



8-bay 12G SAS RAID Tower
ARC-4038

1 | 9.

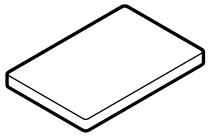
Computer Hardware

Assembly Note
See system setup instructions for details

1x

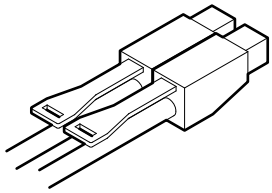
Shunt Jumper
390088-1

8x



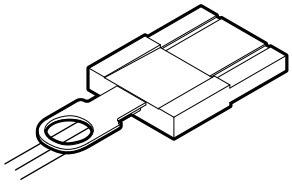
1TB SSD
MZ 7KE1T0BW

2x



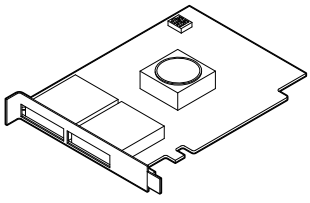
External 4x HD Mini-SAS Cable
SFF-8644 to SFF-8644

1x



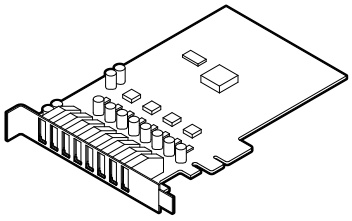
PCIe x8 Active Optical Cable
OSS-PCIe3-CBL-ACT-x8-50M-1x

2x



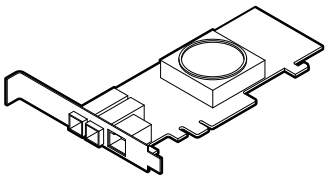
PCIe x8 Gen 3 Cable Adapter
OSS-PCIe-HIB38-x8-DUAL

5x



USB 3.0 Expansion Card
UE-1008 or UE-1004

1x



PCIe 3.0 x8 SAS RAID Adapter
ARC-1883X

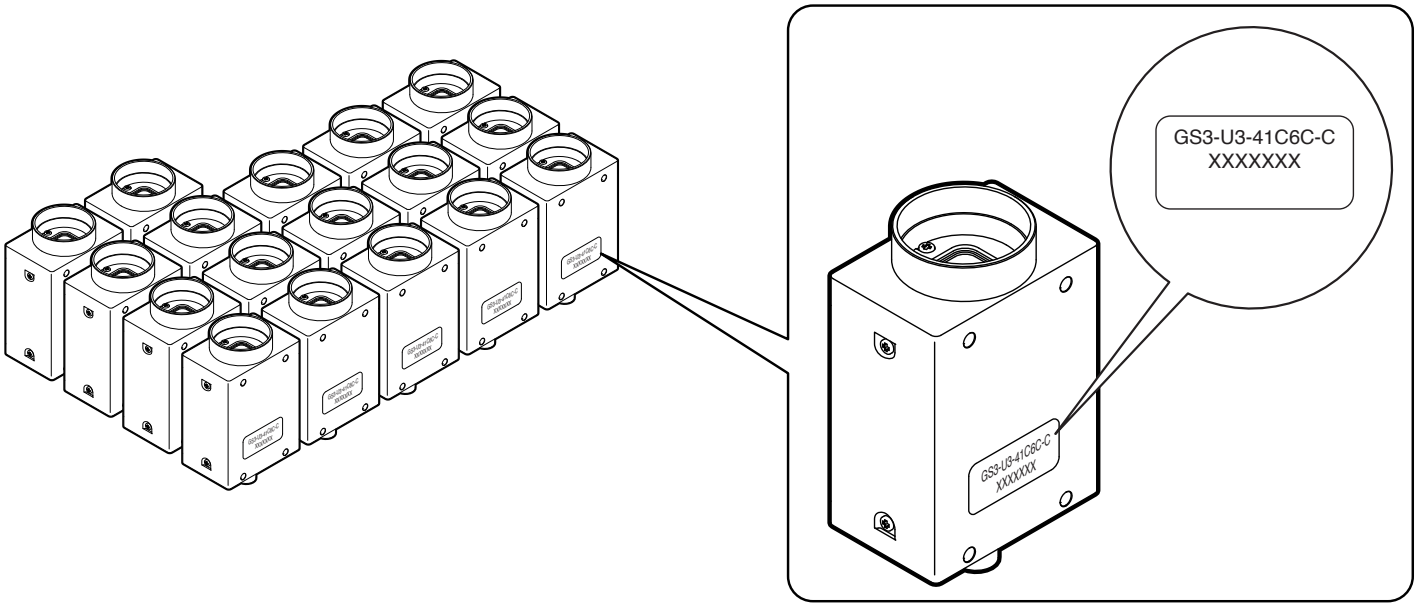
2

Before You Start

2

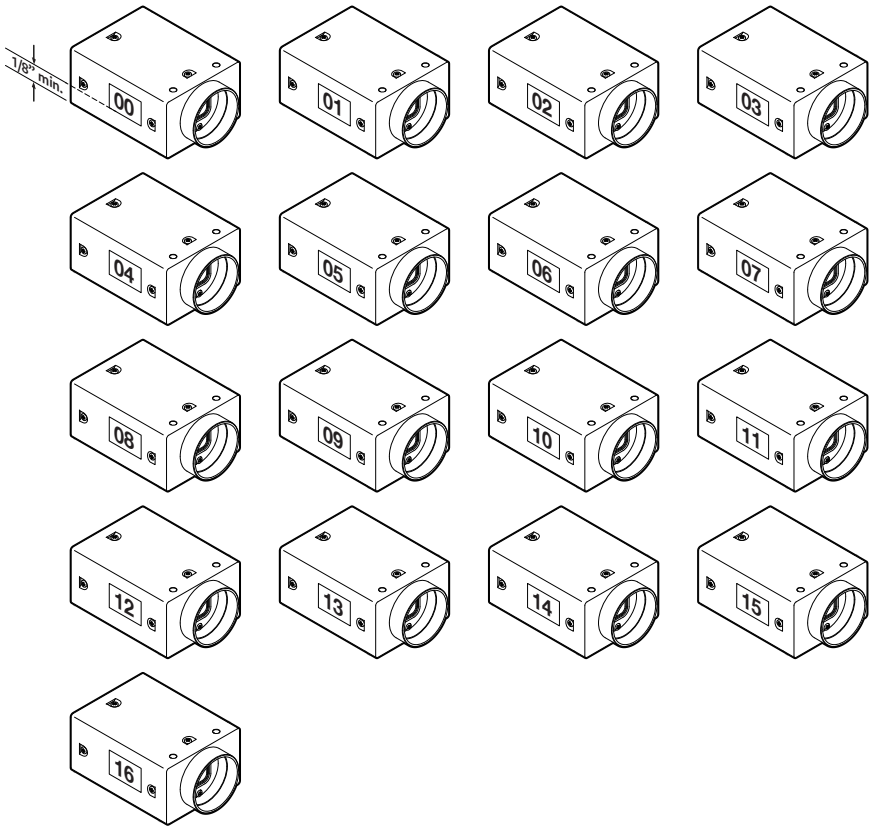
1.

Arrange Cameras in Ascending Sequential Order (Per Serial Number)



2.

Label Cameras # 00 - 16

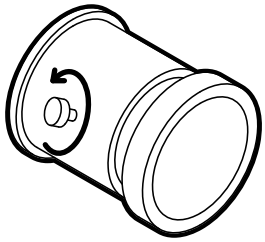


Tip:
Keep labels 1/8” min. away from bottom of camera to clear mounting plate

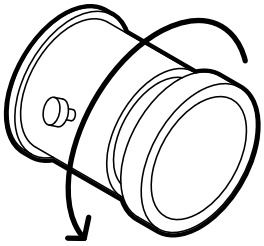
2 | 3.

Replace Focusing Barrel of Wide-Angle Lenses (# 01 - 14)

Step 1:
Remove thumb screw



Step 2:
Remove existing barrel and internal spring (counterclockwise)



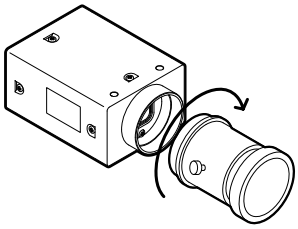
Step 3:
Screw on new focusing barrel* clockwise and tighten thumb screw

Step 4 (Optional):
Tighten optional lock nuts (M3 brass jam nuts)

** Refer to DWG FB360_V1_33*

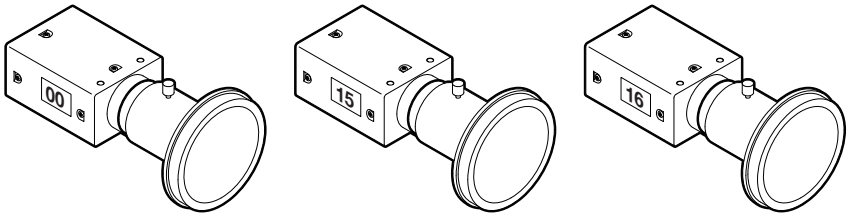
4.

Mount Lenses

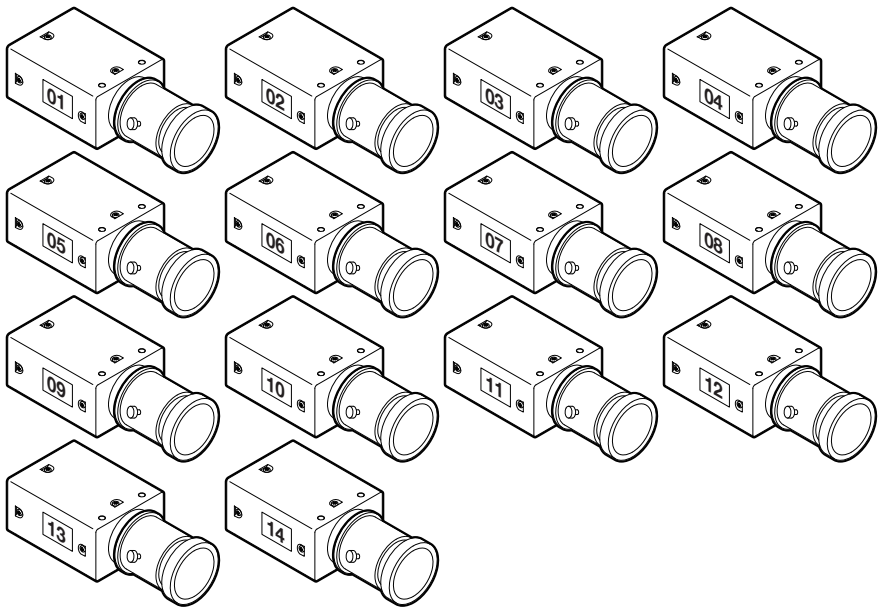


Tip:
Screw lens onto camera clockwise till its snudge

Fujinon Fisheye - # 00, 15, 16



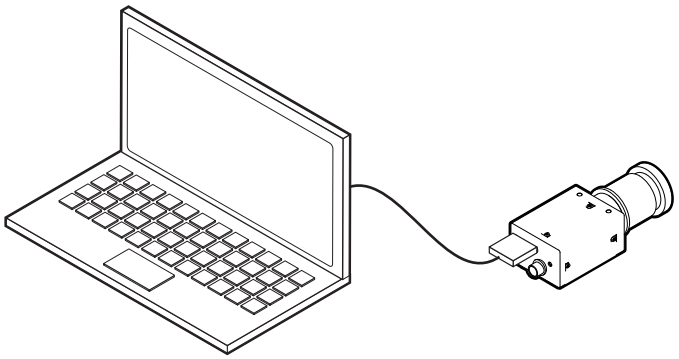
Sunex Wide-Angle - # 01 - 14



2 | 5.

Firmware Update

Firmware Version Required: 2.23.3.0
Follow instructions in camera control README file



6.

Focusing Lenses

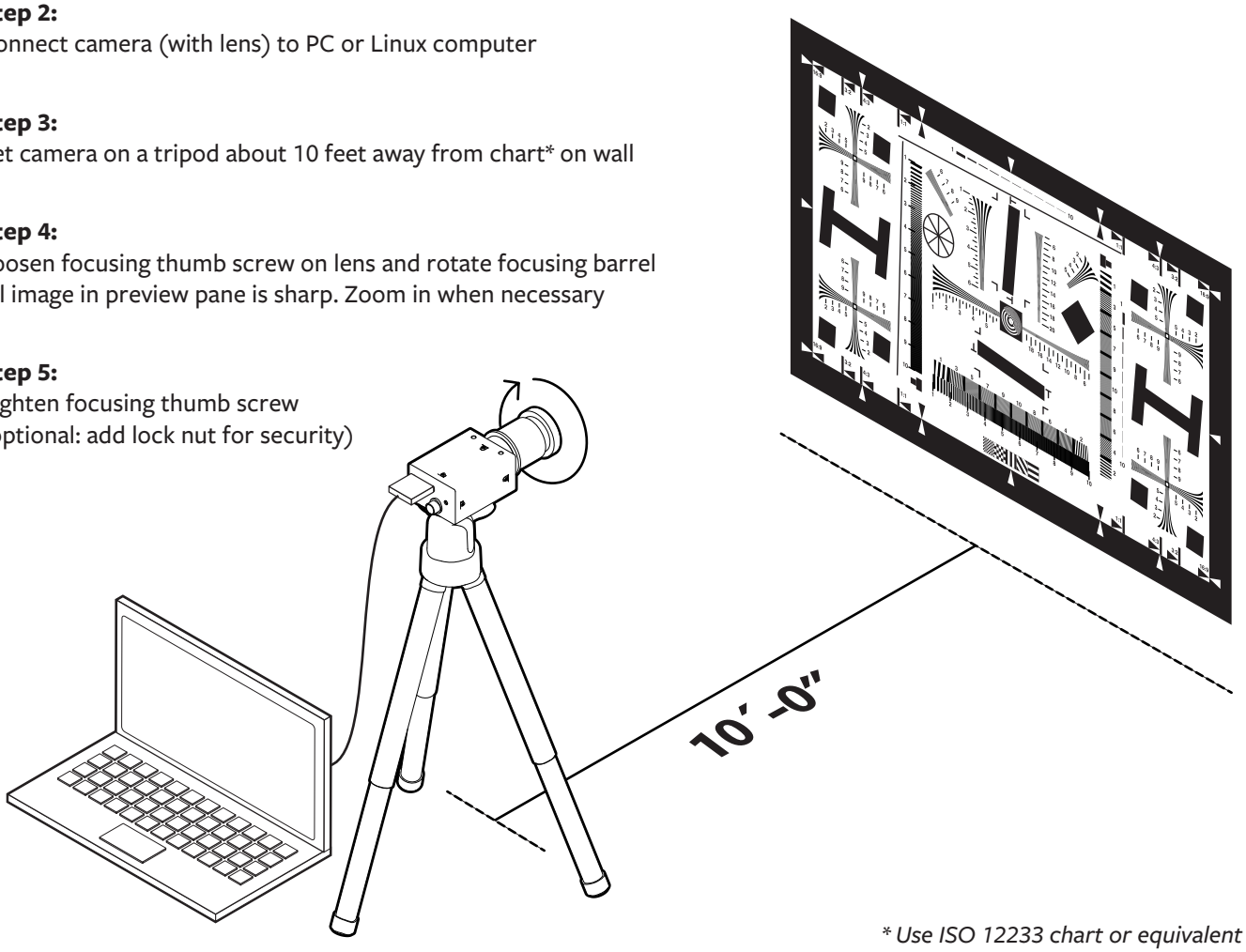
Step 1:
Download and install FlyCapture from Point Grey (https://www.ptgrey.com/support/downloads)

Step 2:
Connect camera (with lens) to PC or Linux computer

Step 3:
Set camera on a tripod about 10 feet away from chart* on wall

Step 4:
Loosen focusing thumb screw on lens and rotate focusing barrel till image in preview pane is sharp. Zoom in when necessary

Step 5:
Tighten focusing thumb screw
(optional: add lock nut for security)

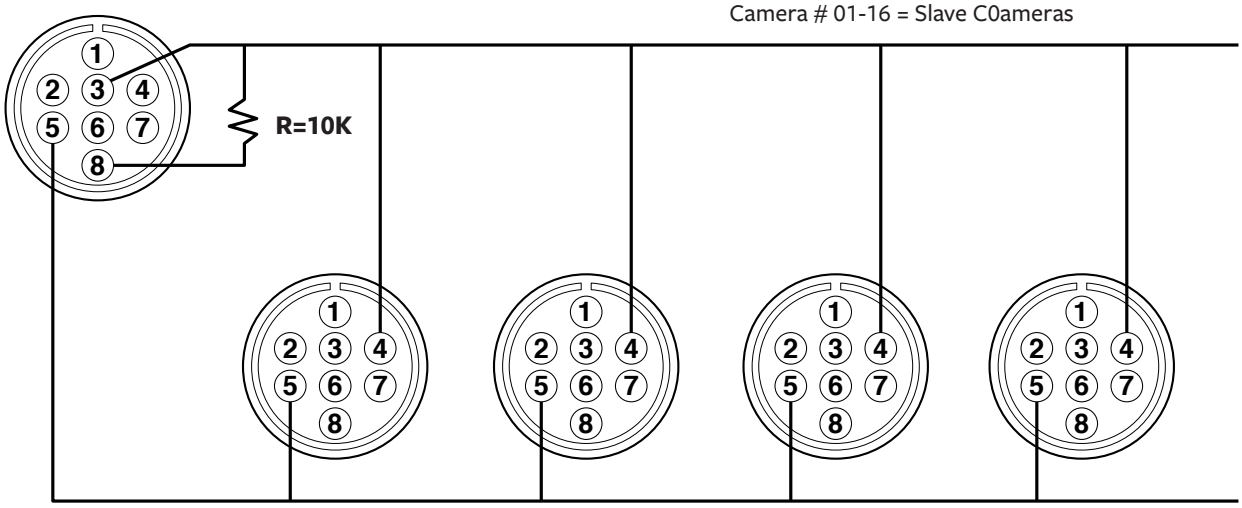


** Use ISO 12233 chart or equivalent*

2 | 7.
GPIO Trigger Cable

A - Wiring Diagram

Camera # 00 (Top) = Master Camera
Camera # 01-16 = Slave C0ameras



PIN ③ IO2 - Input/Output/Serial Transmit (TX)

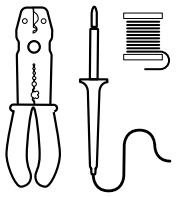
PIN ④ IO3 - Input/Output/Serial Receive (RX)

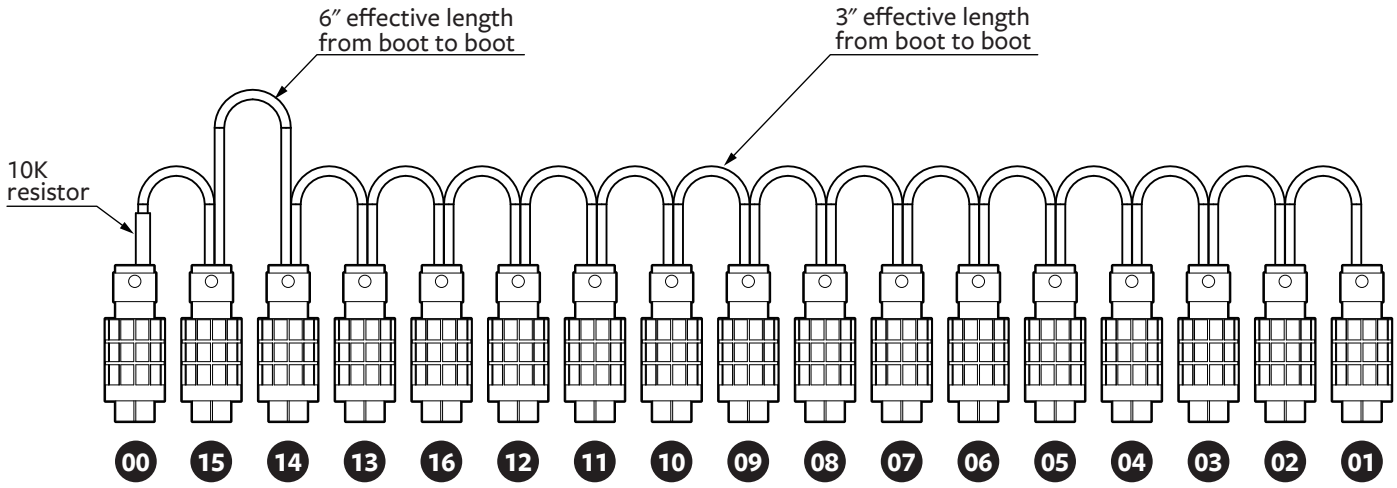
PIN ⑤ GND - Ground for bi-directional IO, V_{EXT}, +3.3V pins

PIN ⑧ + 3.3V - Power external circuitry up to 150mA

** Refer to DWG FB360_V1_34*

B - Cable Lengths From Boot to Boot





** Refer to DWG FB360_V1_34*

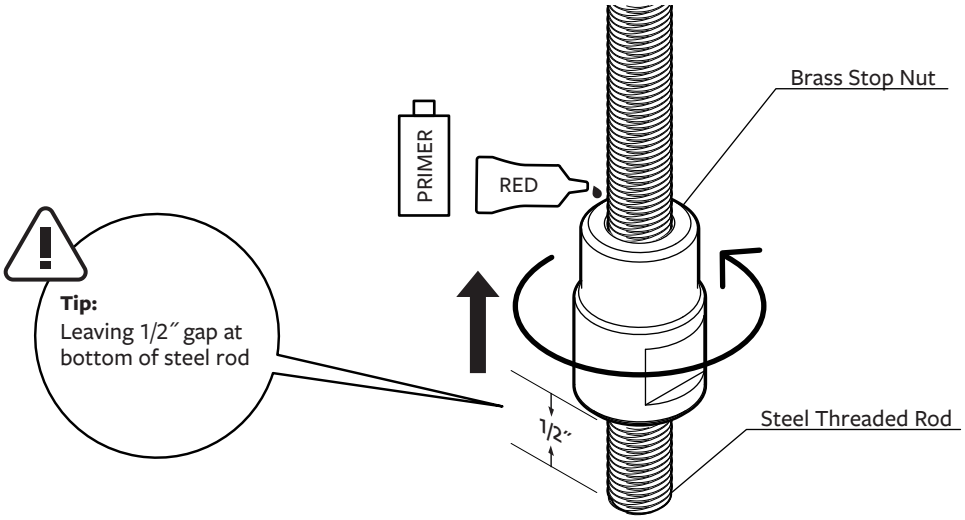
3

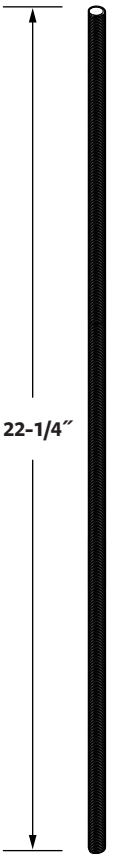
Camera Assembling Instructions

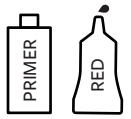
3

1.

Screw Brass Nut Onto Threaded Rod





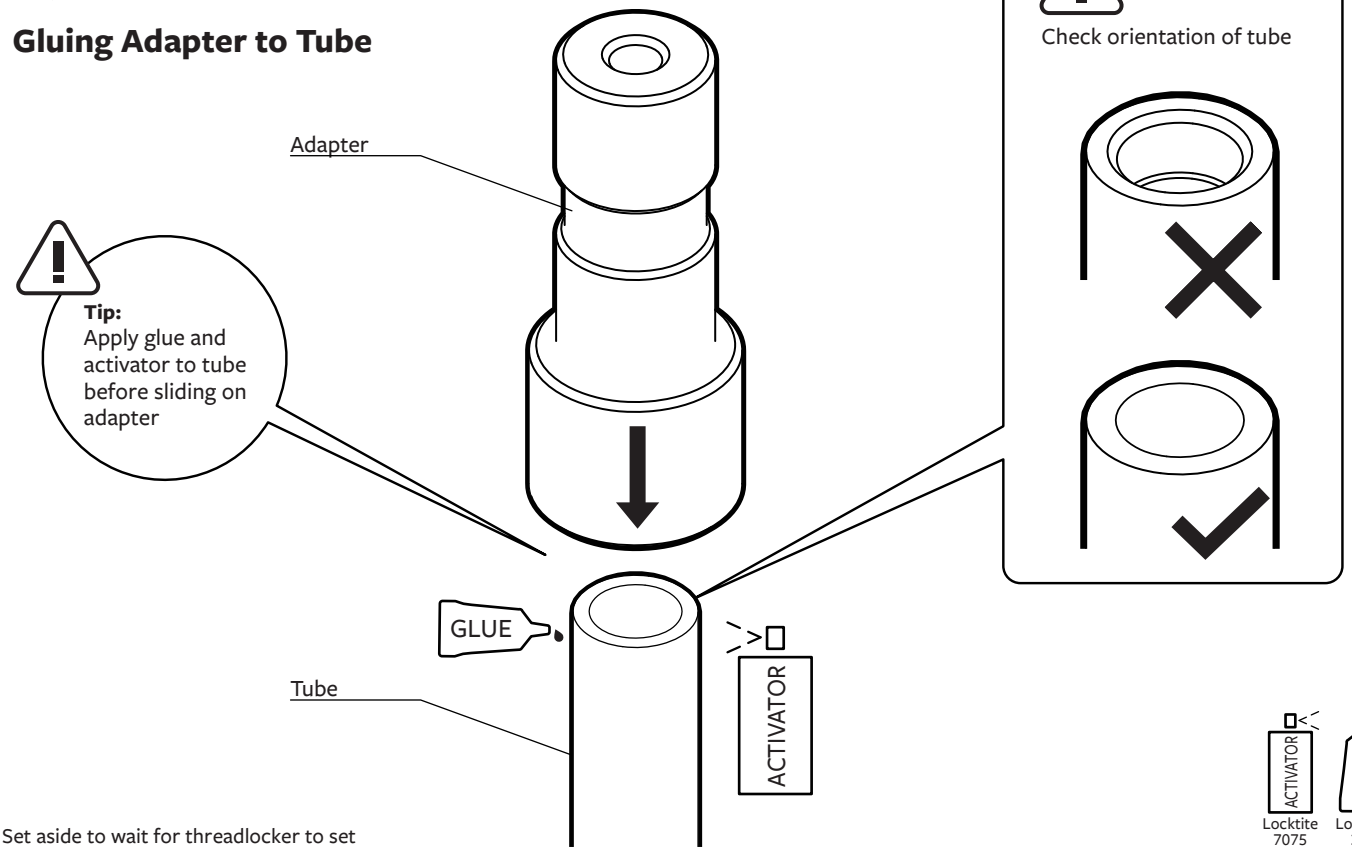


Locktite 7649 Locktite 263

Set aside to wait for threadlocker to set

2.

Gluing Adapter to Tube



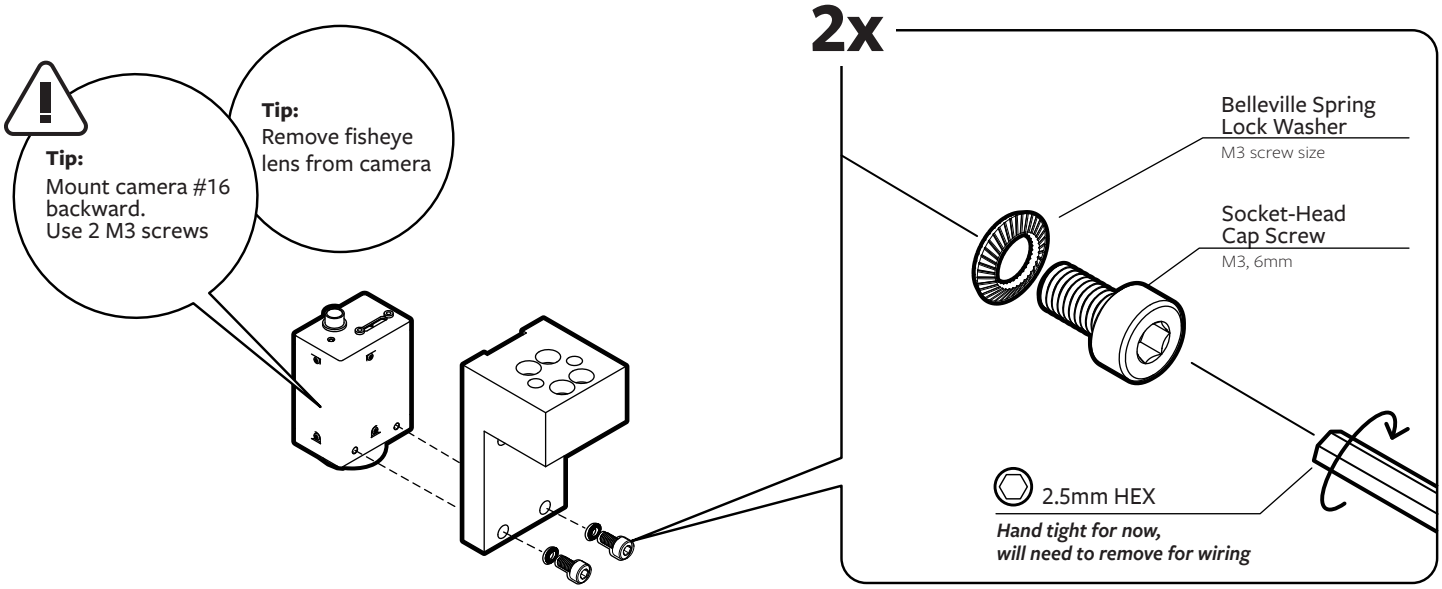
Locktite 7075 Locktite 324

Set aside to wait for threadlocker to set

3

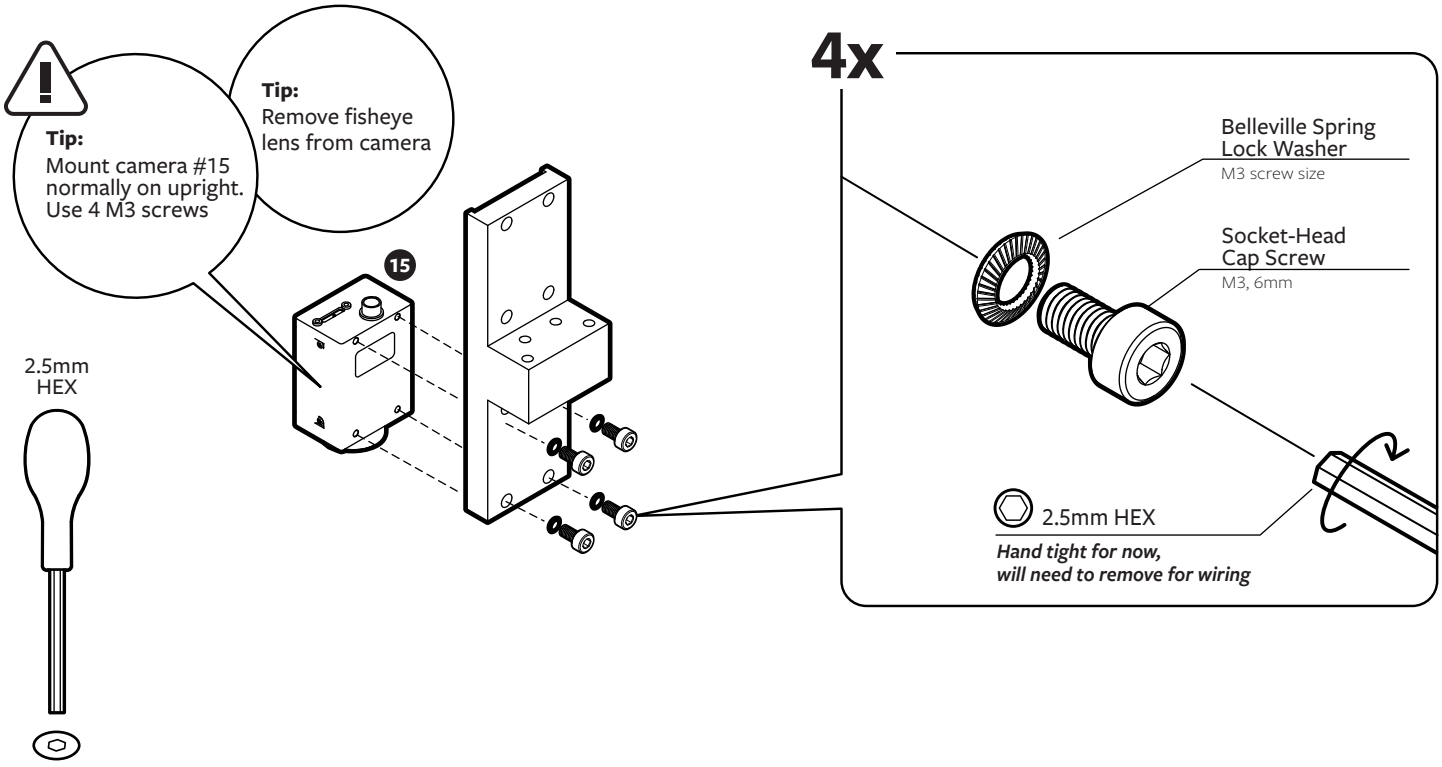
3.

Bottom Camera (#16) & Camera Bracket



4.

Bottom Camera (#15) & Upright

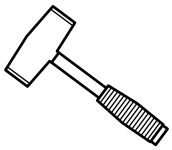


3 | 5.

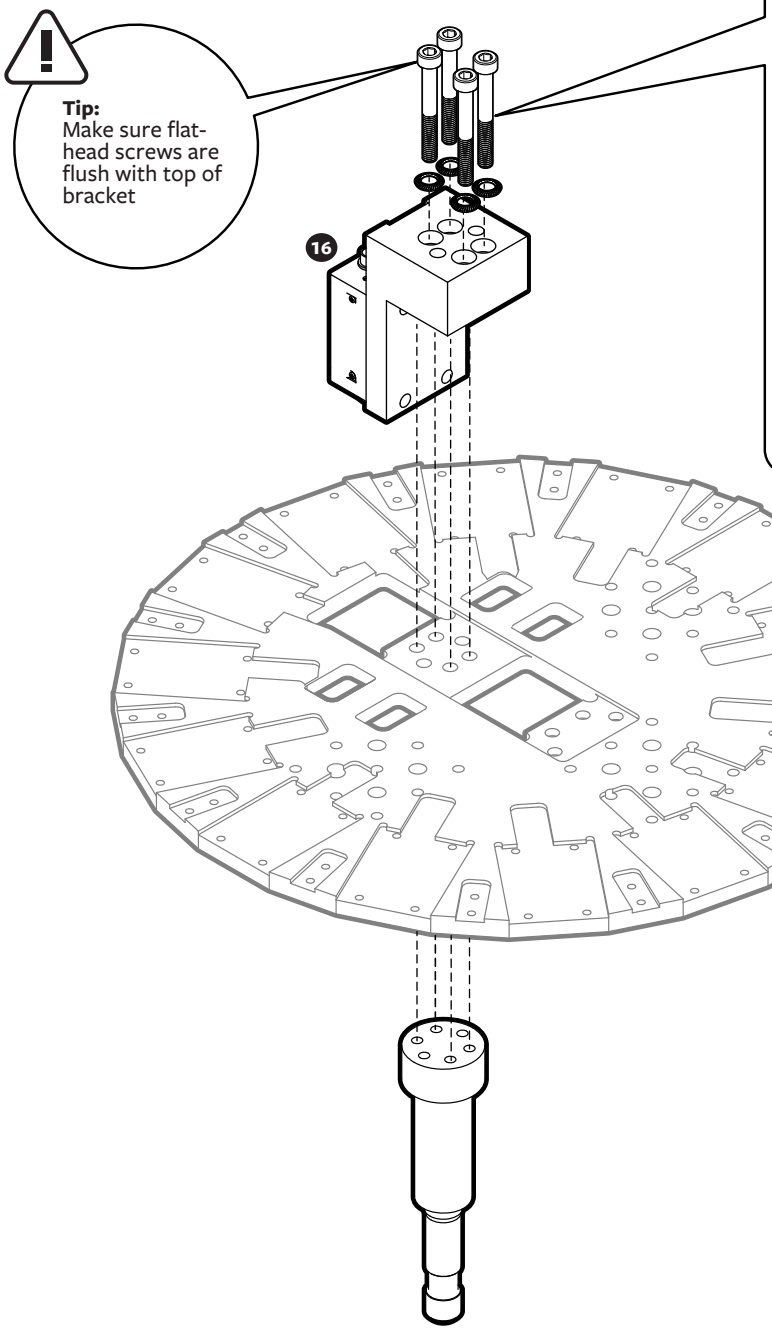
Camera Bracket & Post

Assembly Note

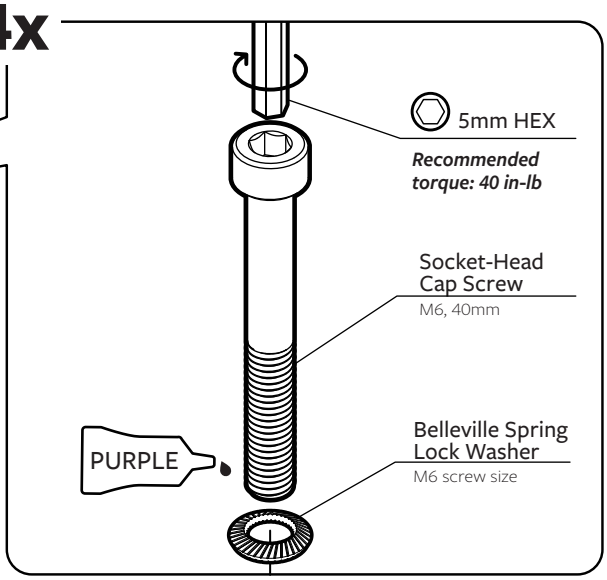
Use a rubber mallet to gently tap upright in place if fitting is too tight



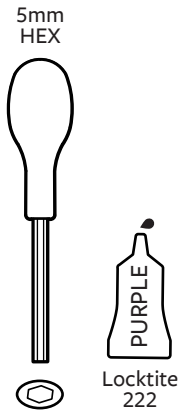
Tip:
Make sure flat-head screws are flush with top of bracket



4x



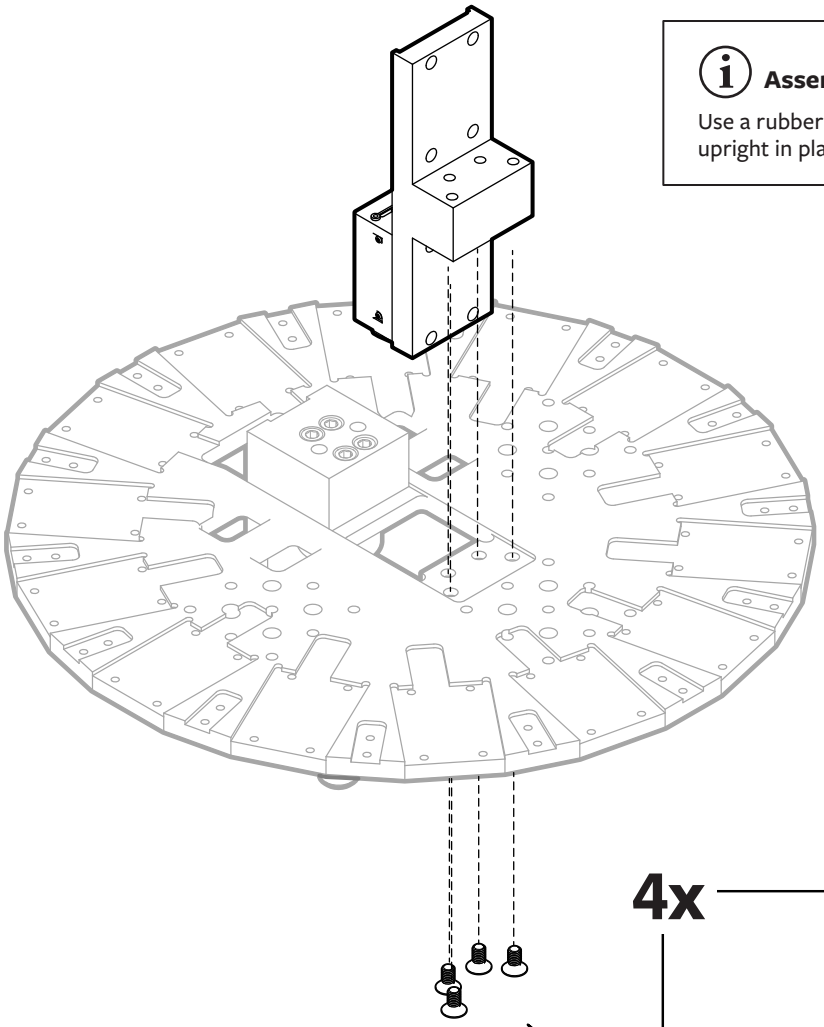
Wait for threadlocker to set



3 | 6.

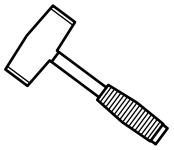
Upright

Assembly Note 180°
Flip 180° to tighten screws

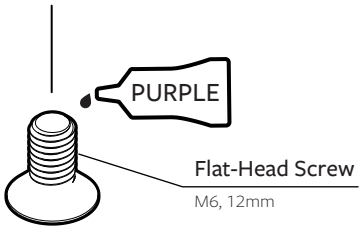


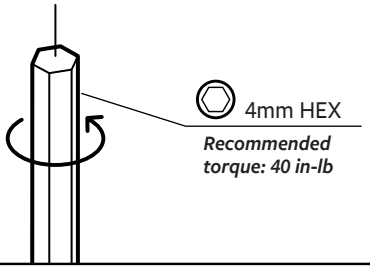
Assembly Note

Use a rubber mallet to gently tap upright in place if fitting is too tight

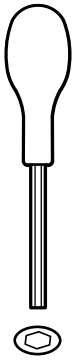


4x





Wait for threadlocker to set



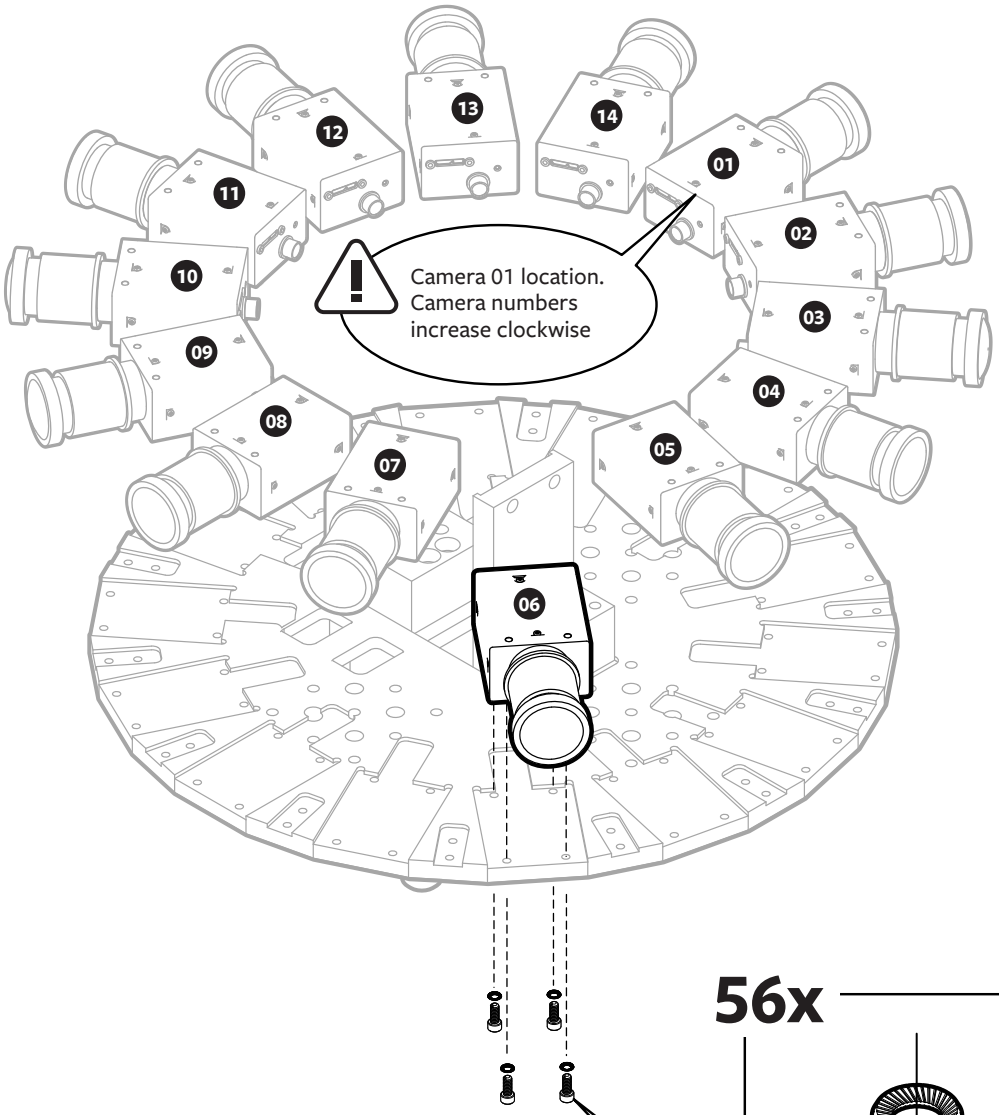
Locktite
222

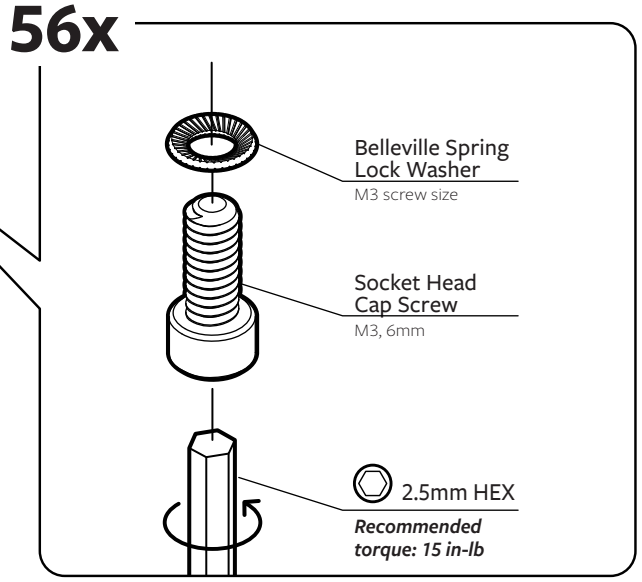
3

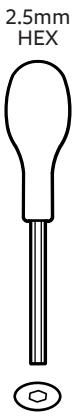
7.

Side Cameras (Camera #01-14)

Assembly Note 180°
Flip 180° to tighten screws



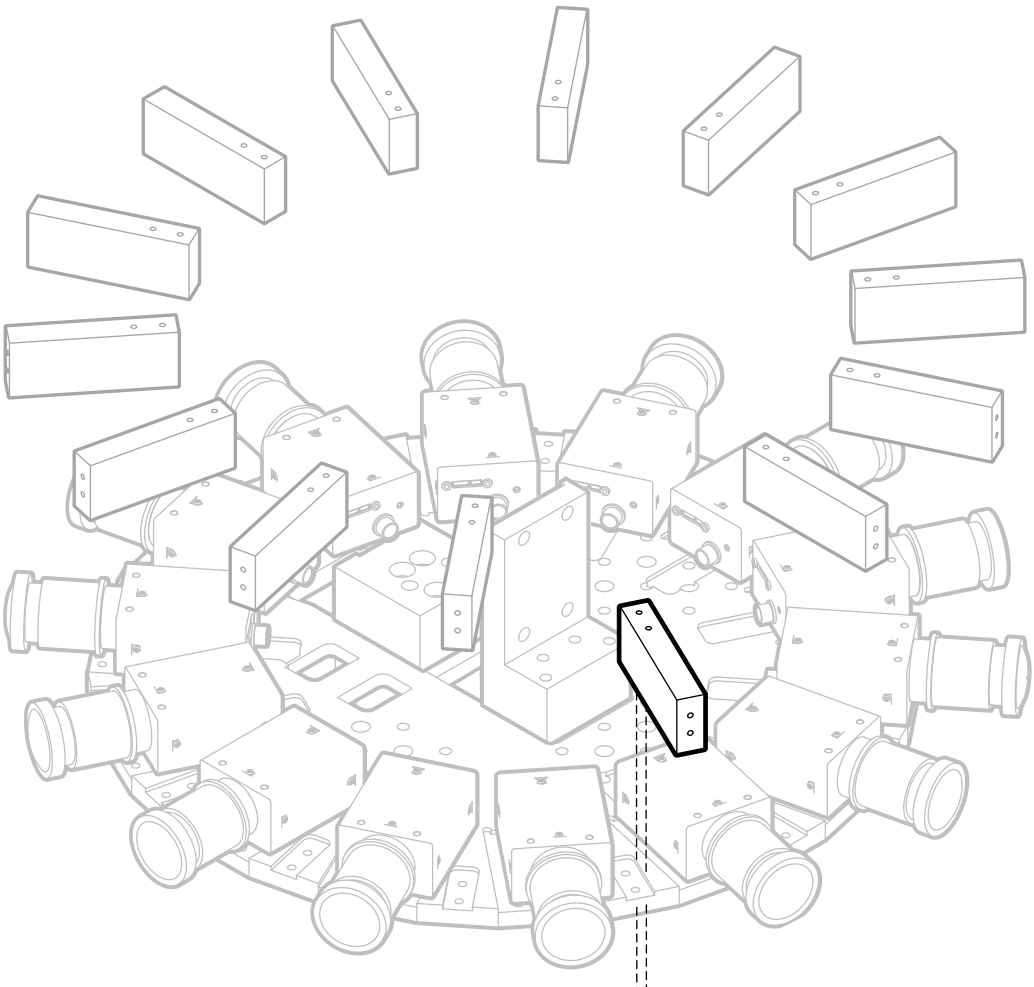


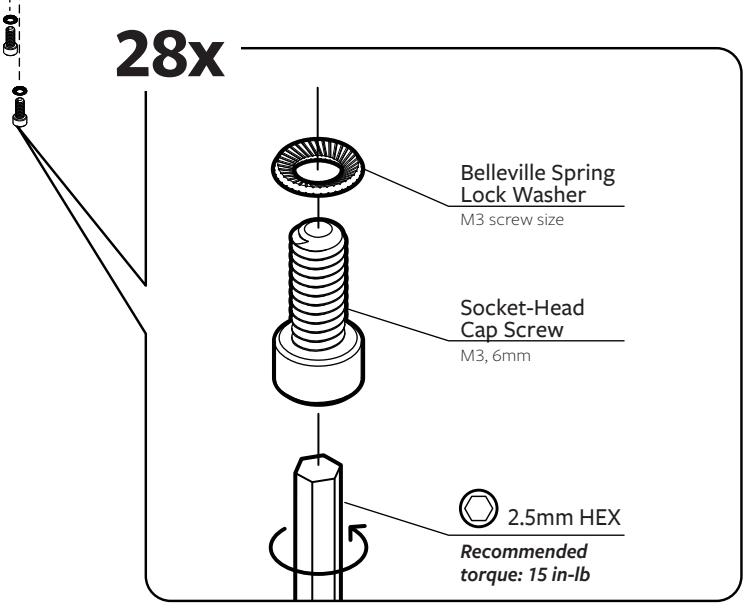


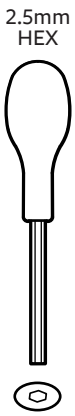
3 | 8.

Blocks

Assembly Note 180°
Flip 180° to tighten screws



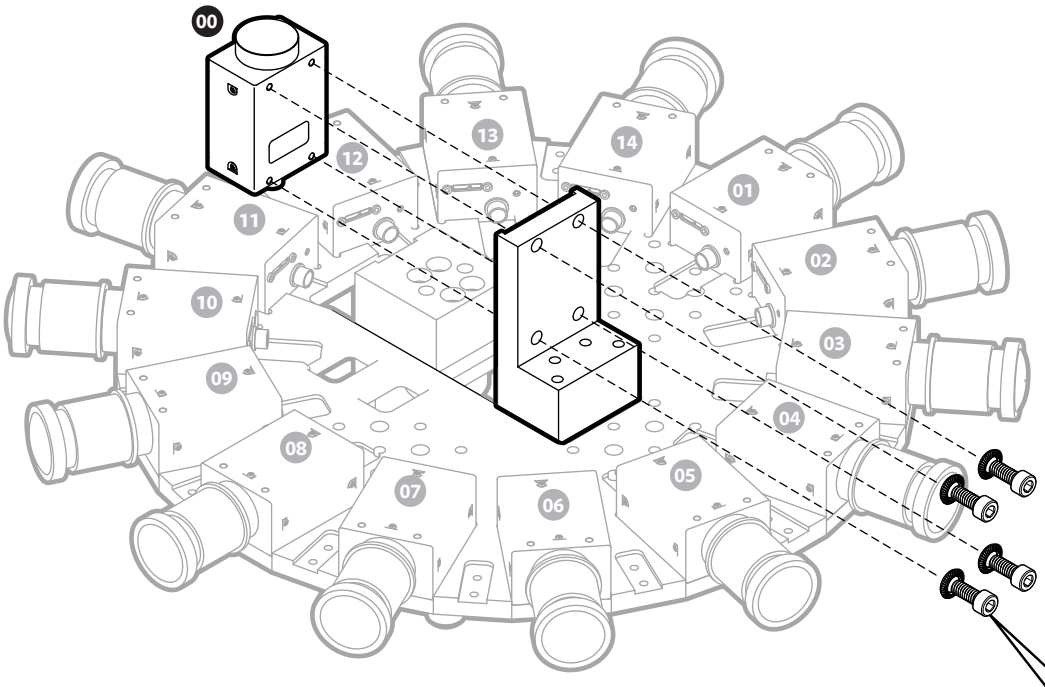


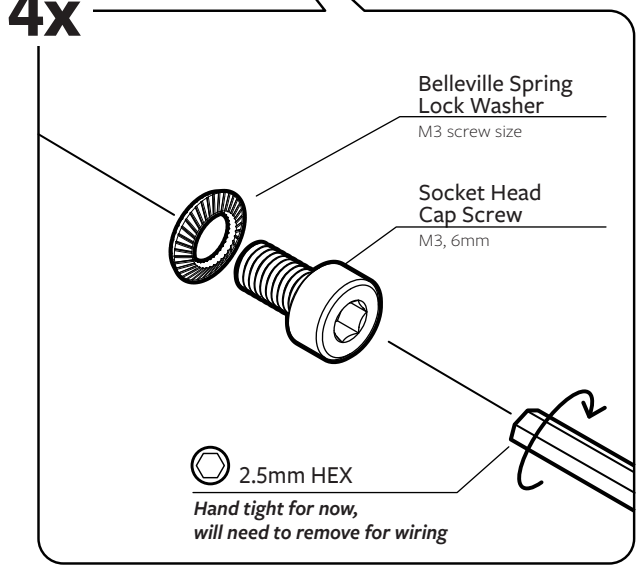


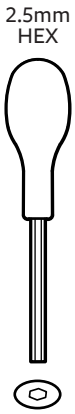
3

9.

Top Camera (Camera #00)







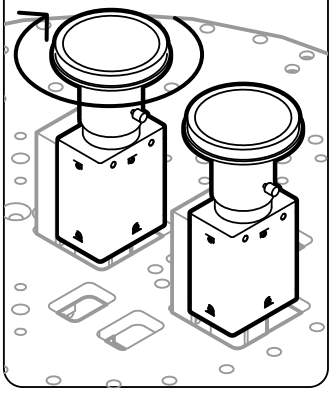
3 | 10.

Threaded Rod & Stop Nut

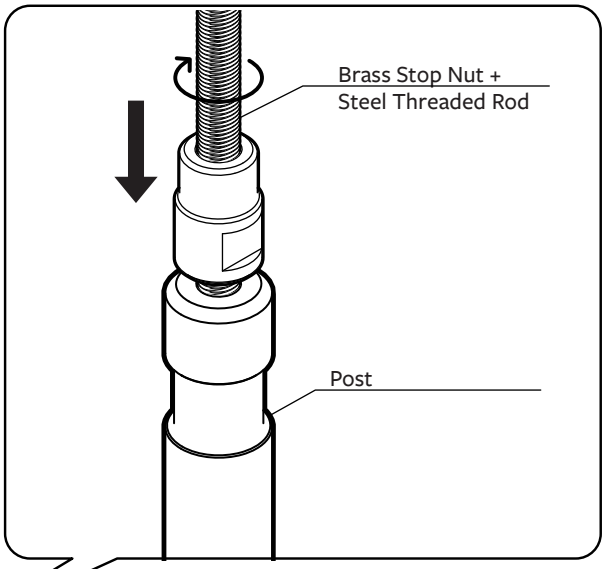
Assembly Note 180°
Flip 180° for this step

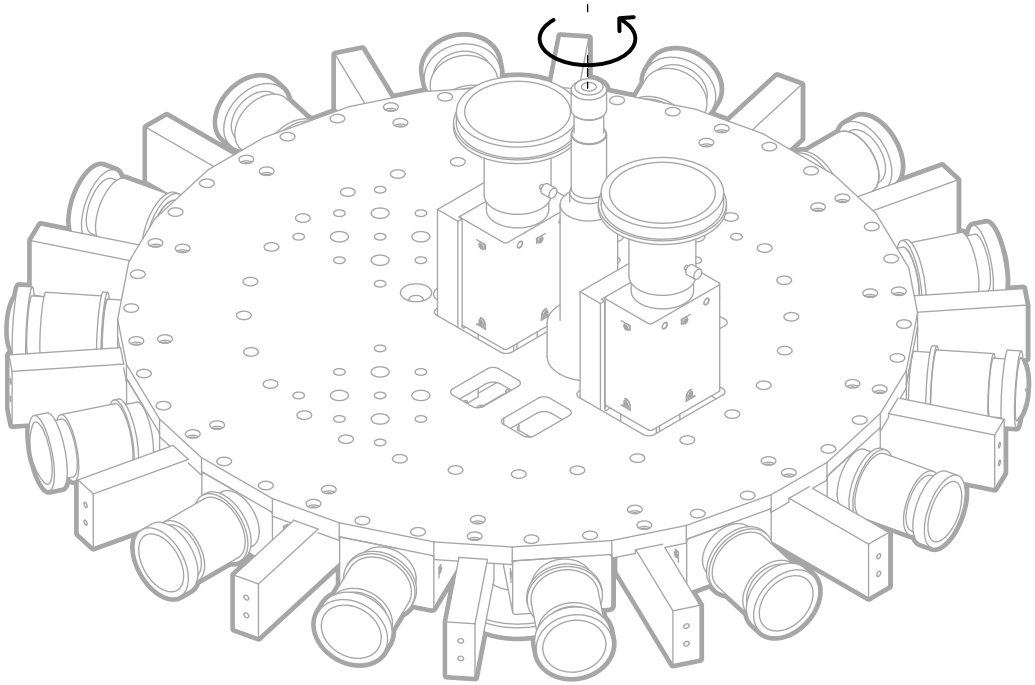
First...

Mount fisheye lens to both bottom cameras



Tip:
Hand tight is enough; do not use excessive force



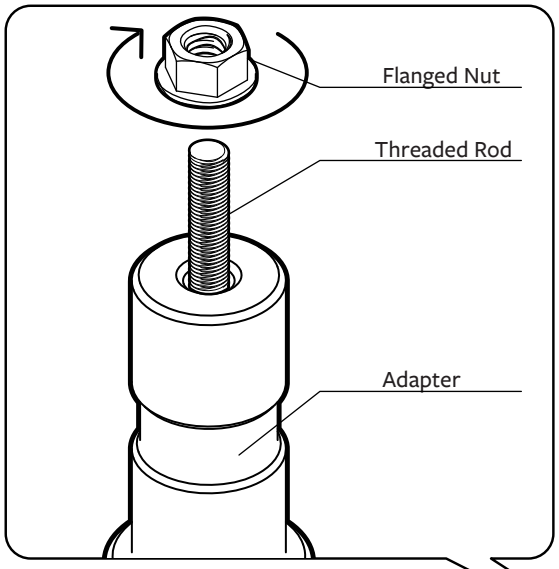


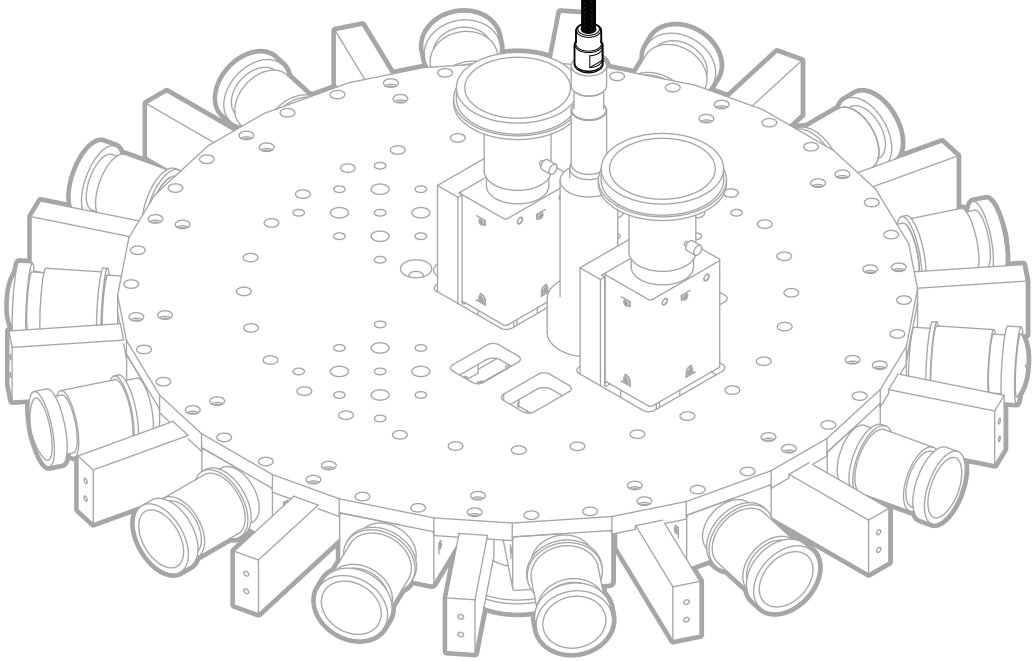
3

11.
Tube & Adapter

Assembly Note 180°
Flip 180° for this step

Tip:
Be careful not to overtighten flange nut



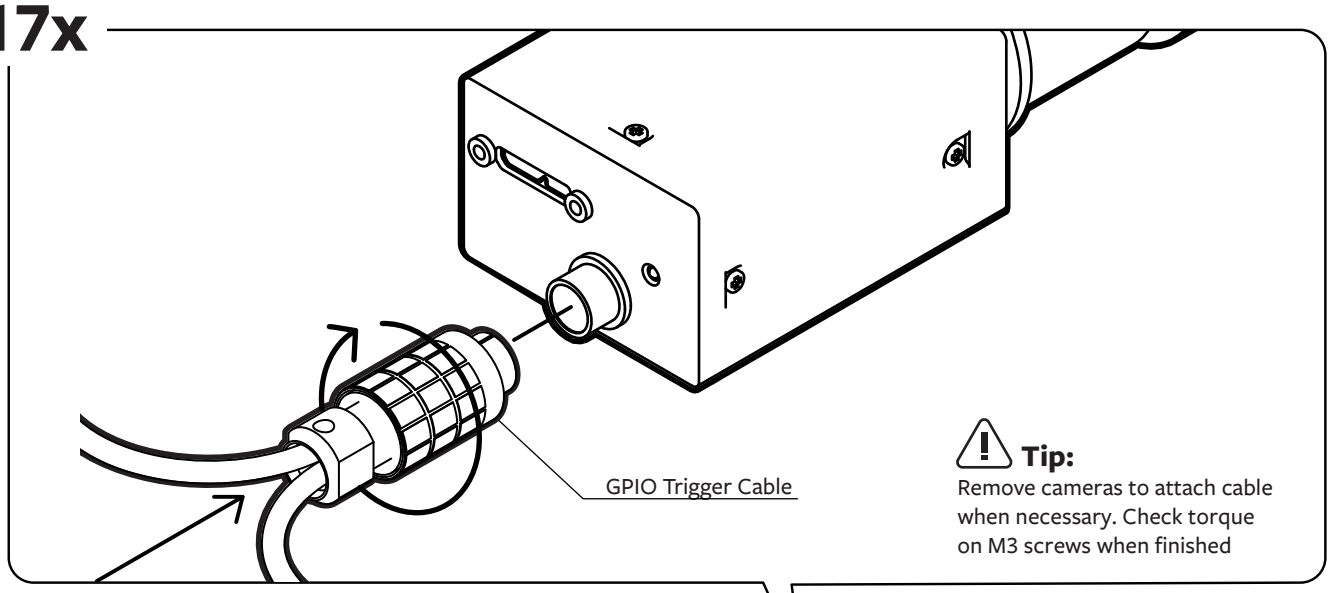


9/16″
Wrench

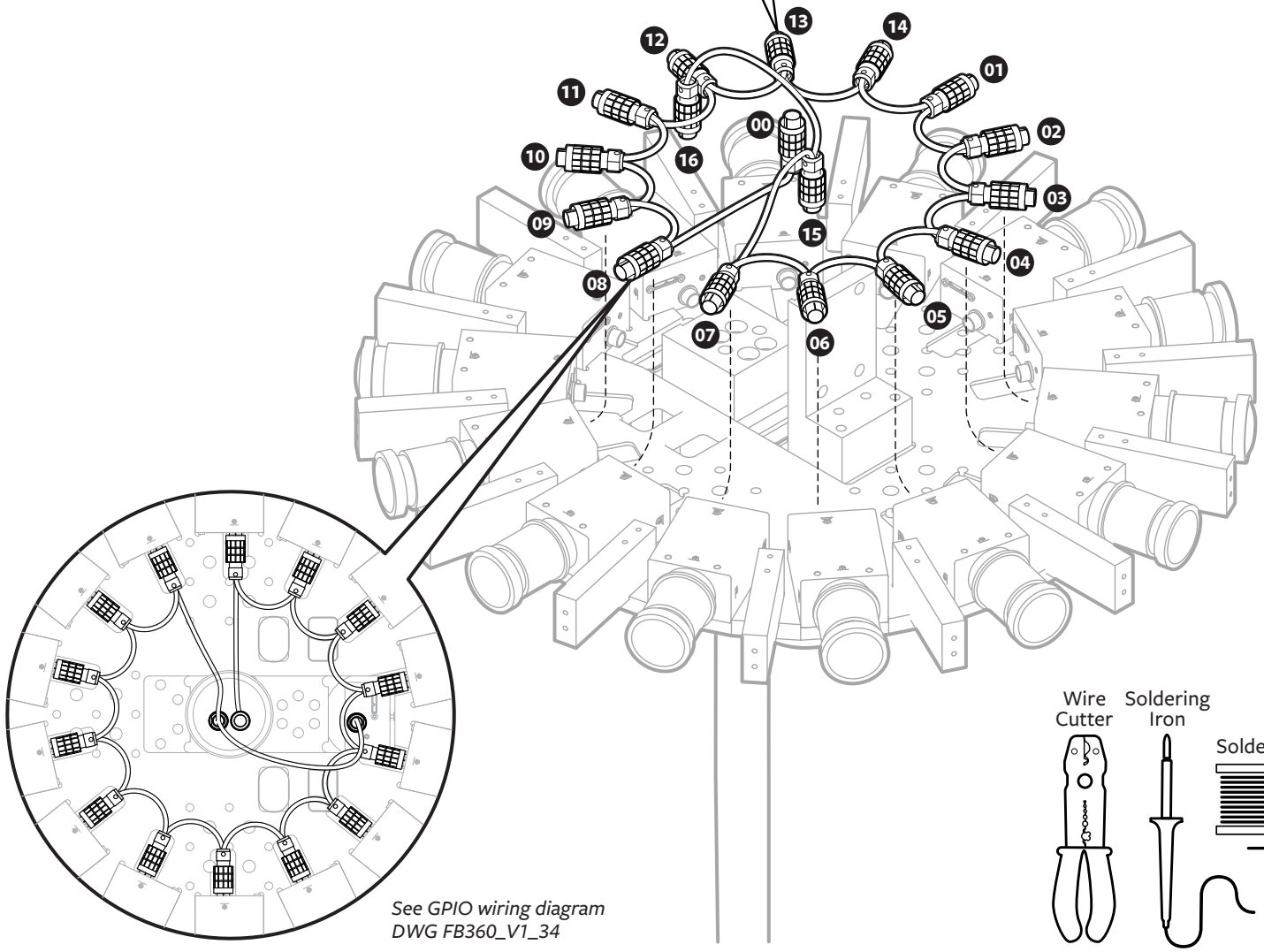
3 | 12.

GPIO Trigger Cable

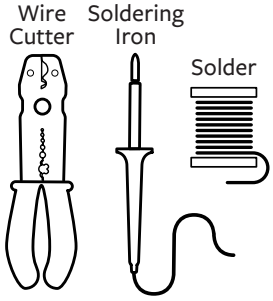
17x



Tip:
Remove cameras to attach cable when necessary. Check torque on M3 screws when finished



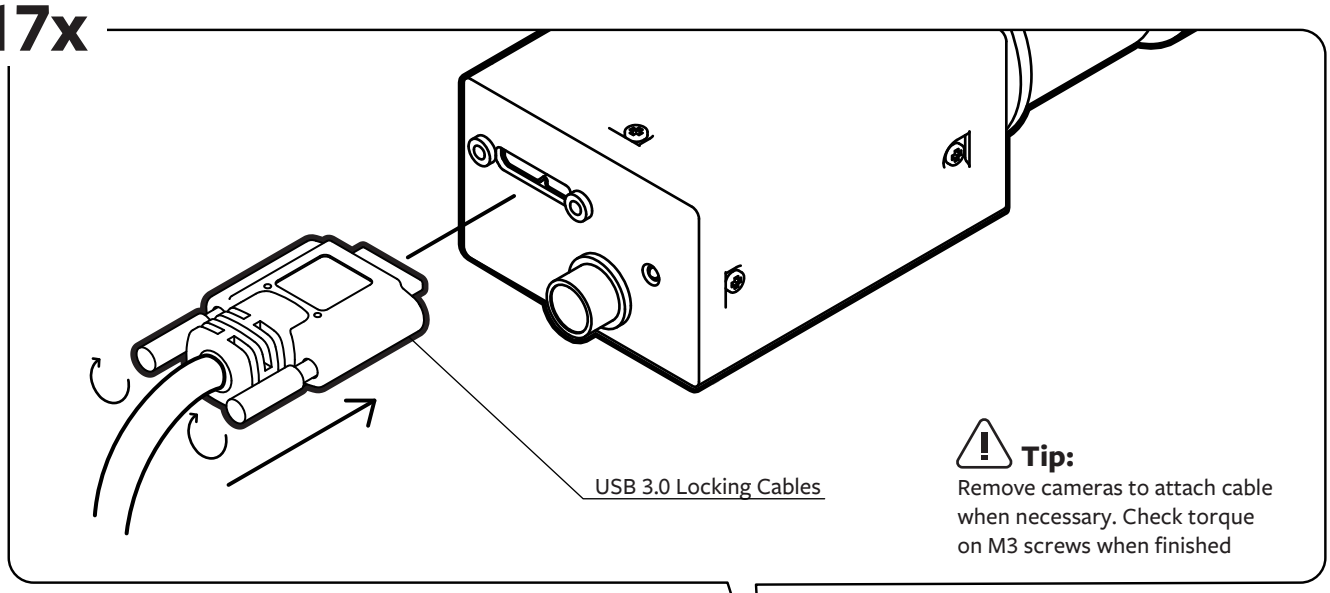
See GPIO wiring diagram
DWG FB360_V1_34



3 | 13.

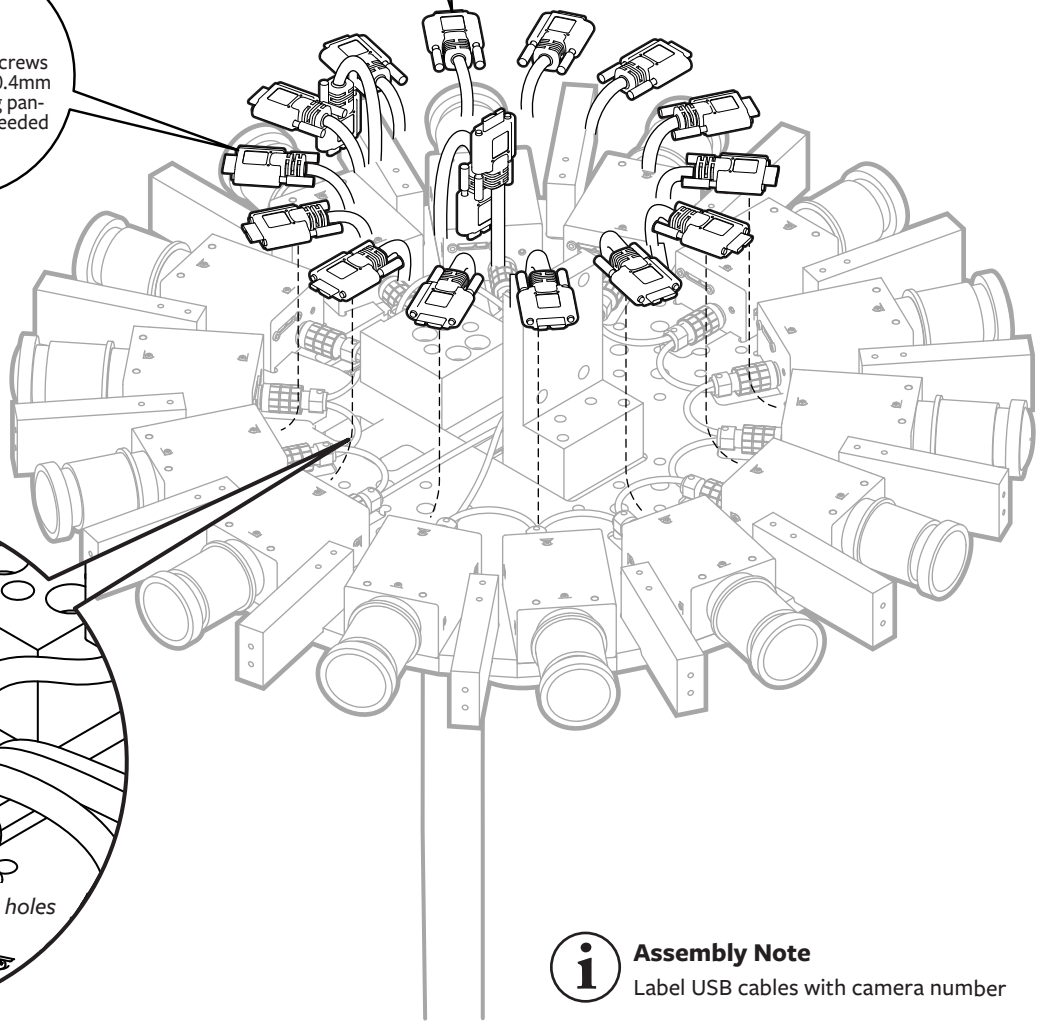
USB 3.0 Cables

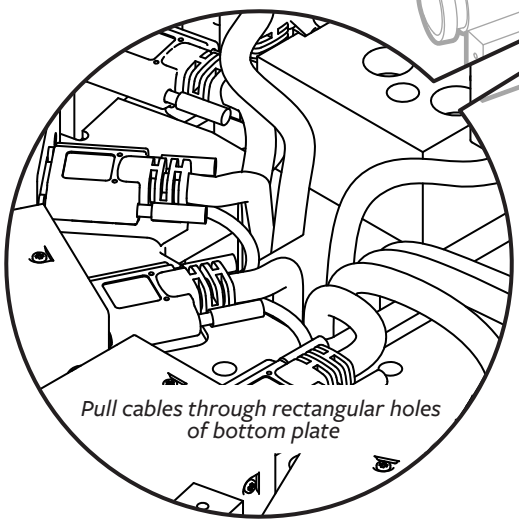
17x



Tip:
Remove cameras to attach cable when necessary. Check torque on M3 screws when finished

Tip:
Replace locking screws with M2 thread, 0.4mm pitch, 20mm long pan-head screws as needed





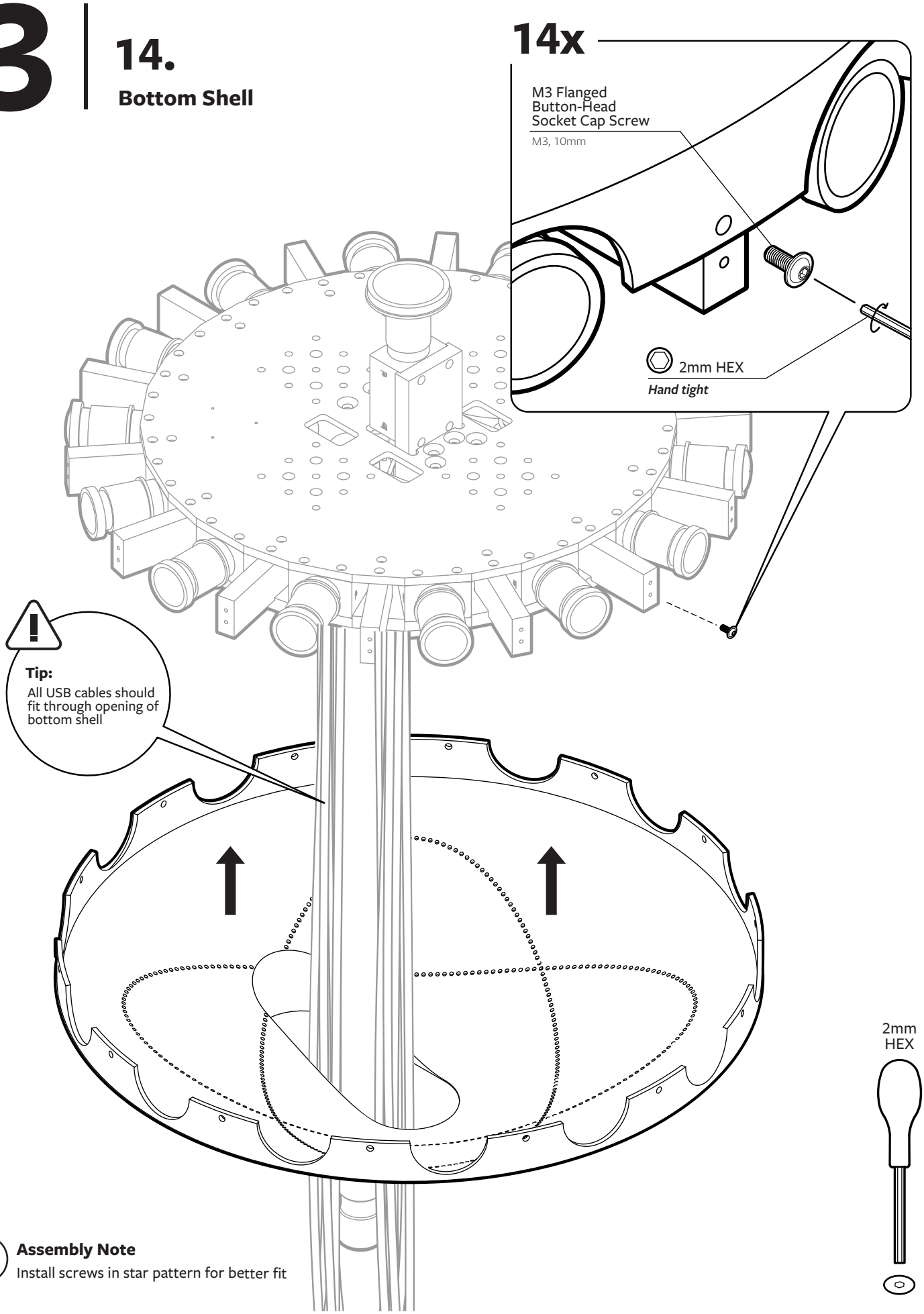
Pull cables through rectangular holes of bottom plate

Assembly Note
Label USB cables with camera number

3

14.

Bottom Shell



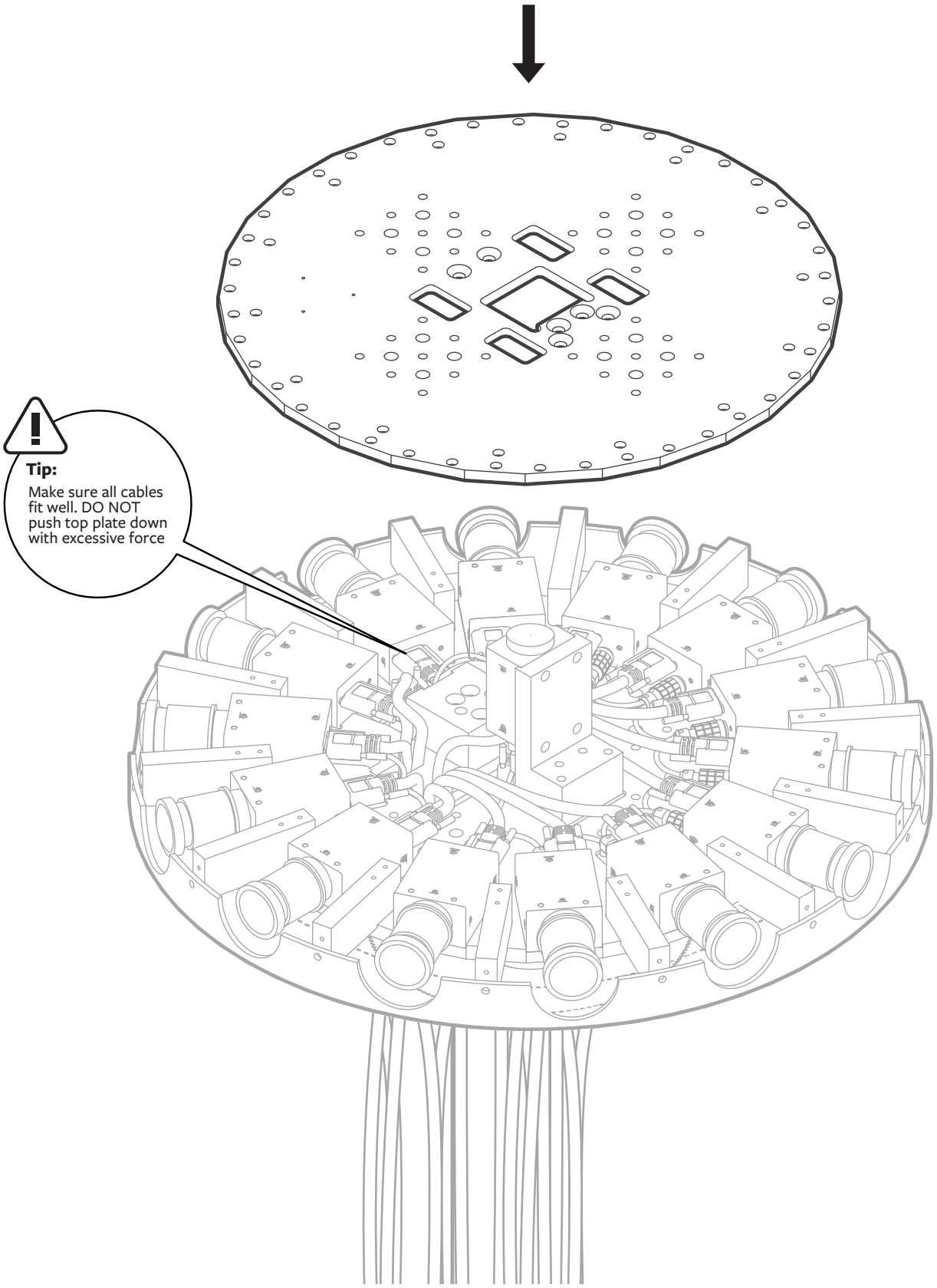
Assembly Note

Install screws in star pattern for better fit

3 | 15.

Top Plate

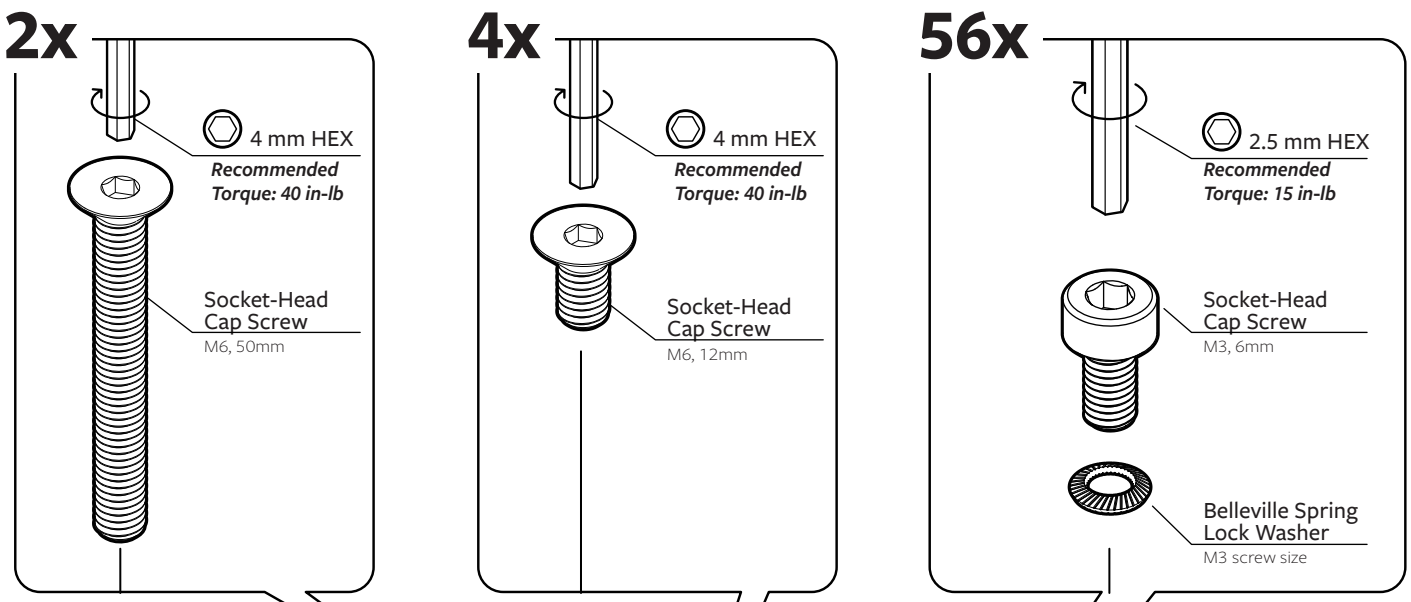
Assembly Note
Optional: Connect cameras to computer to test first

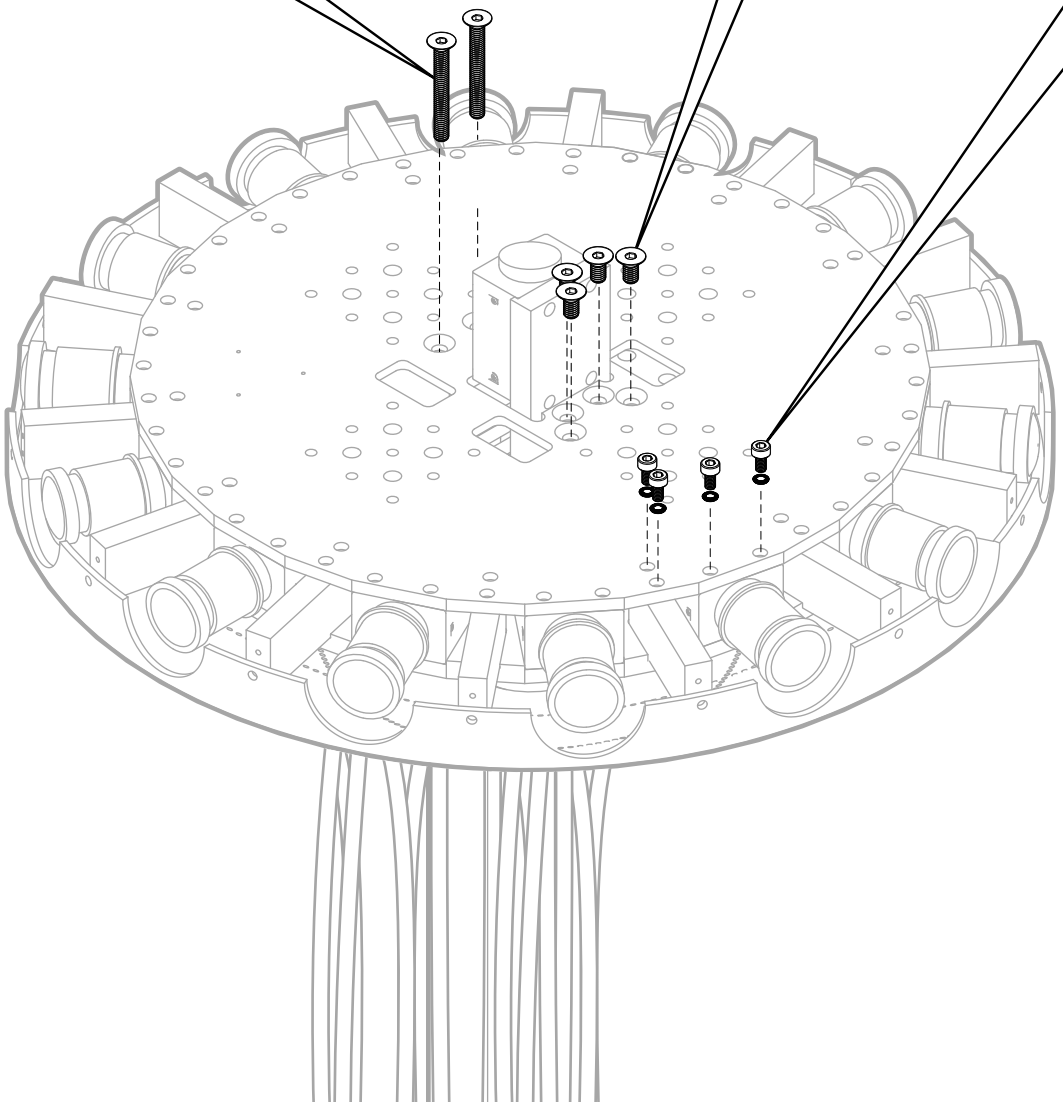


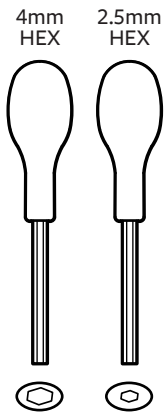
3 | 16.

Top Plate Hardware

Assembly Note
Install M3 socket-head cap screws in star pattern for better fit







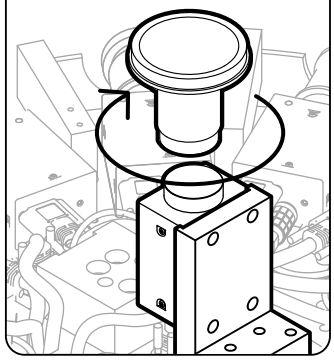
3 | 17.

Top Shell

Assembly Note
Install screws in star pattern for better fit

First...

Mount fisheye lens to top camera before putting shell on



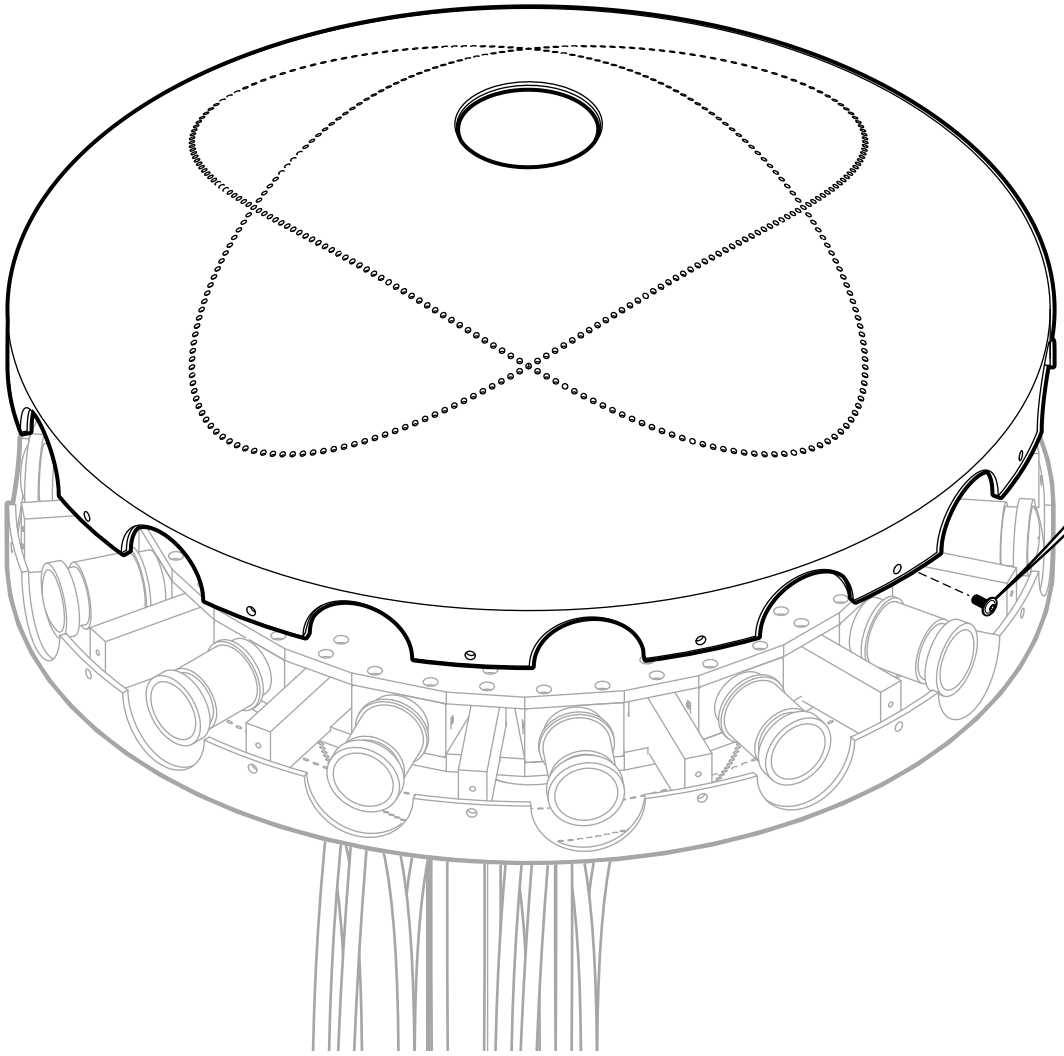
14x

M3 Flanged Button-Head Socket Cap Screw

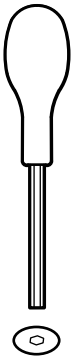
M3, 10mm

2mm HEX

Hand tight

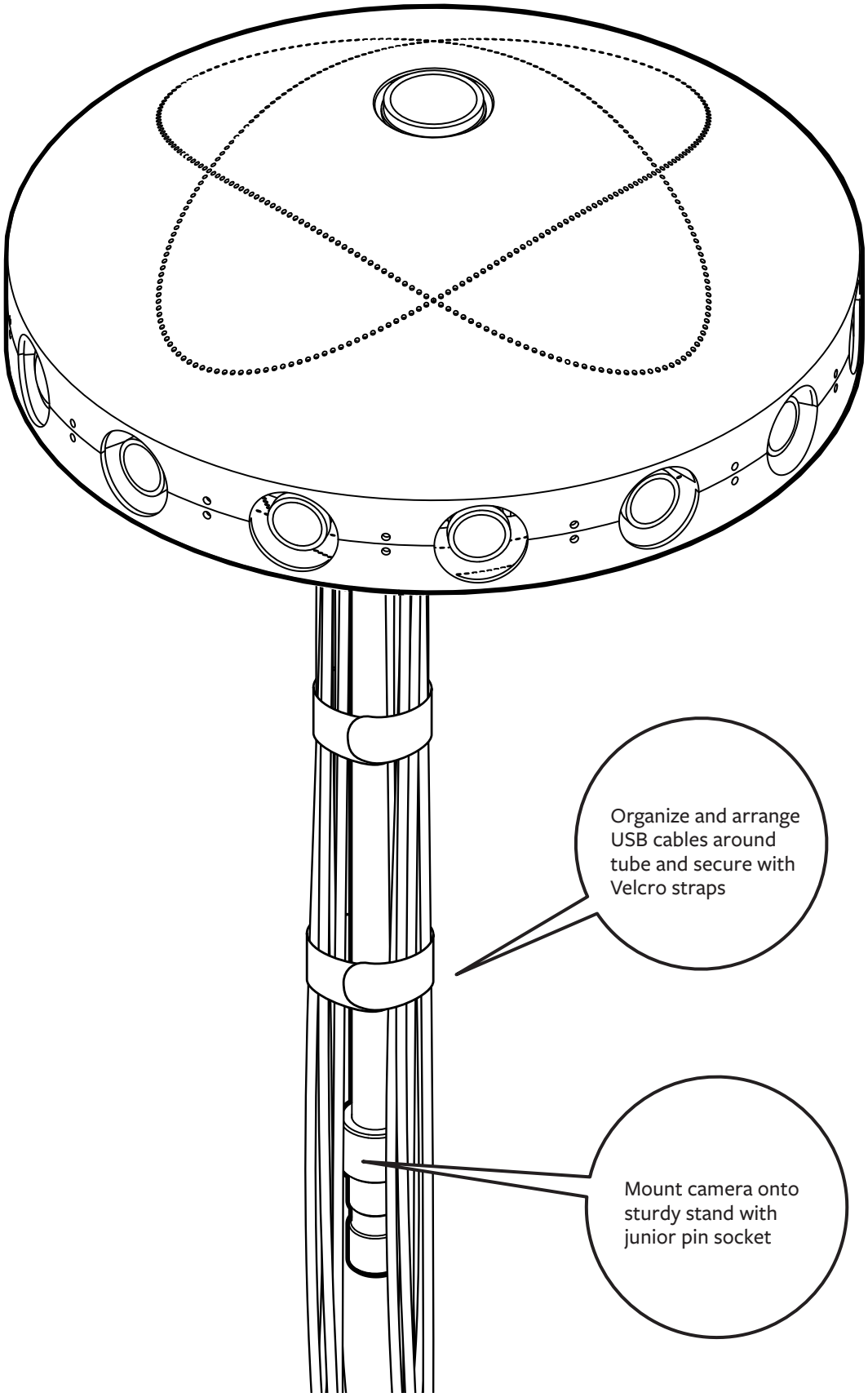


2mm HEX



3 | 18.

Cable Management & Stand



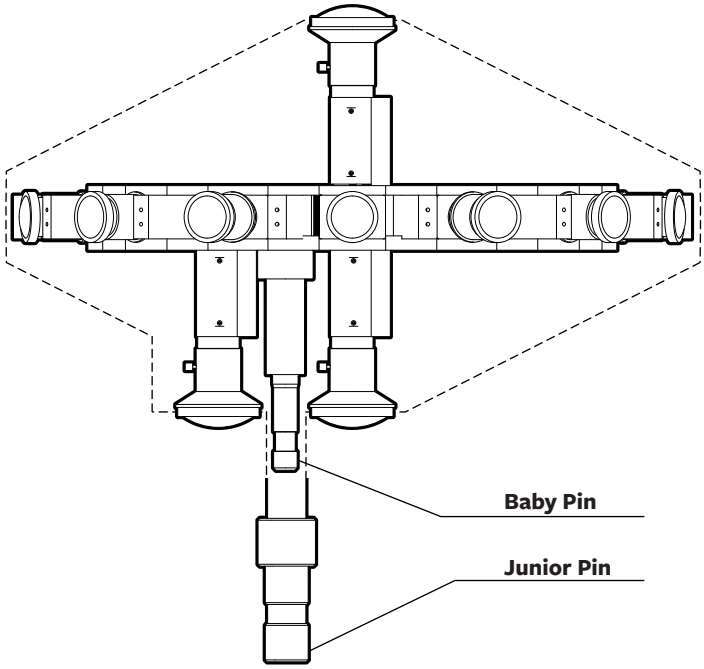
Velcro Straps

4

Camera Info

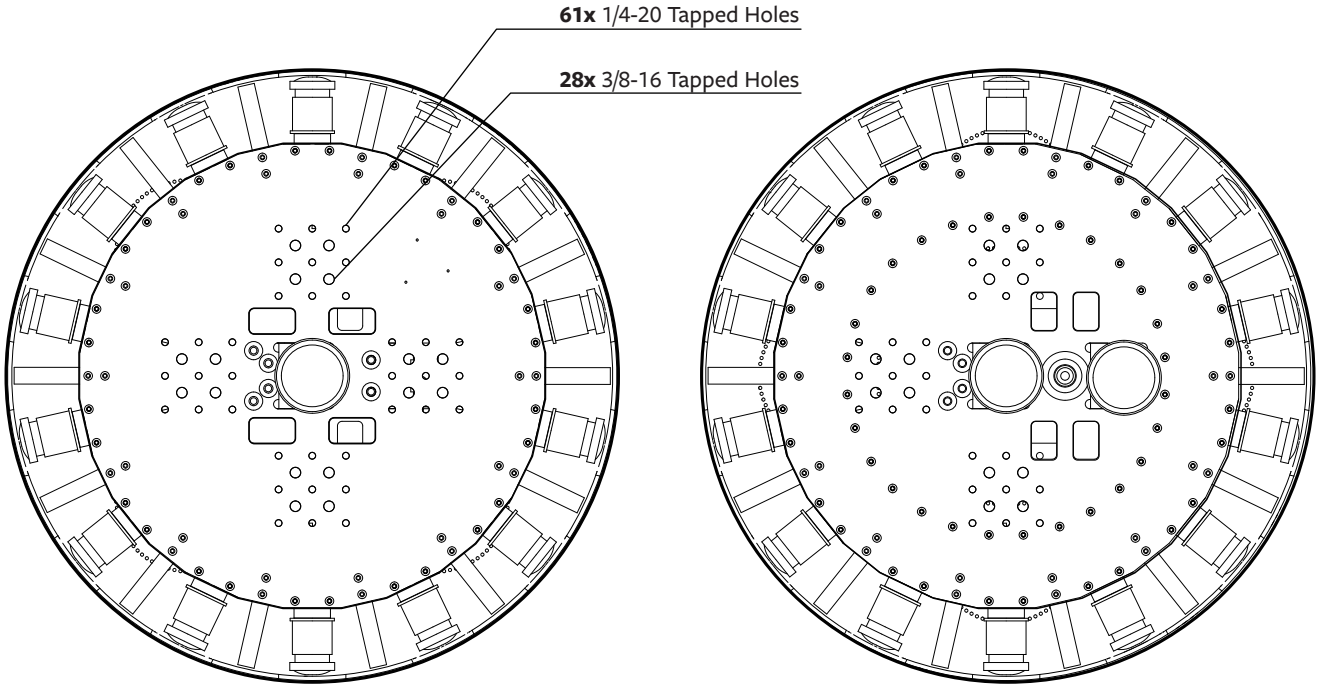
4 | 1a.

Mounting the Camera to Standard Grip Equipment



1b.

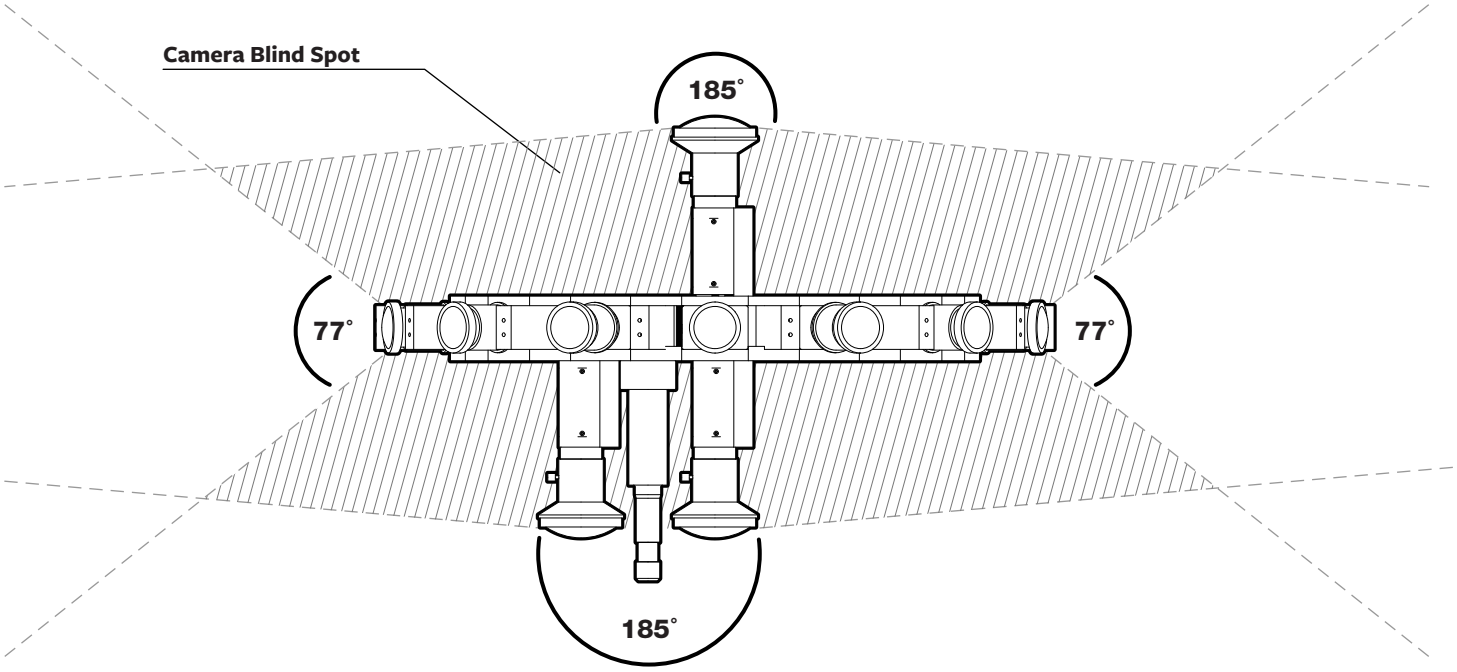
Attachment Options 89 MULTIPURPOSE MOUNTING HOLES



TOP PLATE **BOTTOM PLATE**

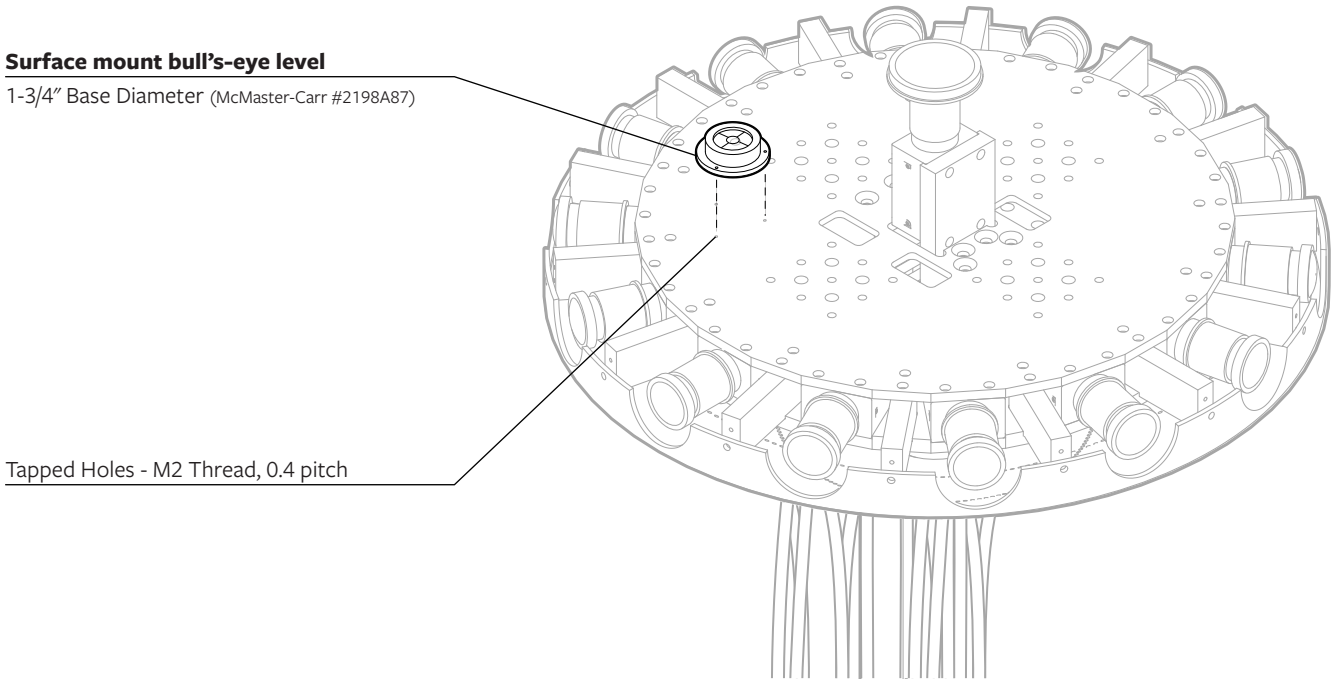
4 | 2a.

Blind Spot for Mounting Accessories



2b.

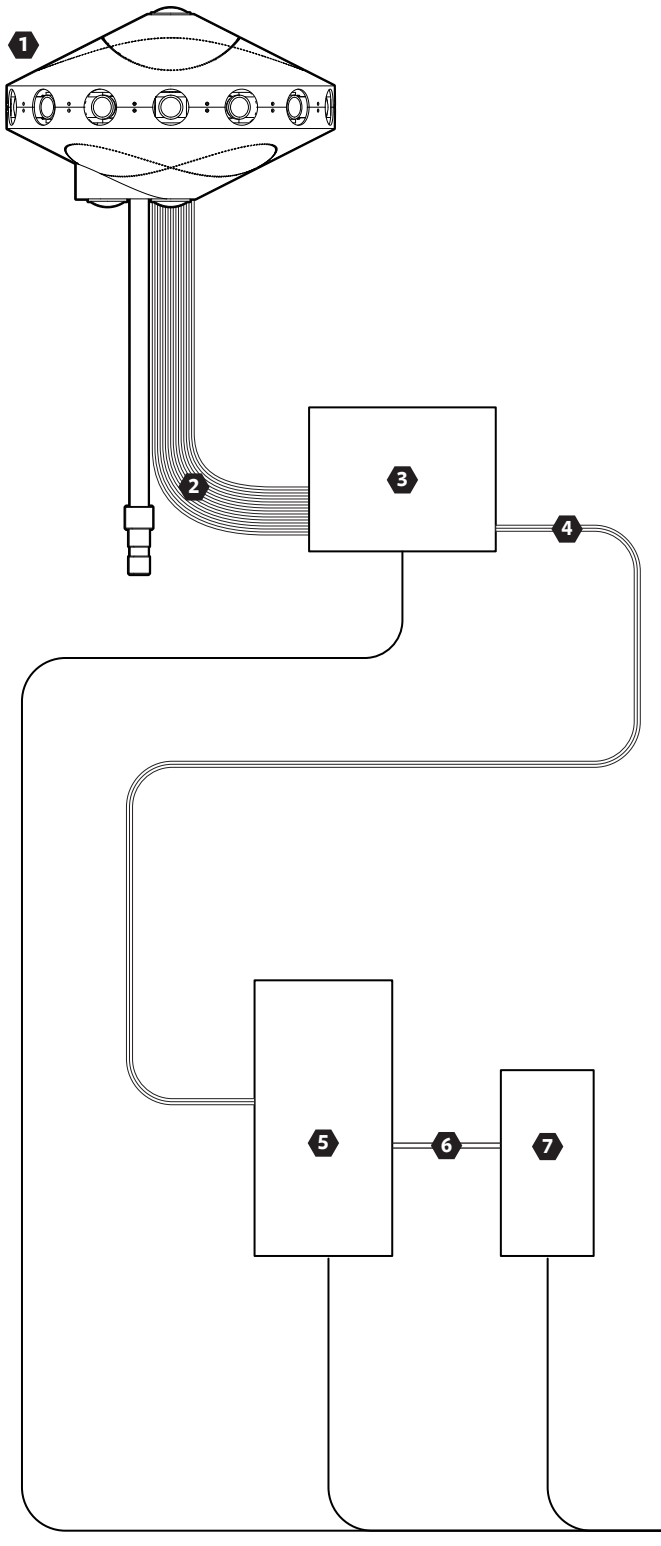
Bull's-Eye Level (Optional)



5

System Set Up Instructions

5 | 1.
Surround 360 System Overview



SURROUND 360 SYSTEM with Fiber Optic Extension

1. **Surround 360 Camera**
2. **USB 3.0 High Speed Type A to Micro-B Cables x17**
3. **Fiber Optic Breakout Box**
4. **PCIe x8 Active Optical Cable**
5. **Lunchbox Computer (Camputer)**
6. **SFF-8644 to SFF-8644 MiniSAS Cable x2**
7. **Raid Tower**
8. **Power Supply**

1 SURROUND 360 Camera

- 17x Point Grey Grasshopper Cameras
- 14x Wide-Angle Lenses
- 3x Fisheye Lenses
- GPIO Trigger Cable

3 FIBER OPTIC BREAKOUT BOX

- 1x Backplane With Power Supply
- 5x PCIe x4, 4 Ports USB 3.0 Expansion Card
- 1x PCIe x16 Host Interface Card

5 CAMPUTER SPECIFICATIONS

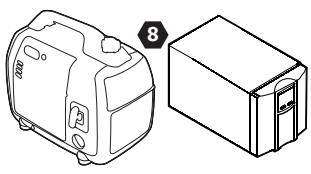
- Intel Core i7-5960X Haswell-E 8-Core 3.0 GHz LGA 2011-v3*
- GIGABYTE GA-X99P-SLI (rev. 1.0) LGA 2011-v3 Intel X99 Motherboard*
- 8GB DDR4 2400 288-PIN Memory ± (64GB of Memory Installed)*
- CPU COOLING FAN FOR LGA 2011-v3*
- 1GB NVIDIA PCIe x16 VIDEO CARD*
- 700 WATT POWER SUPPLY*
- 2.5″ 128GB SSD*
- OPERATING SYSTEM - UBUNTU 14.04 LTS†

7 RAID TOWER SPECIFICATIONS

- 8 x 1TB SSD RAID: 1 Hour of Continuous Raw Video Capture (Bandwidth: 2.1 GB/s)

8 POWER SUPPY

- 110V AC
- 350W max.
- Alternative Options: UPS Backup Battery or Quiet Generator



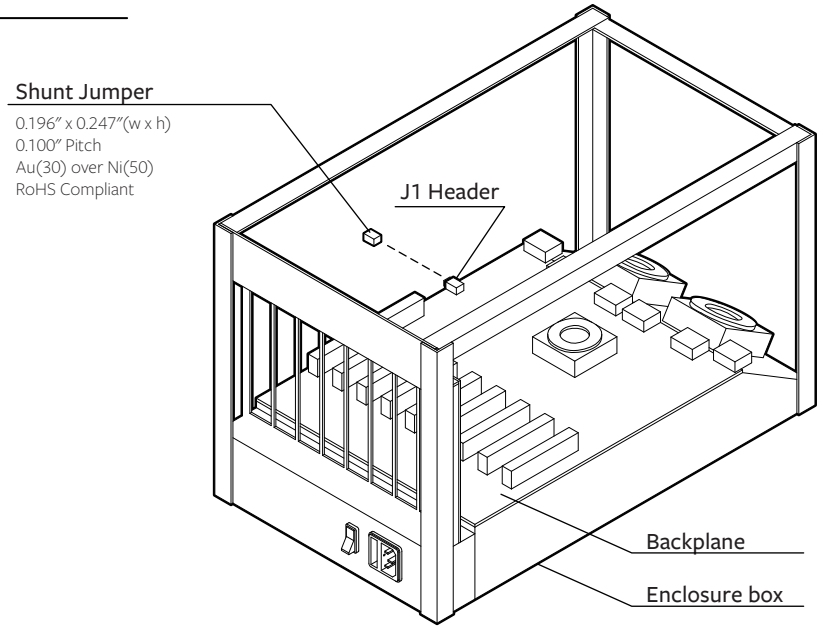
** Recommended*
† Required

5 | 2a.

Fiber Optic Breakout Box

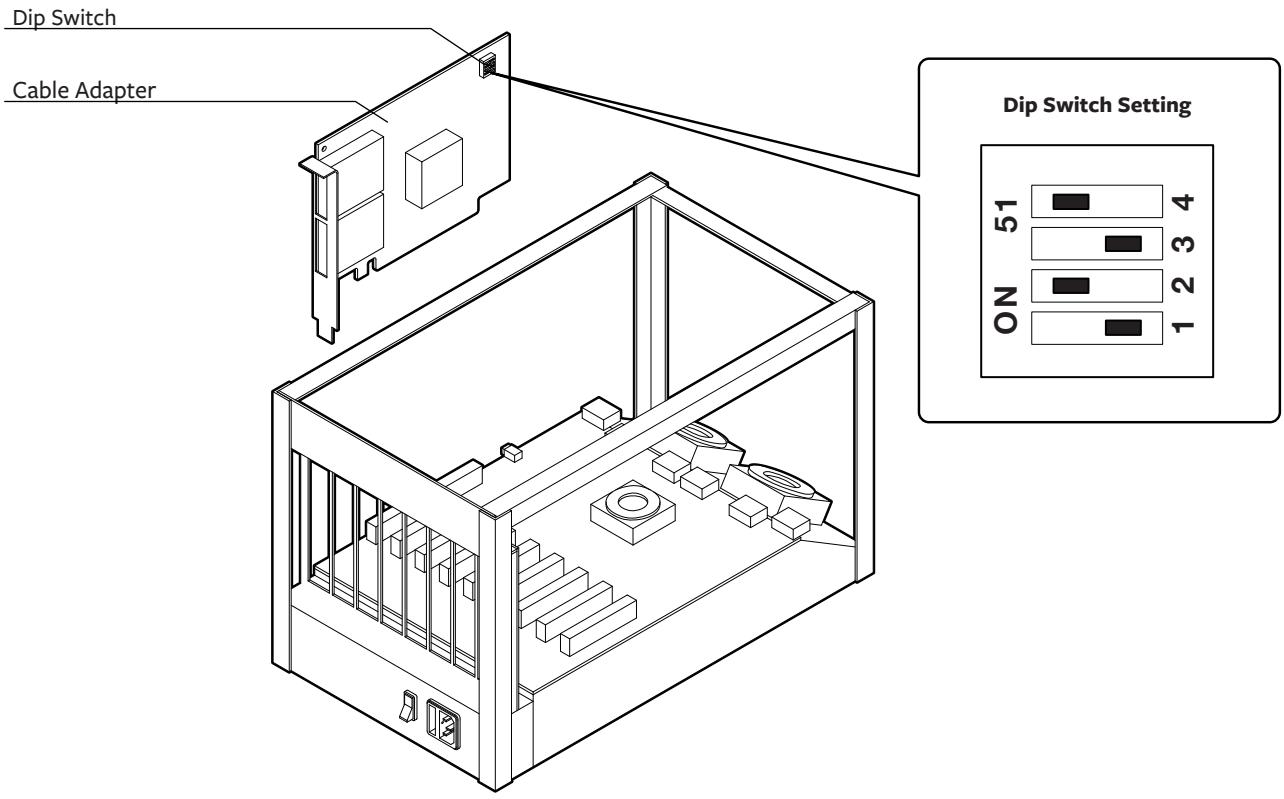
STEP 1

Insert jumper to a header on the backplane labeled J1



STEP 2

Insert PCIe x8 Gen 3 cable adapter and set dip switch setting

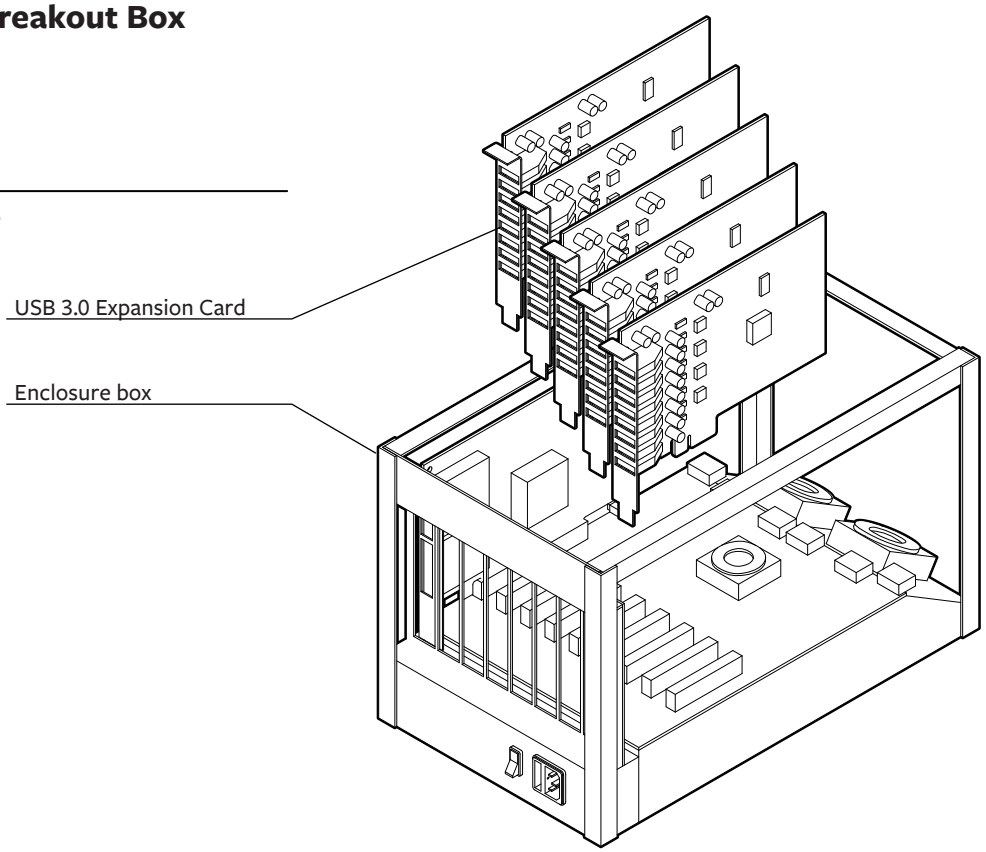


5 | 2b.

Fiber Optic Breakout Box

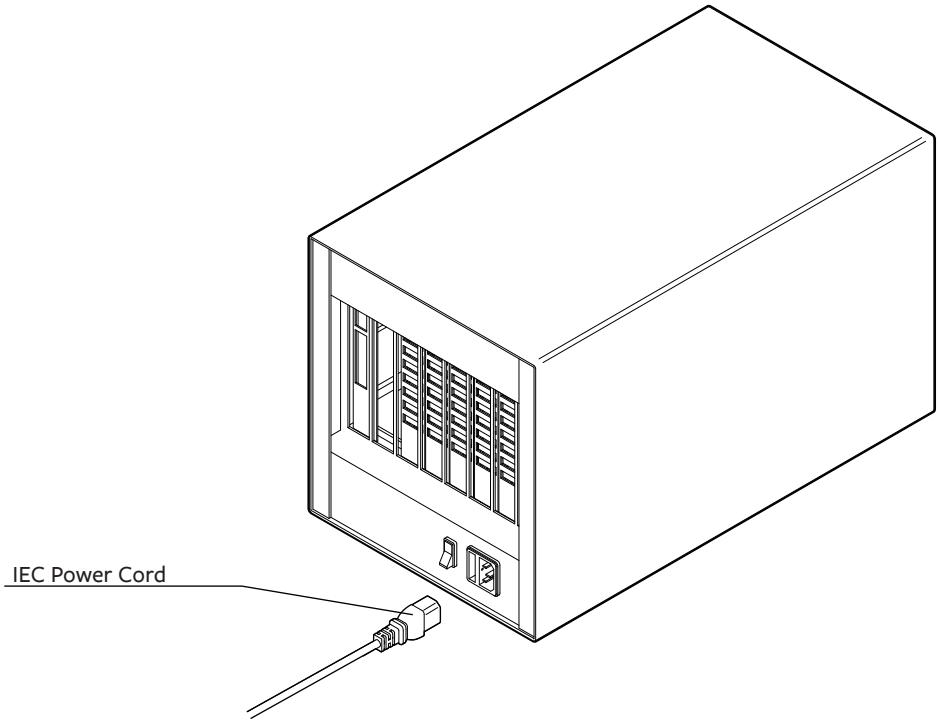
STEP 3

Insert 5 PCIe x4 USB 3.0 Expansion Cards



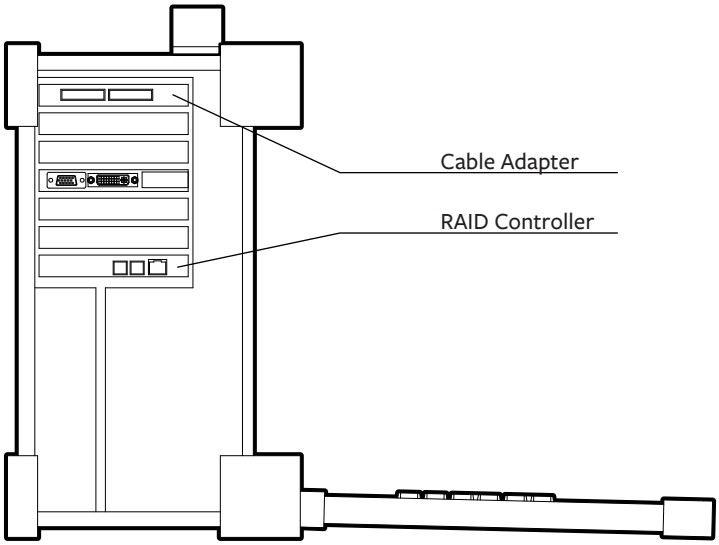
STEP 4

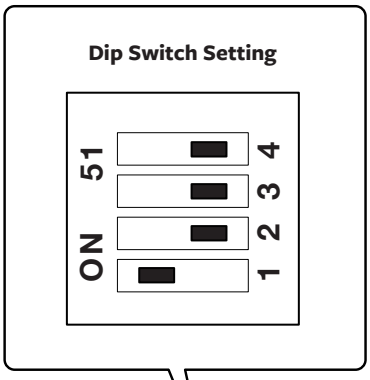
Close enclosure and connect power cable



5 | 3.

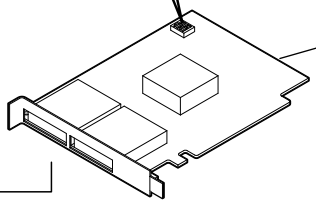
Camputer - Lunchbox PC





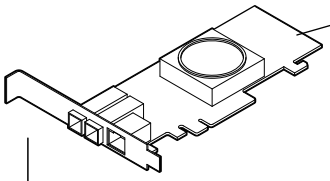
STEP 1

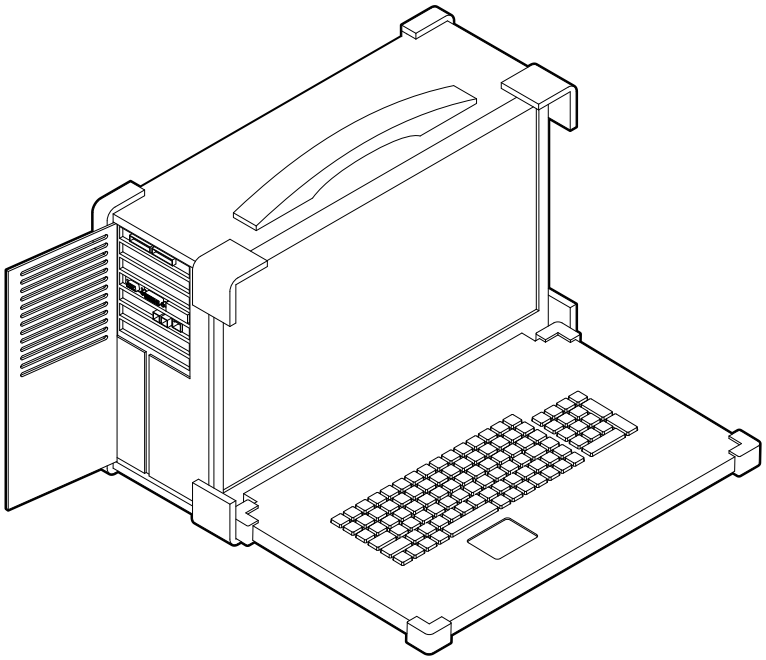
Insert PCIe x8 Gen 3 cable adapter and set dip switch setting



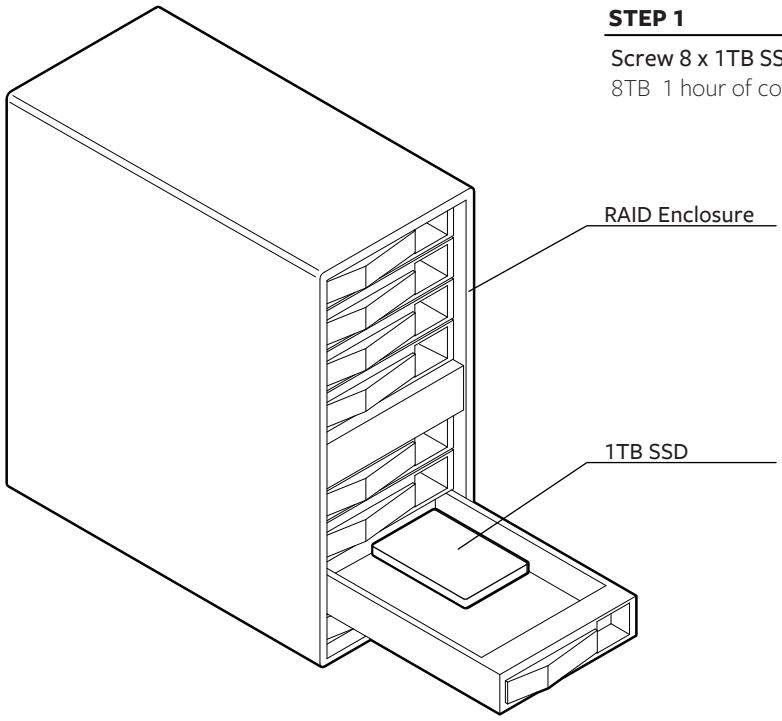
STEP 3

Insert PCIe x8 Gen3 SAS RAID controller





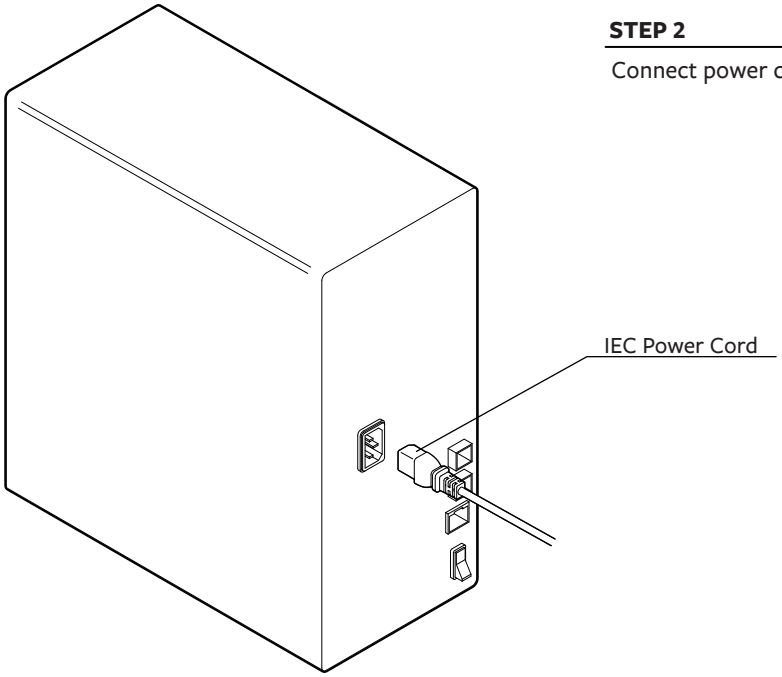
5 | 4.
RAID Tower



STEP 1

Screw 8 x 1TB SSDs into enclosure

8TB 1 hour of continuous raw video capture

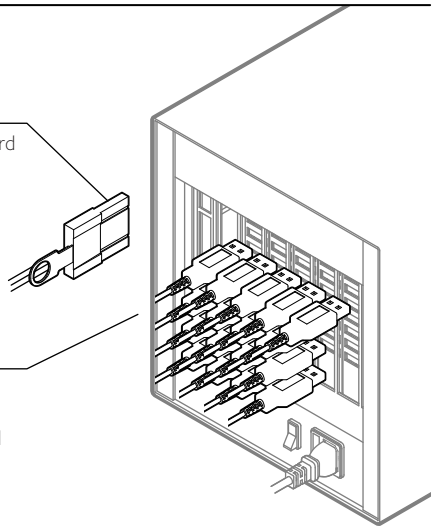


STEP 2

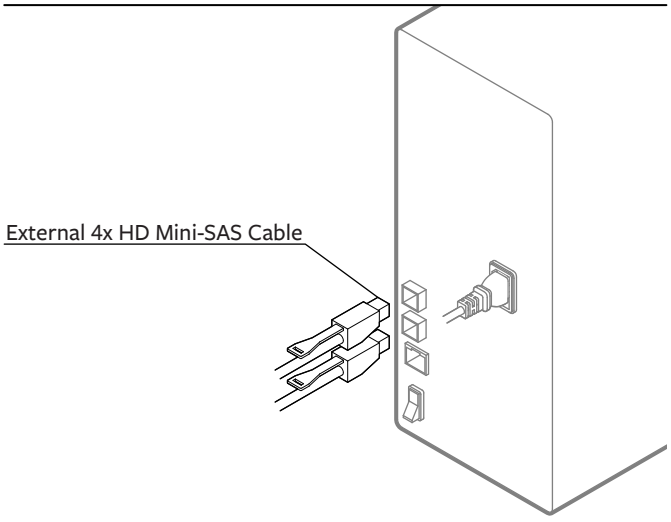
Connect power cable

5 | 5.
Cable Connections

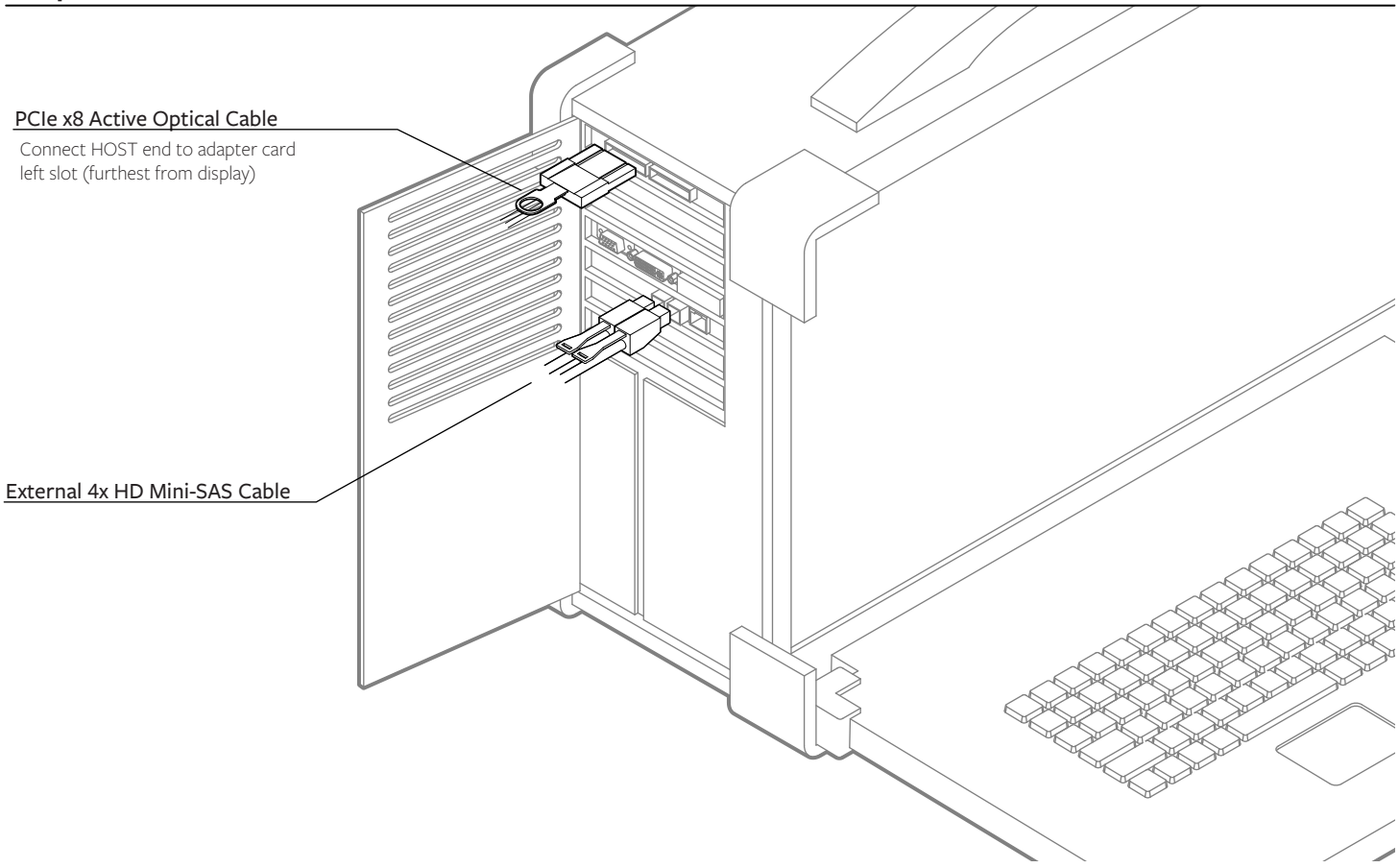
Fiber Optic Breakout Box



RAID Enclosure



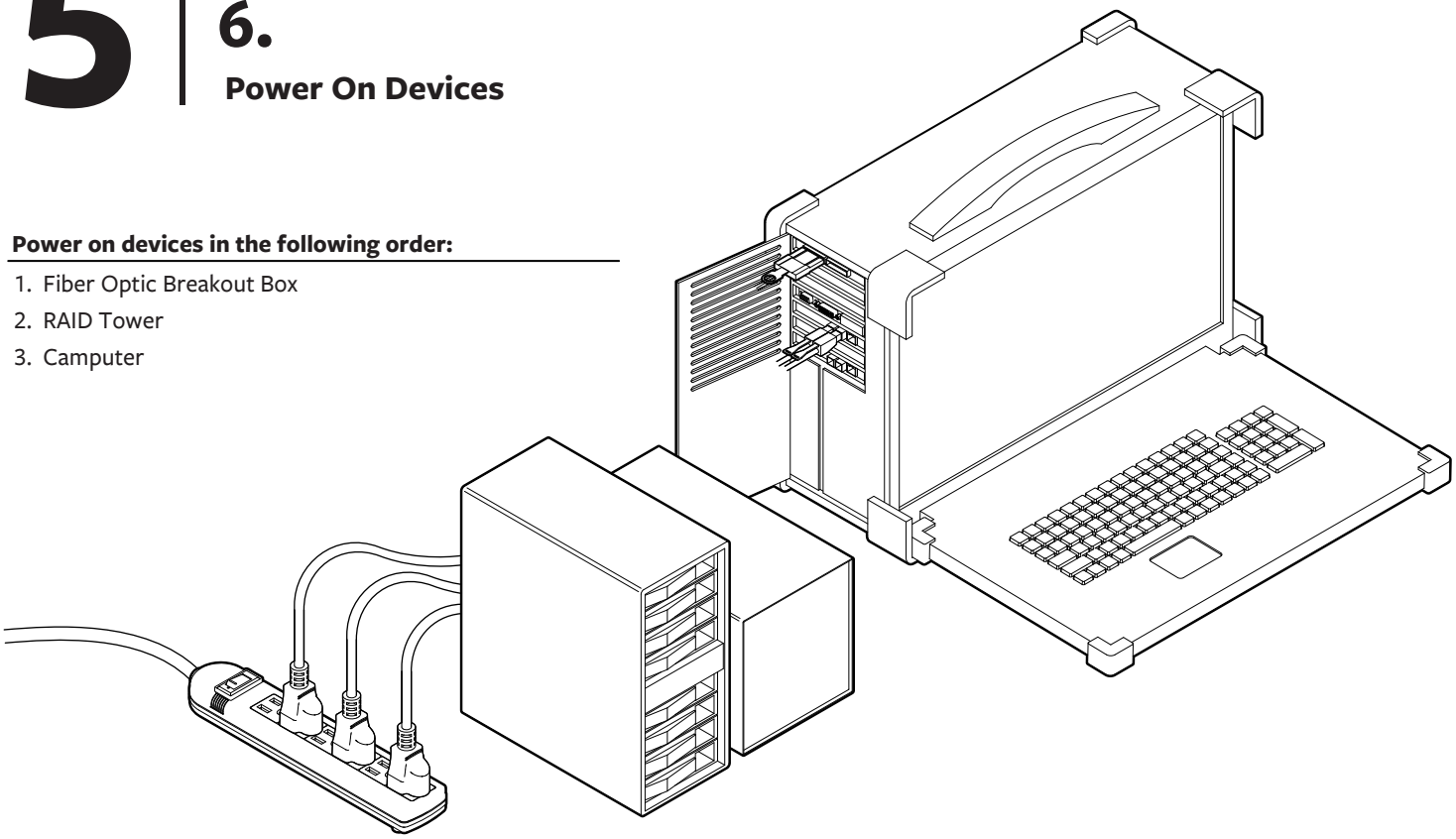
Camputer



6.
Power On Devices

Power on devices in the following order:

1. Fiber Optic Breakout Box
2. RAID Tower
3. Camputer



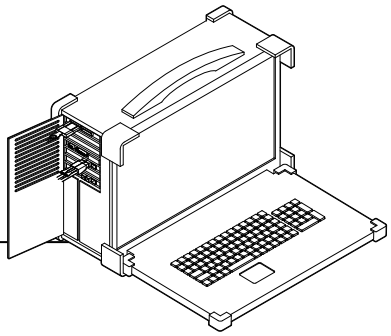
7.
Set Up Surround 360 Camera Control Software

6

Capturing & Rendering Instructions

6 | 1a.

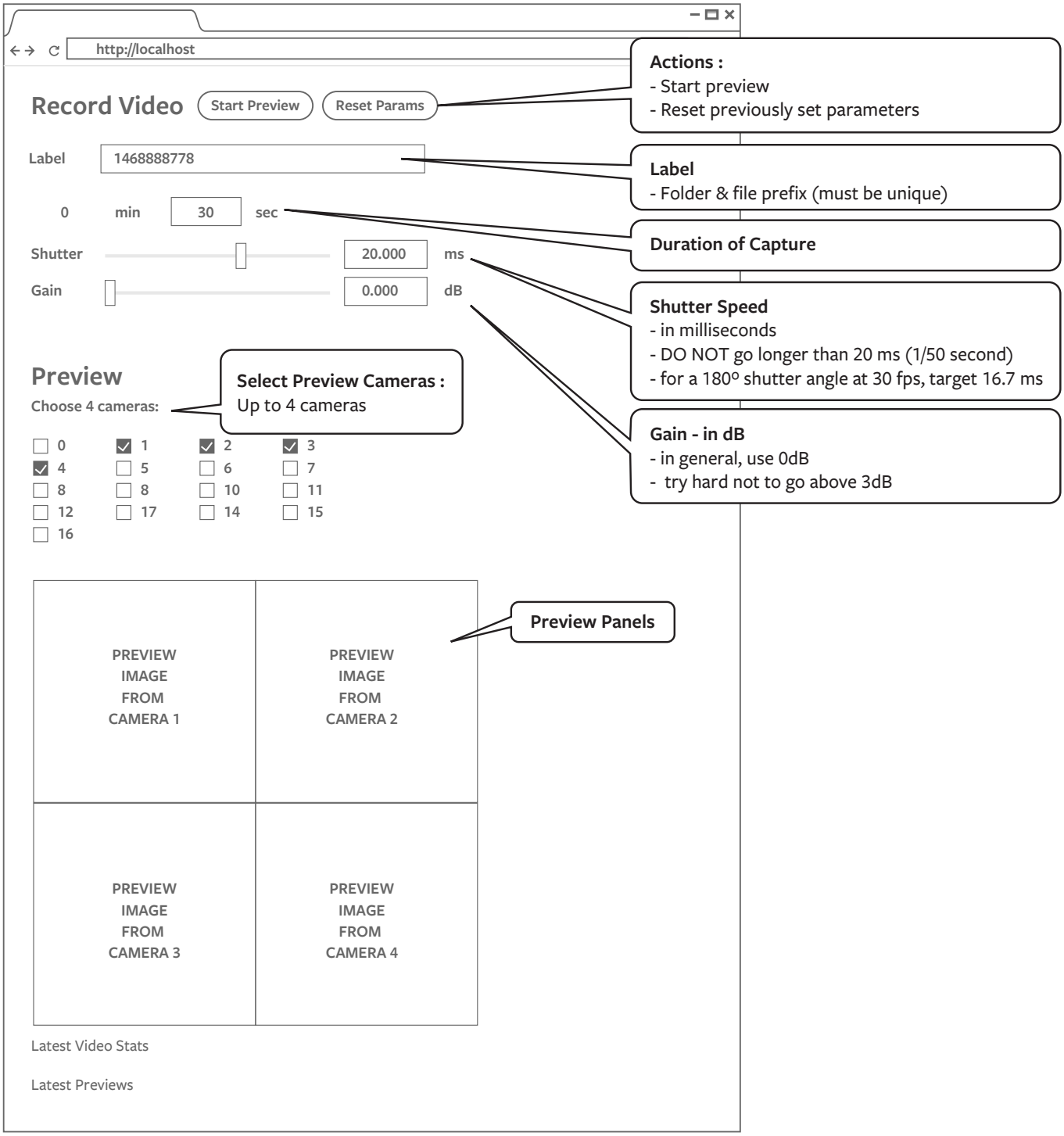
How to Capture with Web GUI



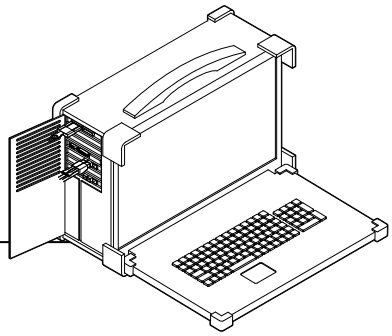
Step 1. Start Preview

To access the web GUI, open a browser (Chrome preferred; no Safari) and navigate to **http://localhost**

The web capture GUI provides all of the controls necessary to operate the Surround 360 and preview live images from any 4 of its 17 camera



6 | 1b.
Start Recording



Step 2. Start Recording

GUI control changes once you hit the start program button.
You will be able to stop the capture and update the live preview while the camera is capturing.

http://localhost

Record Video [Stop Preview] [Start Recording] [Update Preview]

Label 1468888778

0 min 30 sec

Shutter 20.000 ms

Gain 0.000 dB

Actions :
- Stop preview
- Start recording
- Update preview with new parameters

Preview

Choose 4 cameras:

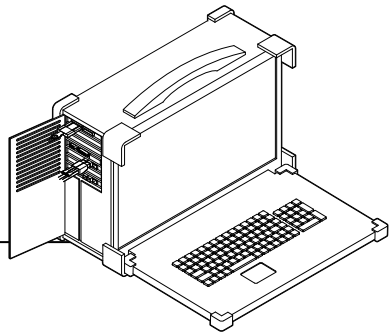
☐ 0	☑ 1	☑ 2	☑ 3
☑ 4	☐ 5	☐ 6	☐ 7
☐ 8	☐ 8	☐ 10	☐ 11
☐ 12	☐ 17	☐ 14	☐ 15
☐ 16			

PREVIEW IMAGE FROM CAMERA 1	PREVIEW IMAGE FROM CAMERA 2
PREVIEW IMAGE FROM CAMERA 3	PREVIEW IMAGE FROM CAMERA 4

Latest Video Stats

Latest Previews

6 | 1c.
Stop recording



Step 3. Recording in Progress

GUI control changes once you hit the start program button.
You will be able to stop the capture and update the live preview while the camera is capturing.

http://localhost

Action:
- Stop recording

Record Video [Stop Recording]

Elapsed time: 00 m 20 s, FPS: 29.9, Dropped frames so far: 0

Progress Info
- Elapsed time, frames per second and drop frames

Shutter 20.000 ms

Gain 0.000 dB

Preview

Choose 4 cameras:

☐ 0	☑ 1	☑ 2	☑ 3
☑ 4	☐ 5	☐ 6	☐ 7
☐ 8	☐ 8	☐ 10	☐ 11
☐ 12	☐ 17	☐ 14	☐ 15
☐ 16			

PREVIEW IMAGE FROM CAMERA 1	PREVIEW IMAGE FROM CAMERA 2
PREVIEW IMAGE FROM CAMERA 3	PREVIEW IMAGE FROM CAMERA 4

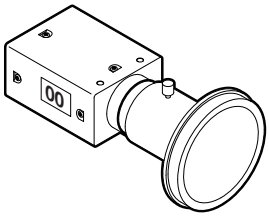
Latest Video Stats

Latest Previews

Tips

It is safe to stop capture midway

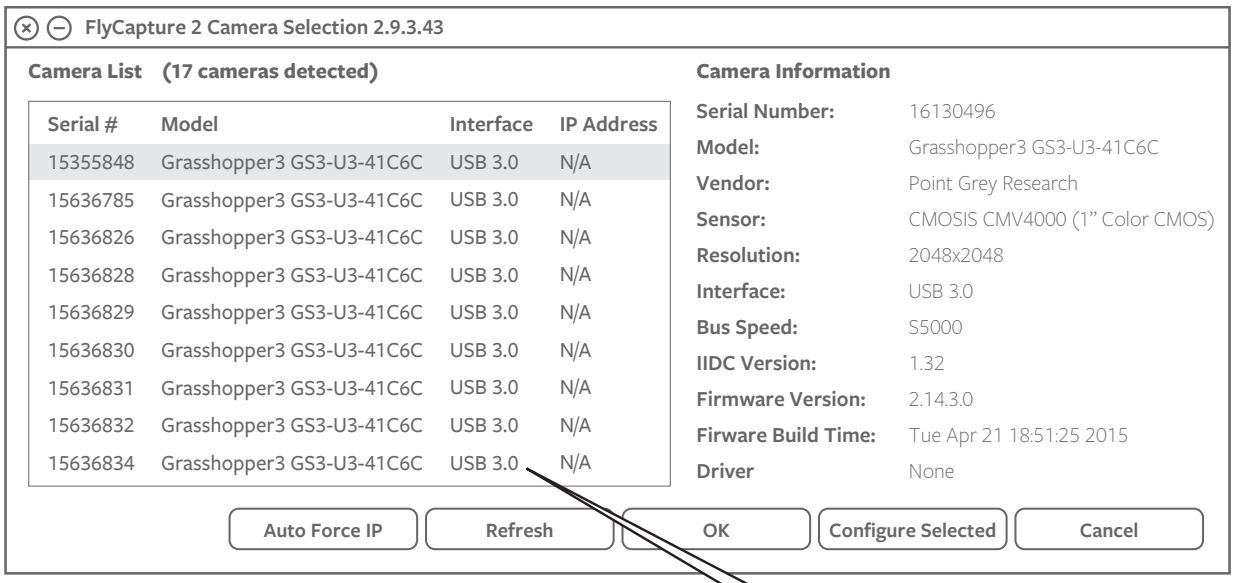
6 | 2. How to Verify Camera Connections (Optional)



Open FlyCapture (Point Grey capturing software)

```
| flycap
```

- Confirm all connections are USB 3.0 and total of 17 cameras detected



Make sure ALL 17 cameras are recognized as USB 3.0

Optional : Check focus by clicking on individual camera

All cameras are pre-focused, but critical focus should be checked before capturing. To check the focus of each lens, open a live preview by double clicking on the camera. Manually adjust the focus of the camera.

Tips

If a camera is not connected, or a camera says it is connected using USB 2.0, unplug cables from the breakout box and re-seat them in different buses until all 17 cameras are connected as USB 3.0. Remember that each camera must be alone in a USB 3.0 port pair in order to satisfy the requirement that there is only 1 camera per bus.

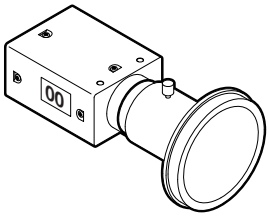
If this doesn't resolve the problem, try the following steps:

1. quit flycap
2. reset physical USB connections
3. run the following USB reset command:

```
| /home/facebook1/vr_camera_hw/scripts/usbreset.sh
```

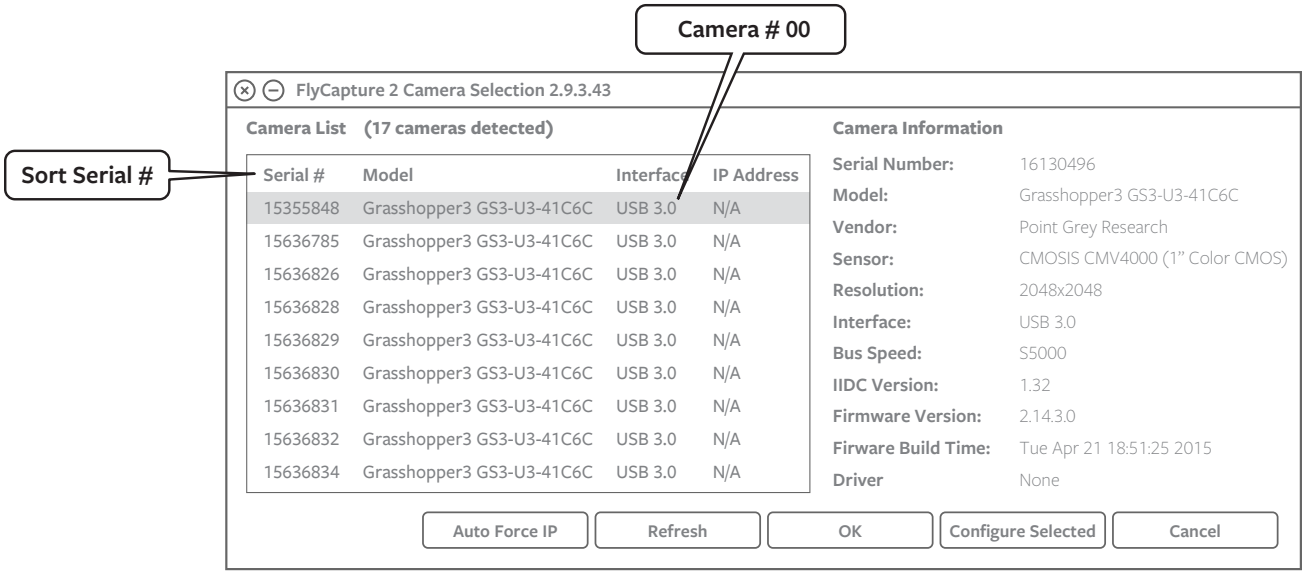
6 | 3a.

How to use Flycapture to Preview (Optional)



Step 1. Select a camera to preview

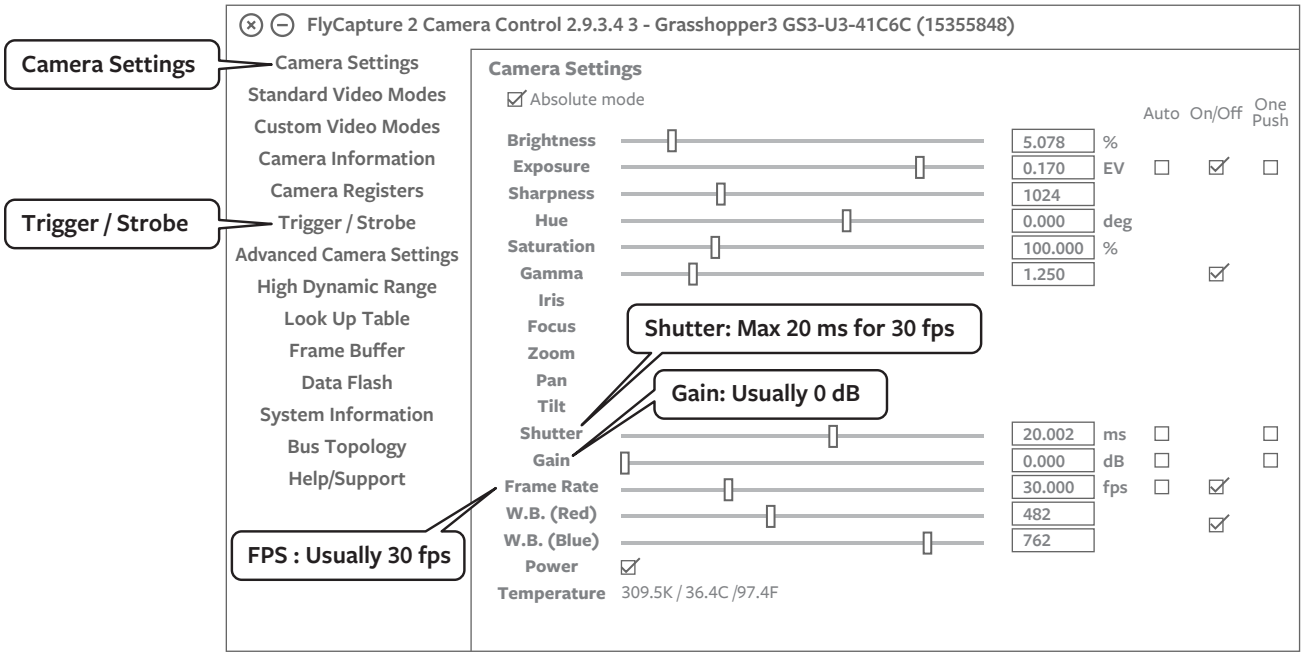
Camera are arranged in ascending sequential order starting with top camera (# 00).



Step 2. Go to setting panel from preview panel

Set the following values:

- shutter speed in milliseconds
- gain (usually 0)
- framerate (usually 30)

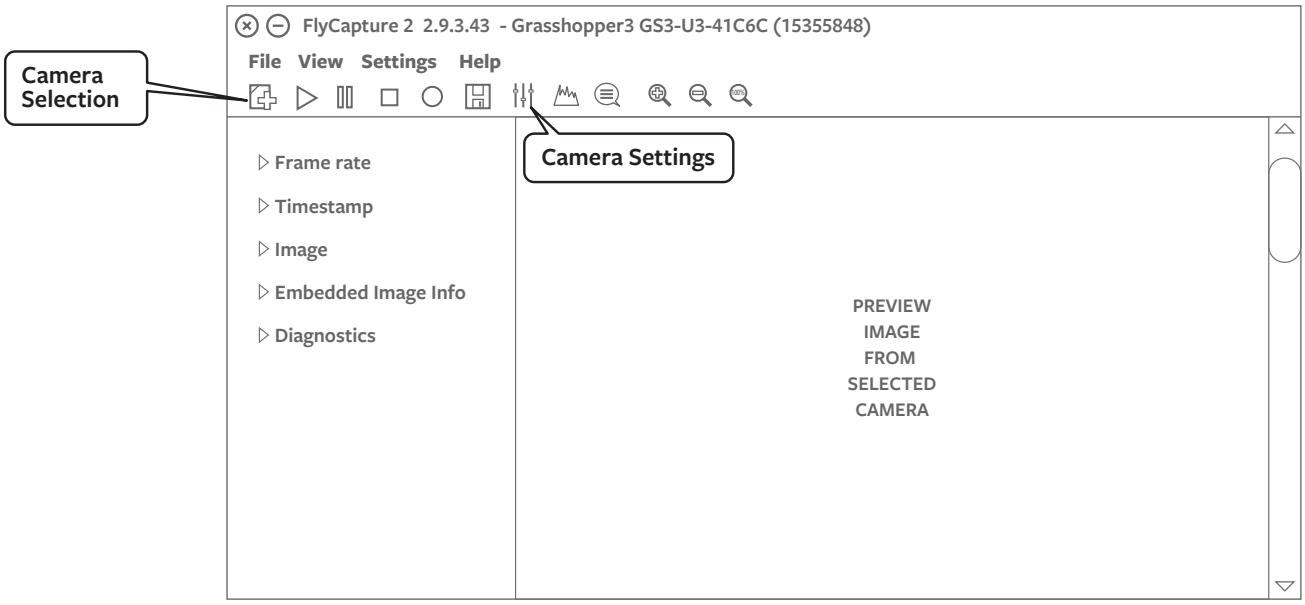


6 | 3b.

How to use Flycapture to Preview (Optional)

Step 3. Back to preview panel to check focus

All cameras are pre-focused, but critical focus should be checked before capturing. To check the focus of each lens, open a live preview by double clicking on the camera. Manually adjust the focus of the camera.



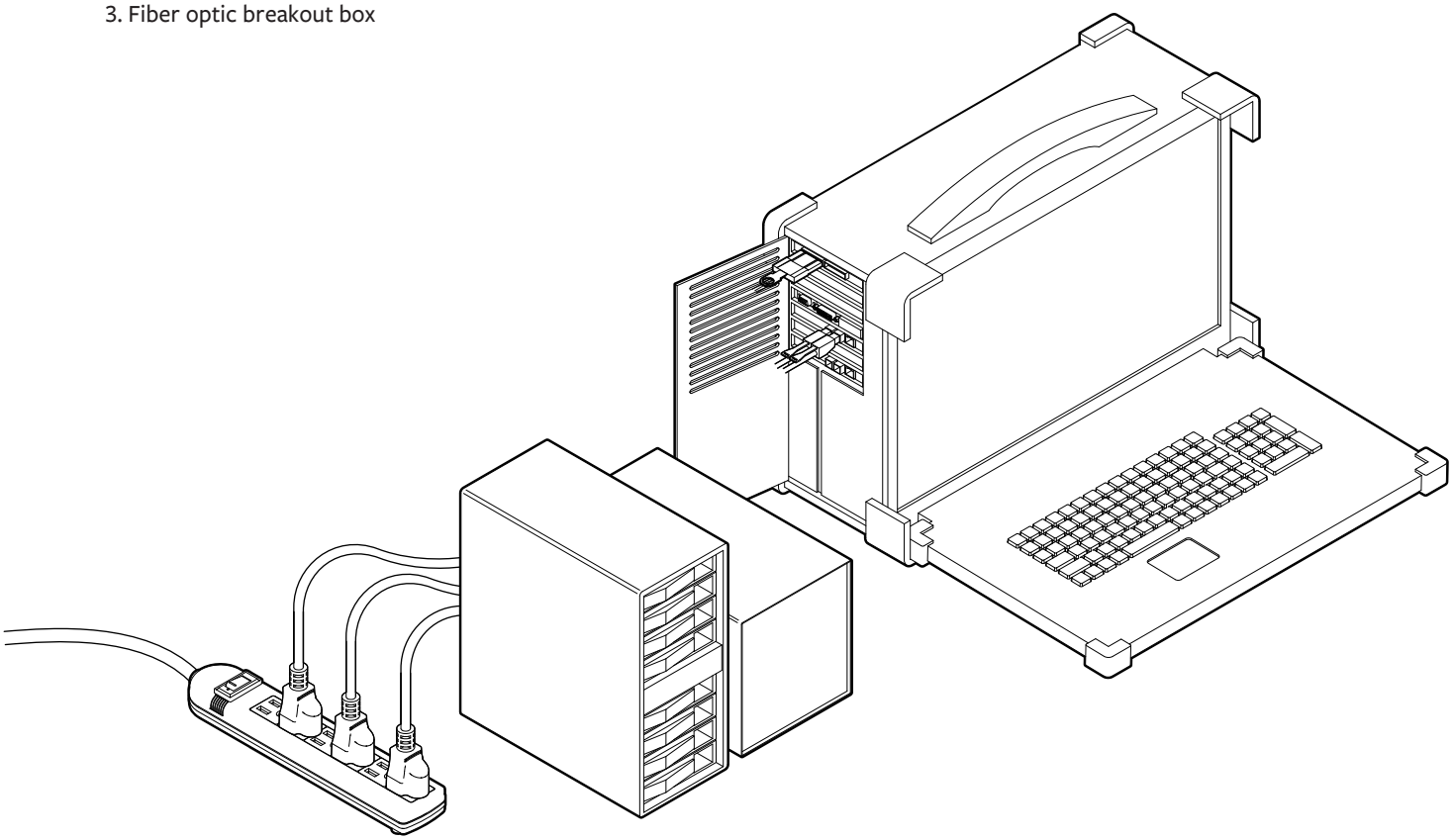
Tips

Each time you capture using the Surround 360, all camera settings are set to capture settings, and preview is disabled. When you load flycap (e.g., to change settings), you'll see the Point Grey logo instead of a preview. To re-enable preview, find the "Trigger/Strobe" settings and uncheck "Trigger Control."

6 | 4. Shutting Down System

Power down devices in the following order:

1. Camputer
2. RAID Tower
3. Fiber optic breakout box



Tips

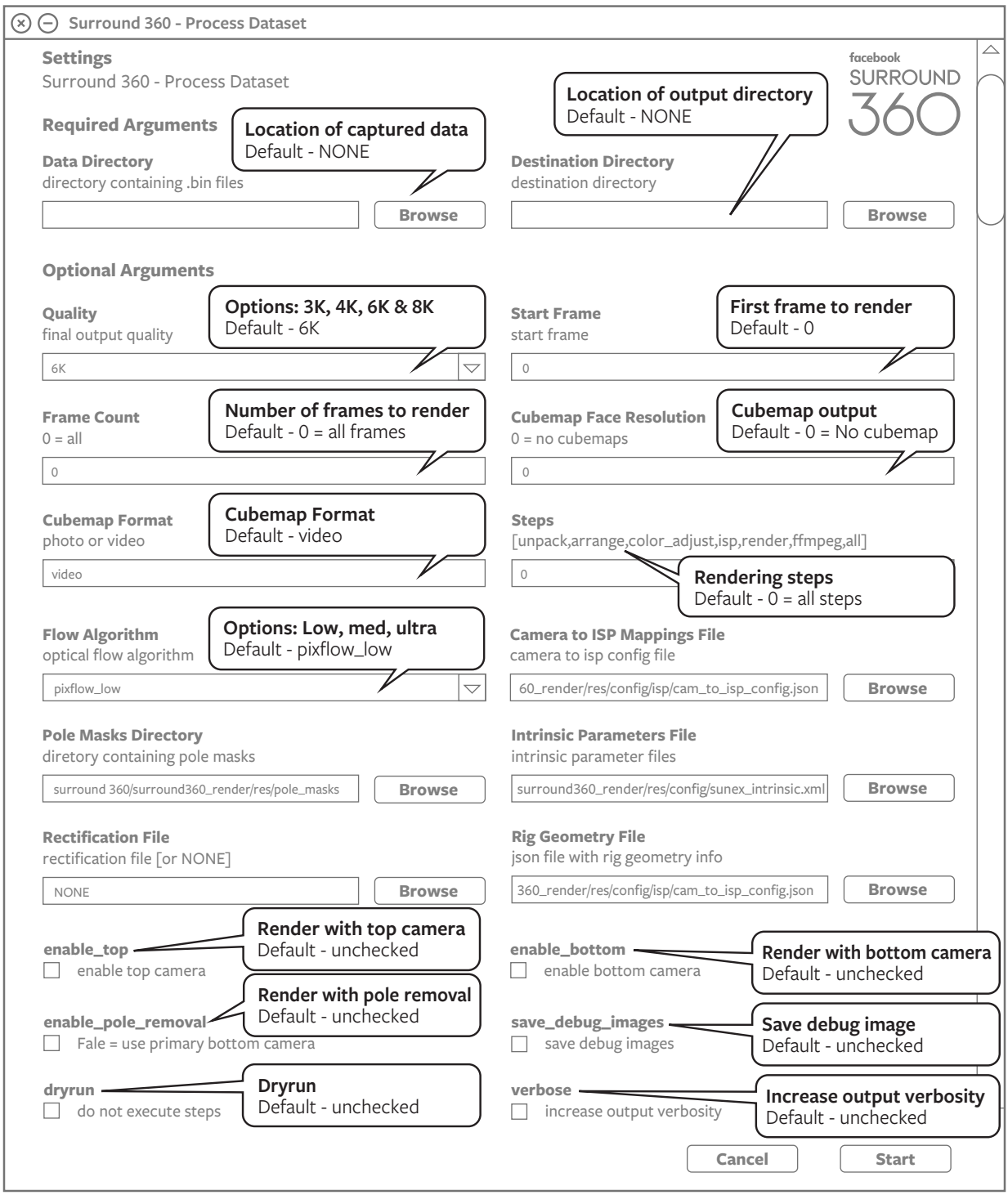
Never disconnect the RAID before unmounting it. The safest way to unmount the RAID is to shut the computer down.

6 | 5. How to Render

To start the rendering process

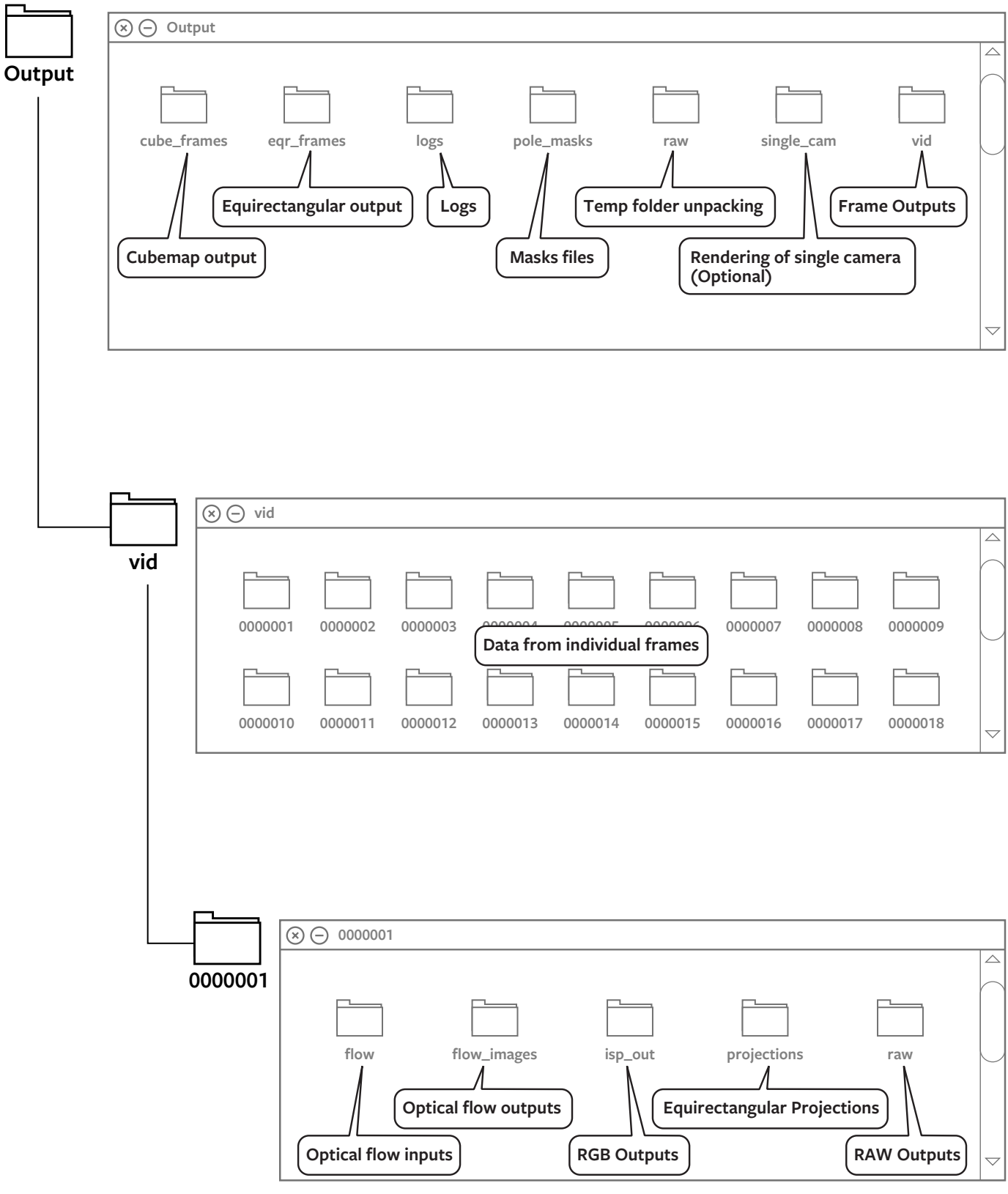
Run the following script

```
cd <path_to_surround360>/surround360_render && python scripts/run_all.py
```



6

6.
Output Directory



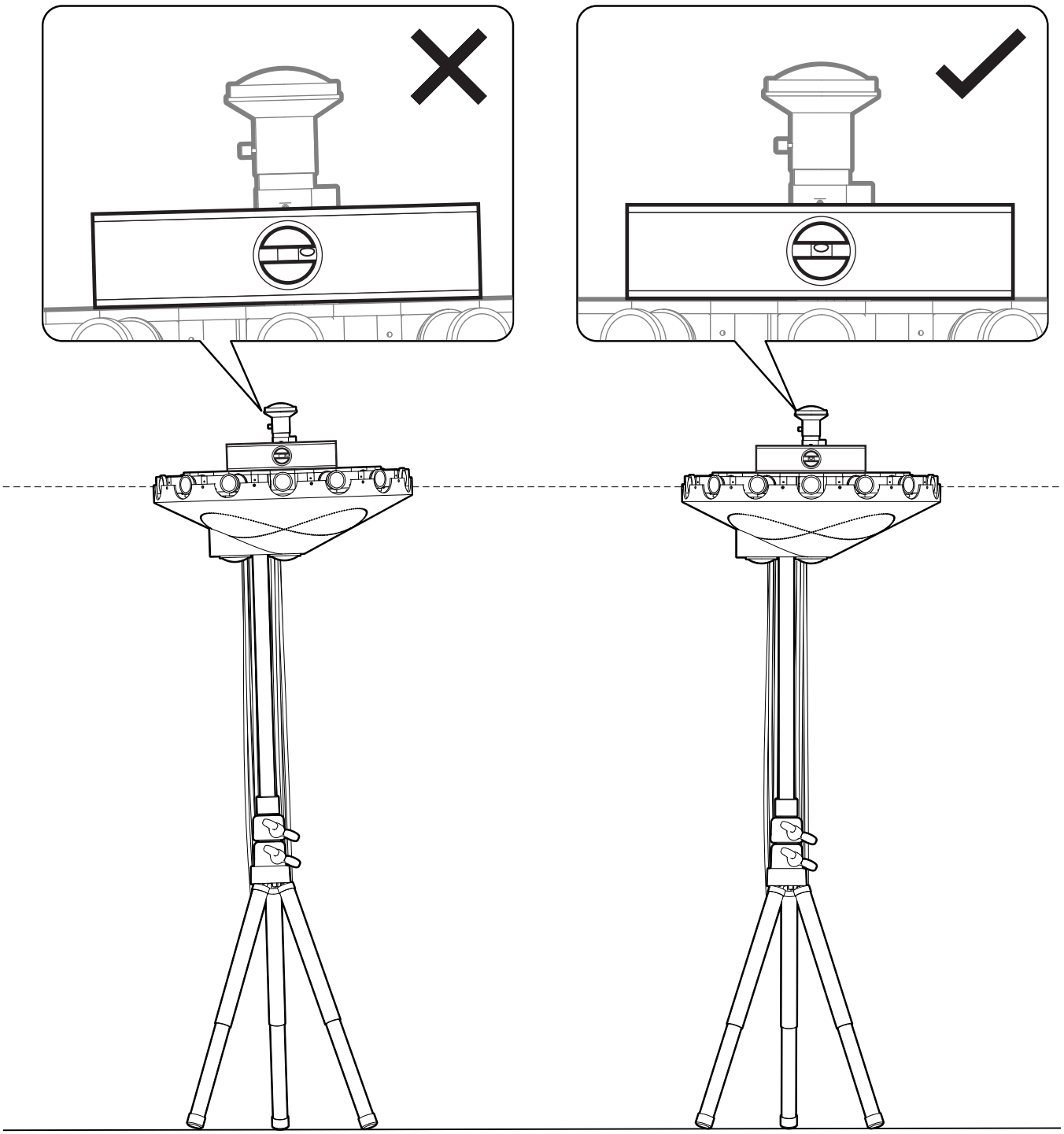
7

Pro Tips

7 | 1.

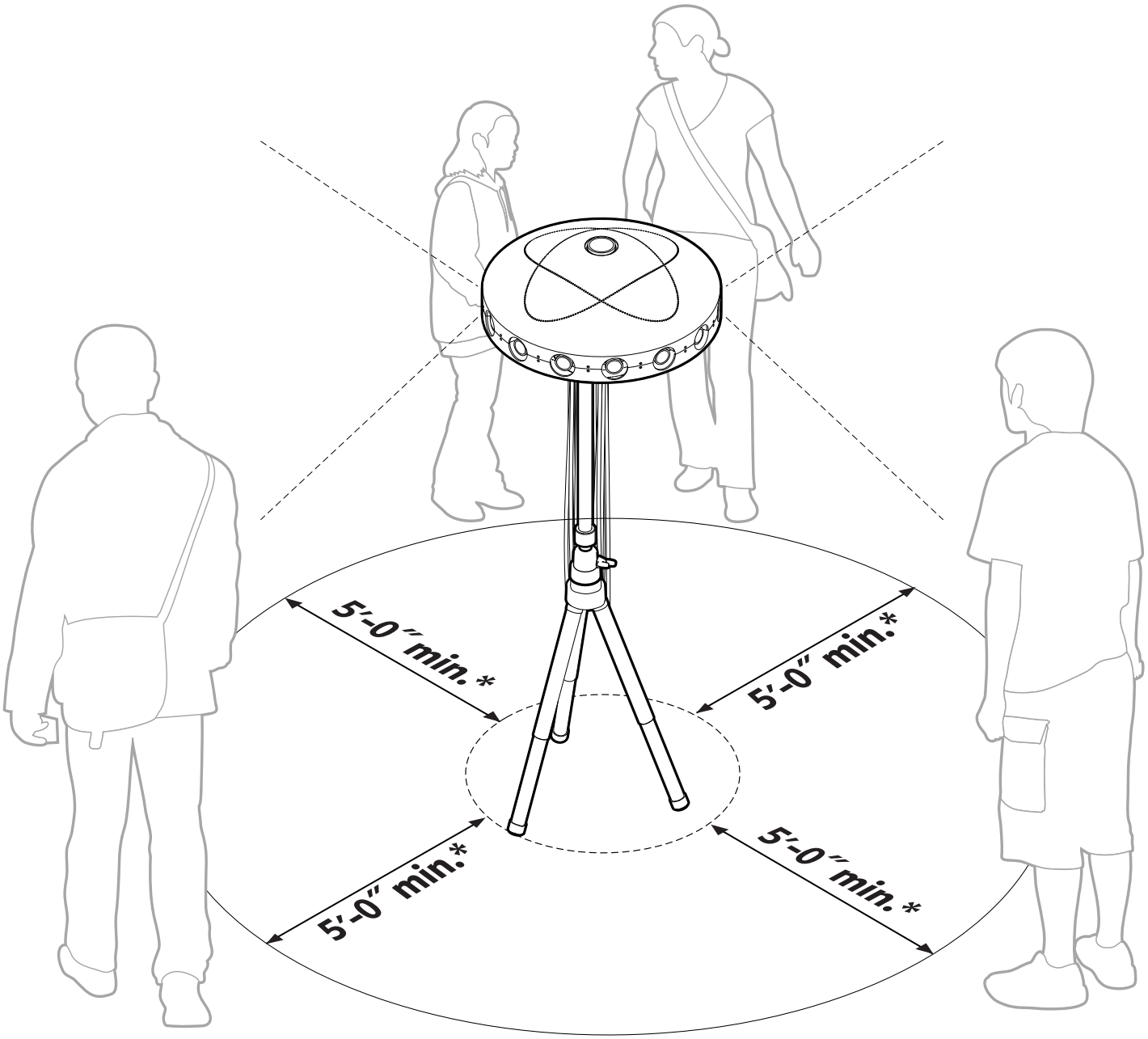
Level Camera for Level Horizon Line

Remove top shell first!



7 | 2.

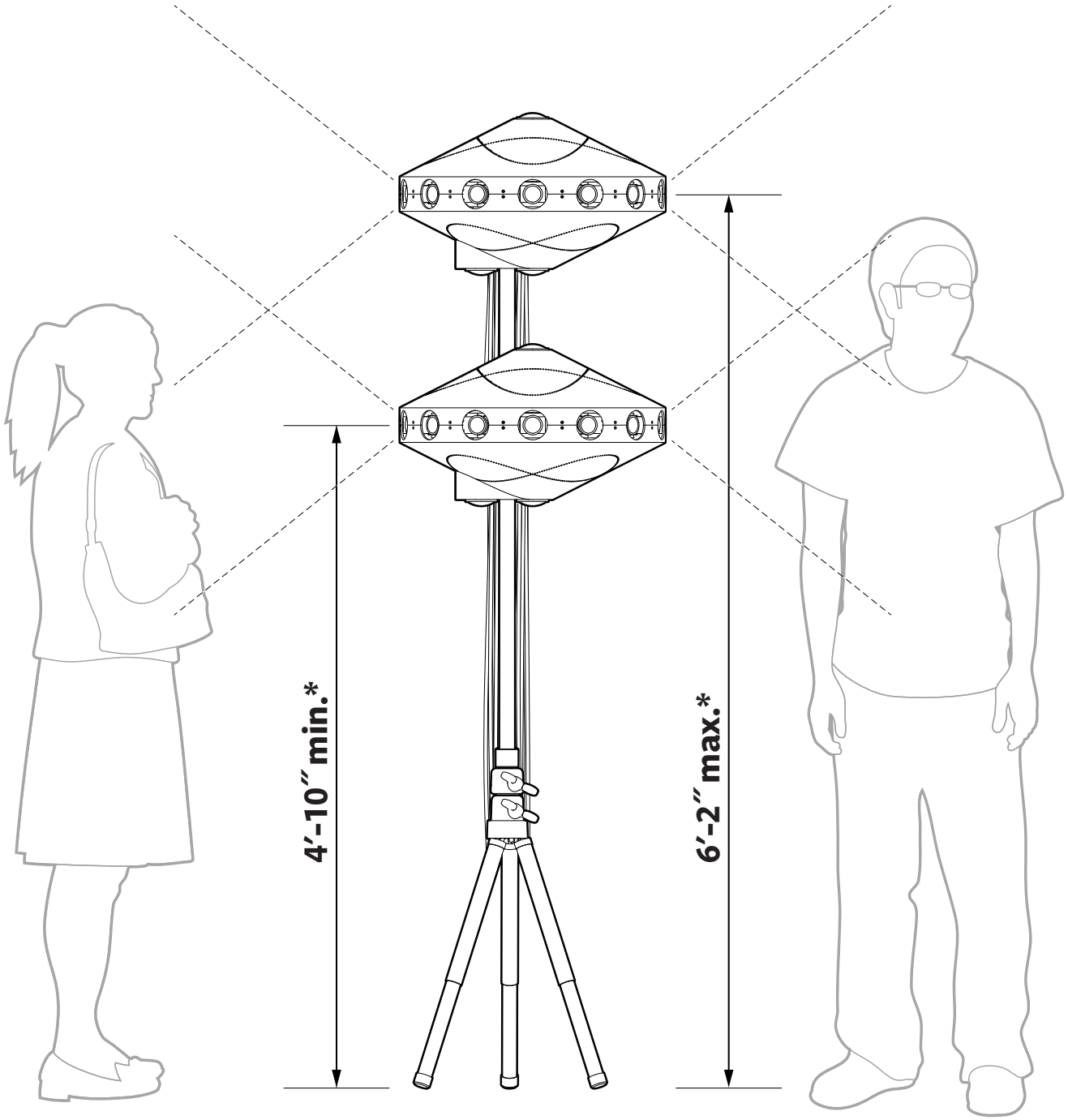
Avoid Close Objects



** recommendation only*

3.

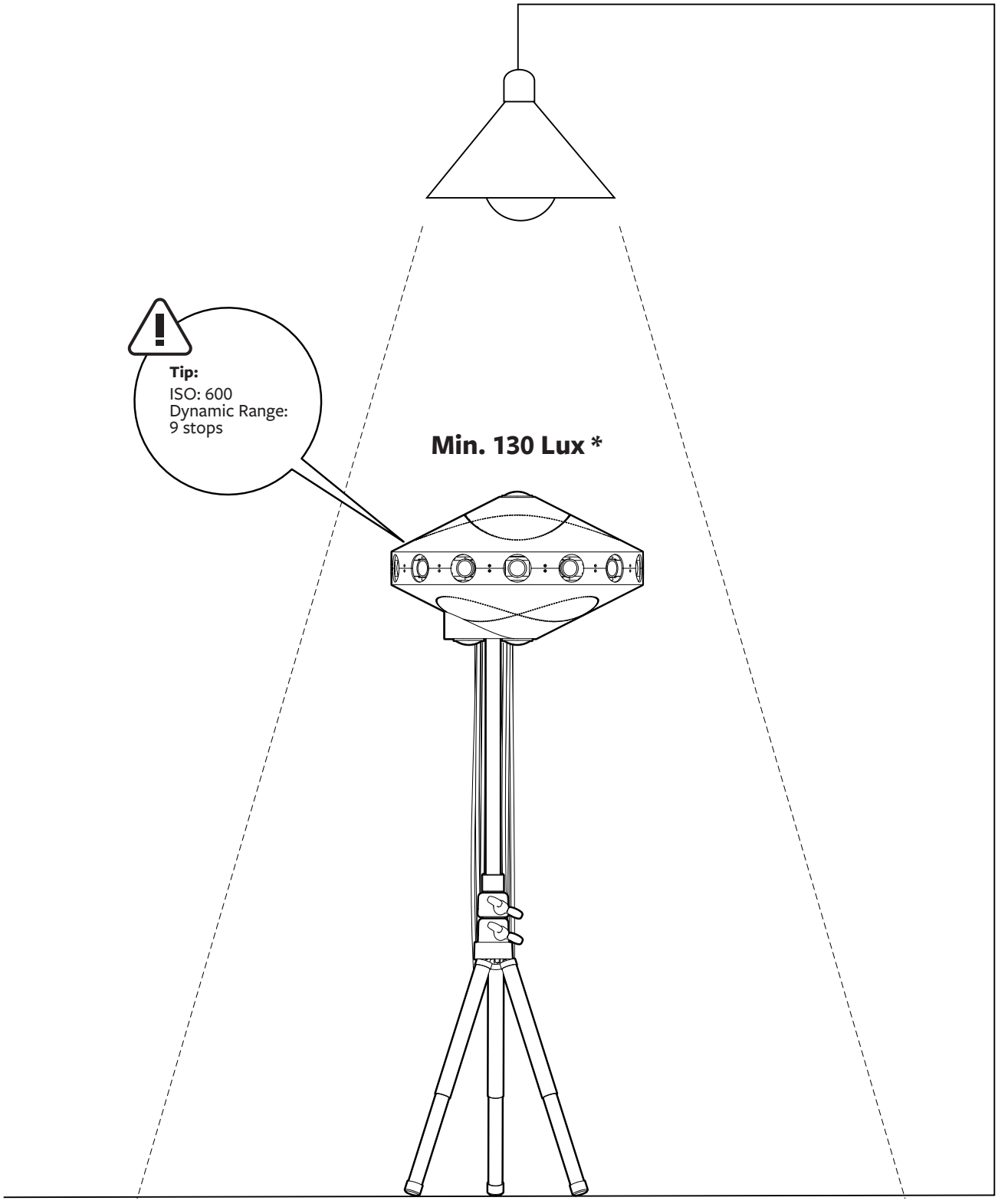
Be Aware of Camera Height for Better VR Experience



** recommendation only*

4.

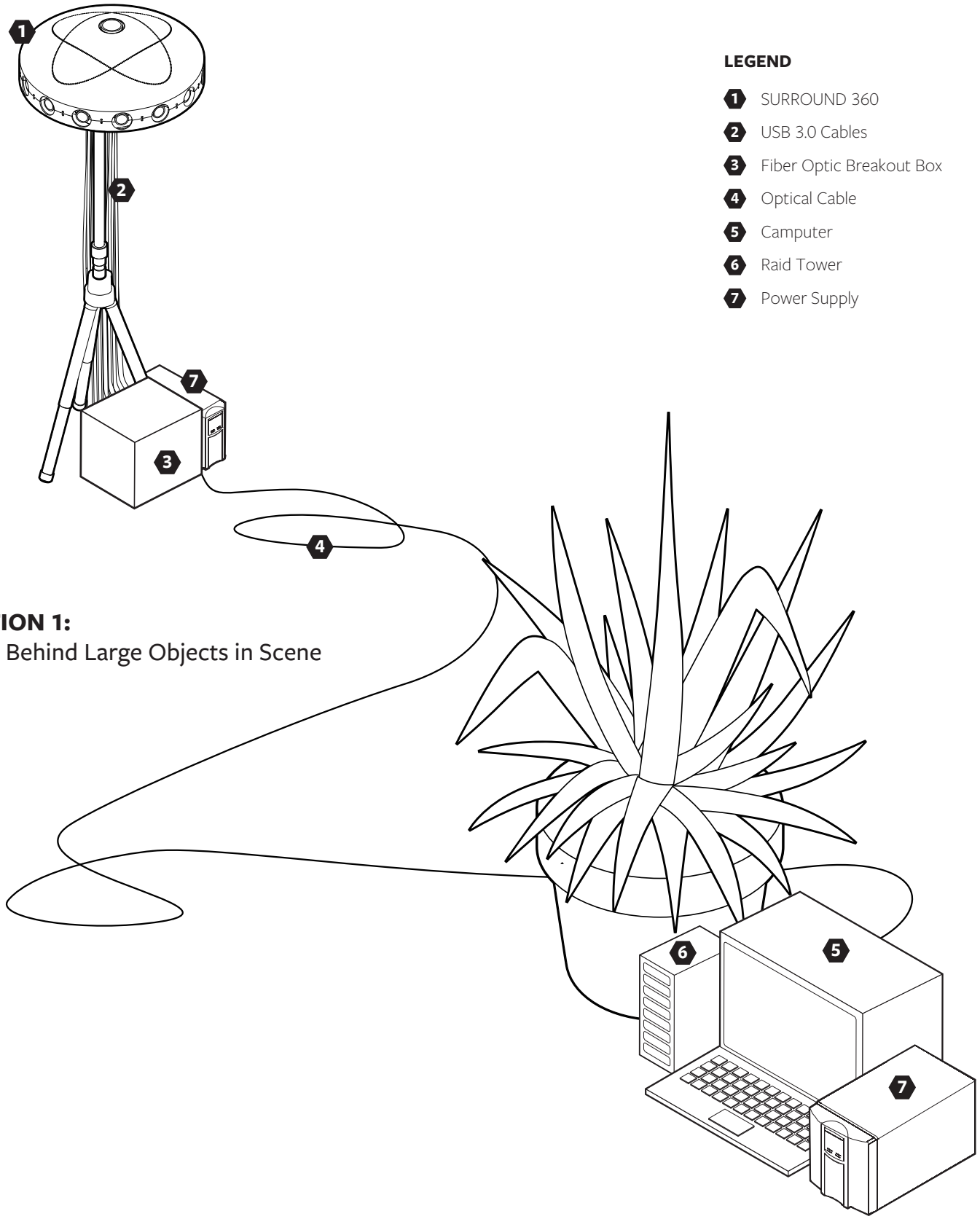
Avoid Low Light Condition to Minimize Noise in Footage



** recommendation only*

7 | 5a.

Placement of Surround 360



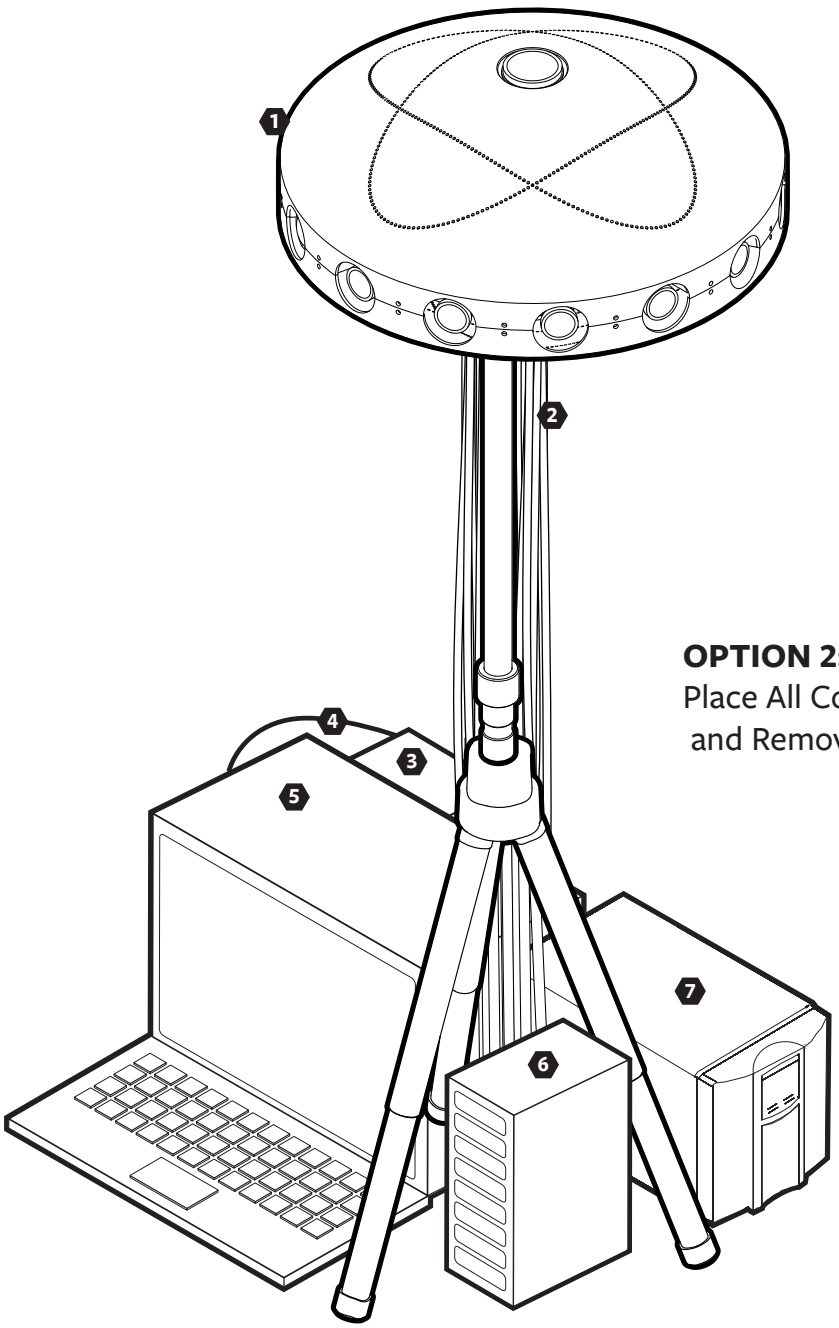
LEGEND

1. SURROUND 360
2. USB 3.0 Cables
3. Fiber Optic Breakout Box
4. Optical Cable
5. Camputer
6. Raid Tower
7. Power Supply

OPTION 1:
Hide Behind Large Objects in Scene

7 | 5b.

Placement of Surround 360



LEGEND

1. SURROUND 360
2. USB 3.0 Cables
3. Fiber Optic Breakout Box
4. Optical Cable
5. Camputer
6. Raid Tower
7. Power Supply

OPTION 2:
Place All Components Beneath Camera and Remove in Post-Production

8

Camera Specifications

8 | 1. Specifications

Type	Spherical and stereoscopic video capture
Sensor Type	CMOS, global shutter
Sensor Name	CMOSIS CMV4000-3E5
Sensor Array	17 synchronized 1” sensors
Pixel Size	5.5 µm
ADC	10 bit
ISO	600
Dynamic range	9 stops
Coverage Area	Full spherical, 360 x 180 degrees
Stereoscopic Coverage	Centerline +/- 144(h), 77(v) degrees
Lens Angle of View	77 degrees diagonal (center), 185 degrees (top and bottom)
Relative Aperture	f/2.4 (wide-angle lens), f/1.8
Lens Mount	C mount
Resolution	8192 x 4096 per eye equirect * 2 = 8192 x 8192
Interface	USB 3.0
Capture Format	RAW
Capture Frame Rate	30 fps (max 60 fps)
Bandwidth	17 Gb/s (@ 30 fps)
Dimensions - camera only	460 x 460 x 307.65mm
Dimensions - with mount	460 x 460 x 796.60mm
Total Weight - camera only	16 kgs
Construction	Milled aluminum alloy & steel
Color	Black
Operating Temperature	0° to 50°C
Storage Temperature	-30° to 60°C
Operating Humidity	20% to 80% (no condensation)
Storage Humidity	20% to 95% (no condensation)
Power Consumption	350 watts max

8 | 2a.

Bill of Material

CAMERA & LENSES		
Point Grey Camera Body	GS3-U3-41C6C-C	17
Sunex Lens	DSL318	14
Fujinon Fisheye Lens	FE185C086HA-1	3
MACHINED PART		
Base Plate	FB360_V1_21	1
Top Plate	FB360_V1_22	1
Upright	FB360_V1_23	1
Camera Bracket	FB360_V1_24	1
Post	FB360_V1_25	1
Support Tube	FB360_V1_26	1
Adapter	FB360_V1_27	1
Shell Support	FB360_V1_28	14
Bottom Cover	FB360_V1_29	1
Top Cover	FB360_V1_30	1
Stop Nut	FB360_V1_31	1
Threaded Rod	FB360_V1_32	1
Lens Mounts/Barrels	FB360_V1_33	14
FASTENER		
M3 X 6 SHCS 18-8		152
M3 Lockwasher		152
M6 X 14 FHCS 18-8		8
M6 X 50 FHCS 18-8		2
M6 X 35 SHCS 18-8		4
M3 X 8 BHCS		28
5/16-18 Flange Nut		1
M3 Jam Nut (Optional)		14
M2 x 20 PHCS (Optional)		10

8 | 2b.

Bill of Material

COMPUTER

Ruggedized “Lunchbox” Computer	APOLLO-A1 MODEL	1

- Intel Core i7-5960X Haswell-E 8-Core 3.0 GHz LGA 2011-v3
- GIGABYTE GA-X99P-SLI (rev. 1.0) LGA 2011-v3 Intel X99 Motherboard
- 8GB DDR4 2400 288-PIN Memory ± (64GB of Memory Installed)
- CPU COOLING FAN FOR LGA 2011-v3
- 1GB NVIDIA PCI-EX16 VIDEO CARD
- 700 WATT POWER SUPPLY
- SAMSUNG 850 Pro Series 2.5″ 128GB SSD
- OPERATING SYSTEM - UBUNTU 14.04 LTS

PCIe x8 Gen 3 Cable Adapter	OSS-PCIe-HIB38-x8-DUAL	1
PCIe 3.0 x 8 SAS RAID Adapter	ARC-1883X	1

FIBER OPTIC BREAKOUT BOX

PCIe Expansion Enclosure	OSS-PCIe3-ENCL-M-CUBE3-8	1
Expansion Backplane	OSS-BP-452	1
PCIe x8 Gen 3 Cable Adapter	OSS-PCIe-HIB38-x8-DUAL	1
USB 3.0 Expansion Card	UE-1008 / UE-1004	5
Shunt Jumper	390088-1	1

RAID TOWER

8-bay 12G SAS RAID Tower	ARC-4038	1
1TB SSD	MZ 7KE1T0BW	8

CABLES

HR25 GPIO connector	FB360_V1_34	17
USB3 type A to micro B locking cables		17
External 4x HD Mini-SAS Cable	SFF-8644 to SFF-8644	2
PCIe x8 Active Optical Cable	OSS-PCIe3-CBL-ACT-x8-50M-1x	1

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